

UNIVERSITY OF THESSALY

SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Separation, Classification, and Localization of Acoustic Events from Multi-channel Audio Signals

Diploma Thesis

Moirotsos Georgios

Supervisor: Potamianos Gerasimos

September, 2026



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ΠΑΝΕΠΙΣΤΗΜΙΟ ΘΕΣΣΑΛΙΑΣ
ΠΟΛΥΤΕΧΝΙΚΗ ΣΧΟΛΗ
ΤΜΗΜΑ ΗΛΕΚΤΡΟΛΟΓΩΝ ΜΗΧΑΝΙΚΩΝ ΚΑΙ ΜΗΧΑΝΙΚΩΝ ΥΠΟΛΟΓΙΣΤΩΝ

Διαχωρισμός, Ταξινόμηση και Εντοπισμός Ηχητικών Γεγονότων από Πολυκαναλικά Ηχητικά Σήματα

Διπλωματική Εργασία

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Acknowledgements

As this Thesis marks the final milestone of my five-year academic journey, I would really like to thank my parents and my brother who always supported me and believed in my effort and choices. Their encouragement and trust have been a source of strength and motivation throughout this journey. I would like to express my deepest appreciation to my supervisor, Prof. Gerasimos Potamianos, for his guidance, support, and patience throughout the duration of this Thesis. I would also like to thank Dr. Katerina Papadimitriou for offering me valuable advice, which has greatly contributed to the development of this research. Finally, I would like to thank all of my close friends for their friendship, encouragement, and for always being by my side. Their presence and support have made these years truly memorable.

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Moirotsos Georgios

Diploma Thesis

Separation, Classification, and Localization of Acoustic Events from Multi-channel Audio Signals

Moirotsos Georgios

Abstract

The analysis and understanding of complex acoustic scenes containing multiple simultaneously active sound sources constitutes a challenging problem in audio signal processing. In multichannel audio scenarios, this requires not only the identification of the active acoustic events, but also the separation of overlapping sources and the estimation of their spatial locations. This Thesis investigates methods for the separation, classification, and localization of acoustic events from multichannel audio mixtures.

The experiments are conducted on multichannel acoustic mixtures from the DCASE2025 Task4 dataset, which contains multiple simultaneously active acoustic events captured by multiple microphones. Source separation is first addressed as a blind source separation (BSS) problem, with limited prior knowledge of the underlying mixing process. The separation stage considers Independent Vector Analysis (IVA). The separated signals are then used for acoustic event classification (AEC) using a Convolutional Neural Network (CNN) architecture, while acoustic source localization (ASL) exploits their spatial information to estimate the directions of arrival (DOAs). The methods are evaluated using task-specific evaluation metrics, followed by a comparative analysis to identify the best-performing conditions.

The proposed framework combines source BSS, AEC, and DOA estimation within a unified processing pipeline. Moreover, it highlights the potential of BSS not only as an independent processing task, but also as a preprocessing stage for the classification and localization of acoustic events in complex multichannel scenes.

Keywords:

Multichannel acoustic mixtures, multichannel blind source separation, acoustic event classification, direction of arrival estimation, independent vector analysis, convolutional neural network, spatial information, DCASE 2025 task 4 challenge.

Διπλωματική Εργασία

Διαχωρισμός, Ταξινόμηση και Εντοπισμός Ηχητικών Γεγονότων από Πολυκαναλικά Ηχητικά Σήματα

Μοιρώτσος Γεώργιος

Περίληψη

Η ανάλυση σύνθετων ακουστικών σκηνών, στις οποίες συνυπάρχουν ταυτόχρονα πολλαπλά ενεργά ηχητικά γεγονότα, αποτελεί ένα απαιτητικό πρόβλημα στο πεδίο της επεξεργασίας σήματος και ήχου. Στα πολυκαναλικά ακουστικά σενάρια, η ανάλυση αυτή απαιτεί όχι μόνο την αναγνώριση των ενεργών ηχητικών γεγονότων, αλλά και το διαχωρισμό των επικαλυπτόμενων πηγών καθώς και την εκτίμηση της χωρικής τους θέσης. Η παρούσα διπλωματική εργασία μελετά μεθόδους για τον διαχωρισμό, την ταξινόμηση και τον εντοπισμό ηχητικών γεγονότων από πολυκαναλικά ηχητικά μίγματα.

Τα πειράματα βασίζονται σε πολυκαναλικά ακουστικά μίγματα, αποτελούμενα από διάφορα ακουστικά γεγονότα και καταγεγραμμένα από πολλαπλά μικρόφωνα. Ο διαχωρισμός αντιμετωπίζεται ως πρόβλημα τυφλού διαχωρισμού πηγών (Blind Source Separation – BSS), με περιορισμένη γνώση της διαδικασίας μίξης. Εξετάζεται η μέθοδος (Independent Vector Analysis – IVA). Τα διαχωρισμένα σήματα χρησιμοποιούνται για την ταξινόμηση ακουστικών γεγονότων με αρχιτεκτονικές βαθιάς μάθησης (Convolutional Neural Network – CNN), ενώ η χωρική πληροφορία αξιοποιείται για την εκτίμηση των διευθύνσεων άφιξης (Directions of Arrival – DOAs). Οι μέθοδοι αξιολογούνται με κατάλληλες μετρικές και συγκρίνονται ως προς την απόδοσή τους.

Το προτεινόμενο πλαίσιο συνδυάζει τον διαχωρισμό ηχητικών πηγών, την κατηγοριοποίηση ακουστικών γεγονότων και την εκτίμηση του προσανατολισμού τους ως μια ενιαία αλυσίδα βημάτων. Παράλληλα, αναδεικνύεται η δυνατότητα αξιοποίησης του BSS όχι μόνο ως ανεξάρτητη διεργασία, αλλά και ως σταδίου προεπεξεργασίας για την ταξινόμηση και τον εντοπισμό ακουστικών γεγονότων.

Λέξεις-κλειδιά:

Πολυκαναλικά ακουστικά μίγματα, διαχωρισμός ηχητικών πηγών, ταξινόμηση ακουστικών γεγονότων, εκτίμηση διεύθυνσης άφιξης, βαθιά μάθηση, χωρική πληροφορία, DCASE 2025.

Table of contents

Acknowledgements **ix**

Abstract **xii**

Περίληψη **xiii**

Table of contents **xv**

List of figures **xix**

List of tables **xxiii**

Abbreviations **xxv**

1 Introduction **1**

1.1 Thesis Focus 1

1.2 Related Work 4

1.2.1 Audio Source Separation 4

1.2.2 Acoustic Event Classification 7

1.2.3 Acoustic Source Localization 9

1.3 Our Contribution 10

1.4 Thesis Structure 11

2 Audio Source Separation **13**

2.1 Introduction 13

2.2 The Mixing Model 14

2.2.1 Multichannel Mixing Model 15

2.2.2 Alternative Representations of the Mixing Process 16

2.2.3 Mixing Scenarios According to the Number of Sources and Microphones . . . 18
2.3 The Separation Model . . . 19
2.4 Independent Vector Analysis . . . 22
2.4.1 Cost Function . . . 24
2.4.2 Optimization Techniques . . . 24
2.4.3 IVA Algorithmic Framework . . . 28

3 Acoustic Event Classification **29**
3.1 Introduction . . . 29
3.2 General Workflow of an Audio Classification System . . . 31
3.3 Signal Representation Techniques for Audio Classification . . . 33
3.3.1 Sound Waveform . . . 34
3.3.2 Fourier Transform . . . 35
3.3.3 Spectrograms . . . 36
3.3.4 Filter Bank Coefficients . . . 37
3.3.5 Mel Frequency Cepstral Coefficients . . . 39
3.4 Convolutional Neural Networks . . . 40

4 Acoustic Source Localization **47**
4.1 Introduction . . . 47
4.2 Spatial Information in the Multichannel Mixing Model . . . 49
4.3 Spatial Analysis of Separated Sources . . . 52
4.3.1 Channel-Wise Power Analysis . . . 52
4.3.2 Direction-of-Arrival Estimation . . . 56

5 Dataset **61**
5.1 Development Dataset . . . 61
5.2 Evaluation Dataset . . . 65
5.3 Evaluation Ground Truth . . . 66
5.4 Recording Details . . . 67

6 Experimental Setup **69**
6.1 Experimental Setup for Independent Vector Analysis . . . 69

6.1.1 AuxIVA 69
6.1.2 GradIVA 71
6.2 Experimental Setup for Convolutional Neural Network 74

7 Evaluation **77**
7.1 Evaluation in Blind Source Separation 77
7.1.1 Performance Metrics 77
7.1.2 Experimental Evaluations of AuxIVA 79
7.1.3 Experimental Evaluation of GradIVA 87
7.2 Evaluation in Acoustic Event Classification 90
7.2.1 Performance Metrics 90
7.2.2 Experimental Evaluation of CNN 91

8 Experimental Results **95**

9 Conclusions **99**
9.1 Future Work 101

Bibliography **105**

List of figures

1.1 Overview of the audio source separation, classification, and localization workflow, illustrated for a mixture containing two sources. 3

2.1 Overview of the multichannel mixing model. The following microphone signals $x_1(t), \cdots, x_4(t)$ jointly constitute the multichannel mixture $\mathbf{x}(t)$. . . . 16

3.1 Types of classification. 30

3.2 Different steps of the AEC [1]. 33

3.3 Time-domain representations of multiple acoustic event classes, illustrating differences in their amplitude and temporal structure. 35

3.4 Fourier transform of the acoustic event waveforms presented in Fig. 3.3. . . 36

3.5 Time–Frequency representations (spectrograms) of acoustic events. 37

3.6 Filter bank on a Mel-Scale [2]. 38

3.7 Mel-scaled filter banks representations of acoustic events. 39

3.8 MFCC representations of acoustic events. 40

3.9 Sample spectrogram representation of an audio signal. 41

3.10 Structure of CNN [3]. 41

3.11 ReLU activation function [4]. 42

3.12 Max pooling layer and average pooling layer. 43

3.13 Flattening operation in CNNs [5]. 43

3.14 A fully connected layer [6]. 44

3.15 Softmax function. 45

3.16 Final stage of a CNN [5]. 46

4.1 Acoustic signal paths for a particular microphone in the x–y plane. r_s and r_m denote the locations of the source and the m-th microphone, respectively. Each path from the source to the microphone m may be represented by an attenuation and a delay of the source signal. The direct path possesses the least delay. The sum of all the paths constitutes the impulse response of the room for the particular source and microphone position [7]. 49
4.2 Spatial analysis of the separated “Buzzer” source from the first mixture in the DCASE2025 Task4 evaluation dataset, showing the relative channel-wise power distribution and the corresponding inter-channel covariance phase relationships. 55
4.3 Spatial analysis of the separated “Typing” source from the first mixture in the DCASE2025 Task4 evaluation dataset, showing the relative channel-wise power distribution and the corresponding inter-channel covariance phase relationships. 56
4.4 Example results in a two-source scenario from the first mixture in the DCASE2025 Task4 evaluation dataset: “Buzzer” and “Typing”. The estimated DOAs are marked by a blue circle for source “Buzzer” and an orange circle for source “Typing”. 60

5.1 Distribution of isolated target training samples across the 18 sound event classes of the DCASE2025 Task4 Development Dataset. 62
5.2 Distribution of audio samples across the 18 sound event classes of the DCASE2025 Task4 Development Dataset on validation and test subset. 63
5.3 High-energy transient sound event classes of the development training subset. 63
5.4 Continuous noise-like sound event classes of the development training subset. 64
5.5 Periodic sound event classes of the development training subset. 64
5.6 Four-channel waveform representation of an isolated “BicycleBell” target sound event from the DCASE2025 Task4 Development Dataset. 65
5.7 Waveforms of the first 6 evaluation mixtures from the evaluation set. 66
5.8 Relationship between evaluation mixtures and their corresponding isolated ground truth target source signals. 67
5.9 The microphone configuration used to record isolated sound events in an anechoic chamber [8]. 68

6.1 Convergence behavior of the AuxIVA cost function for different values of β. 70
6.2 GradIVA Cost Function Evolution for $\mu = 0.1$ and Different Values of α . . 71
6.3 GradIVA Cost Function Evolution for $\mu = 0.01$ and Different Values of α . 72
6.4 GradIVA Cost Function Evolution for $\mu = 0.001$ and Different Values of α 72

7.1 Mean SDR, SNR, and SAR values obtained by AuxIVA for the three evaluated STFT configurations. The left bars correspond to SDR, the middle bars to SNR and the right bars to SAR. 80
7.2 Distribution of the SDR values obtained by AuxIVA for all the separated target sources (398 sources). The plot corresponds to the best evaluated STFT configuration (FFT = 2048, 50% overlap, Hamming window). The red dashed horizontal line indicates the mean SDR value, highlighting the variability of the separation performance across different target sources. 82
7.3 Distribution of AuxIVA SDR values across acoustic event classes. 83
7.4 Comparison of the SDR values obtained by GradIVA for $\alpha = 2, e, 10$, with a fixed value of $\mu = 0.1$ and $N = 500$. 88
7.5 Comparison of the SDR values obtained by GradIVA for $\mu = 0.01$ and $\mu = 0.001$, with a fixed value of $\alpha = 2$ and $N = 500$. 89
7.6 Confusion Matrix 93
7.7 Waveforms and log-Mel spectrograms of the clean “Typing” source (7.7.1) and the estimated separated source by AuxIVA (7.7.2), illustrating changes in the TF representation associated with residual interference and separation artifacts. 94

9.1 Complete processing pipeline of the Thesis in practice. The two separated sources are classified correctly as “Buzzer” and “Dishes”, and a DOA is estimated for each. 101

List of tables

6.1 STFT parameters for AuxIVA 70
6.2 Configuration of CNN. 75

7.1 Mean values (dB) of SDR, SNR, SAR for AuxIVA, evaluated on the same 100 multichannel mixture signals. 81
7.2 Number of separated target sources across different SDR performance ranges for AuxIVA. 83
7.3 Selected Best-Performing AuxIVA Mixtures. 85
7.4 Selected Worst-Performing AuxIVA Mixtures. 86
7.5 Average AuxIVA SDR According to the Number of Active Sources. 87
7.6 Confusion Matrix. 90
7.7 Classification performance of the CNN model on the held-out test set. . . . 92
7.8 Classification performance on clean and estimated separated sources. . . . 94

Abbreviations

AEC	Acoustic Event Classification
ASL	Acoustic Source Localization
ASS	Audio Source Separation
BSS	Blind Source Separation
CNN	Convolutional Neural Network
DNN	Deep Neural Network
DOA	Direction of Arrival
FD-ICA	Frequency-Domain Independent Component Analysis
FOA	First-Order Ambisonics
ISTFT	Inverse Short-Time Fourier Transform
ICA	Independent Component Analysis
IVA	Independent Vector Analysis
NMF	Non-Negative Matrix Factorization
ReLU	Rectified Linear Unit
RIR	Room Impulse Response
RNN	Recurrent Neural Network
SAR	Signal-to-Artifacts Ratio
SCM	Spatial Covariance Matrix
SDR	Signal-to-Distortion Ratio
SED	Sound Event Detection
SELD	Sound Event Localization and Detection
SNR	Signal-to-Noise Ratio
STFT	Short-Time Fourier Transform
TDOA	Time Delay of Arrival
TF	Time-Frequency

Chapter 1

Introduction

Audio processing constitutes an important branch of digital processing, focusing on the analysis, transformation, and interpretation of acoustic signals. In real-world environments, multiple audio sources are often active simultaneously, producing complex acoustic mixtures. Similarly to the human auditory system, which can often distinguish different sounds and perceive their spatial origin, computational systems can be developed, requiring specialized signal processing and machine learning methods to extract such information automatically.

A central challenge in this field is the recovery of the individual audio signals contained in an observed acoustic mixture. Beyond recovering these signals, an audio analysis system should also be able to identify the acoustic event represented by each source and estimate its direction of arrival. Therefore, this Thesis focuses on the separation, classification, and localization of acoustic events from multichannel input signals that contain multiple acoustic events recorded by multiple microphones.

1.1 Thesis Focus

Humans can focus on a particular sound and separate it from others, a skill called selective hearing, and have the ability to distinguish and localize different sounds coming from several directions. By mimicking this behavior, computational methods have been developed for applications such as human-computer interaction, robot audition, and acoustic scene understanding. With their popularity growing recently, many have started developing models so that machines can have the ability to extract a sound event, identify the corresponding acoustic event class, and estimate the direction of arrival of this acoustic source from a mix

of multiple acoustic events in the same way that human listeners do. However, this has proved to be a challenging task. The variability of acoustic events along with their spatial locations add complexity to the problem.

The acoustic scenario considered in this Thesis consists of a multichannel audio mixture signal. Multiple acoustic events originating from different sound sources may occur simultaneously, causing their corresponding signals to overlap within the recorded mixture. Since the same acoustic scene is captured by several microphones, each channel provides a different observation of the source signals due to differences in propagation delay, signal level, and phase. More specifically, the multichannel spatial audio mixture in the considered scenario contains, at most, three simultaneous target sound events, additional non-target interfering events, and non-directional background noise, recorded by a four-microphone array in indoor environments. The differences among the microphone channels provide valuable spatial information that can be exploited for the separation and localization of the active audio sources.

As already mentioned, the problem considered in this Thesis is divided into three main modules. The first task addresses audio source separation (ASS) that aims to recover a set of audio source signals from the observation of the multichannel mixture signal. In our case, the decomposition of the mixture into its individual source signals takes place without any prior knowledge about the mixing process or the existing active source signals, thus constituting a blind source separation (BSS) problem. The second module concerns the classification of the separated audio source signals. Each isolated target sound event will be assigned a predicted label of an acoustic event class. Finally, the third stage estimates the spatial direction of each separated source signal, by exploiting the spatial information captured by the microphone array. Since the source separation precedes the other two stages and is performed with little prior knowledge about the mixture signal, the performance of the audio classification and localization is closely related to the quality of the separated source estimates. The following figure illustrates the overall workflow of this Thesis, showing how the mixture is separated into individual source signals, each of which is subsequently classified and assigned an estimated DOA.

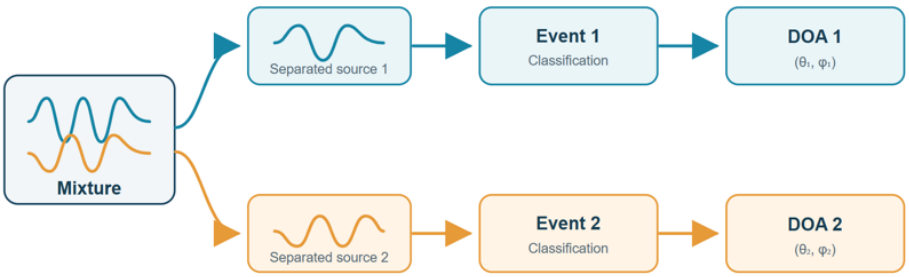


Figure 1.1: Overview of the audio source separation, classification, and localization workflow, illustrated for a mixture containing two sources.

The considered problem involves several technical challenges arising from both the acoustic environment and the sequential nature of the processing framework. First, multiple sound sources may be active simultaneously, causing their signals to overlap in both time and frequency. As a result, each microphone records a complex mixture rather than an isolated source signal. In addition, the variability of the acoustic events further complicates the problem, as the acoustic events might differ in duration, intensity, spectral content, and temporal structure. Moreover, the separation stage must estimate the individual source signals under uncertain and changing conditions of the mixing process. The recovered signals may also contain residual interference, background noise, or processing artifacts, making acoustic event classification even more challenging. This indicates that audio classification is applied to separated source estimates rather than perfectly clean signals. Therefore, the extracted output signals of the separation stage may contain distorted or misleading information, resulting in incorrect class predictions.

Our work in this Thesis is inspired by the DCASE2025 Challenge Task 4, entitled "Spatial Semantic Segmentation of Sound Scenes (S5)" [9]. This challenge aims to enhance technologies for sound event detection (SED) and separation from multichannel input mixture signals. The S5 task requires systems to detect the active target events and recover their corresponding source signals from the multichannel mixture simultaneously. In our case, the Thesis adopts a different processing order of the task's baseline approach [9], by having audio source separation precede acoustic event classification, with the resulting separated source estimates used to determine their corresponding acoustic event classes and spatial directions.

1.2 Related Work

Overall there exists a significant number of works on ASS, AEC and ASL. In this Thesis ASS and AEC are treated as distinct stages. ASL, however, is closely related to the spatial information extracted during the multichannel separation process. Therefore, the related literature is reviewed separately for ASS and AEC, while localization methods are discussed in relation to the employed separation framework.

1.2.1 Audio Source Separation

Source separation is a recent field of research within audio signal processing. Yet, some established techniques have gradually matured and have begun to be integrated into professional or general consumer audio applications. Before reviewing specific source separation methods, it is important to note that different approaches rely on different assumptions regarding the observed mixtures, the source signals, and the available prior information. Some methods are designed for single-channel recordings, whereas others exploit multichannel mixtures and the spatial information captured by multiple microphones. Furthermore, source separation problems can be categorized according to the relationship between the number of sources and the number of recording channels. Another important distinction concerns the amount of prior knowledge available. BSS methods assume little or no prior information about the source signals and the mixing process, while informed or supervised approaches incorporate additional knowledge to guide the separation procedure.

One of the earliest approaches to multichannel audio source separation is beamforming [10]. Beamforming is a signal processing technique that can be used mostly to separate a source from an audio mixture and estimate the direction of arrival (DOA) of a sound. Beamforming techniques approach the problem primarily from a spatial point of view. Specifically, beamforming can separate sources with overlapping frequency content that originate at different spatial locations. A microphone array is used to form a spatial filter to extract a signal from a specific direction and reduce the contamination of signals from other directions. The spatial filters selectively enhance sounds arriving from a desired direction while suppressing signals originating from other directions. In general, the separation problem is more difficult when the reverberation time of the acoustic environment is large, and when the number of sources is larger than the number of microphones. As the number of microphones increases,

beamforming performance generally improves. Therefore, beamforming techniques require information about the microphone array configuration and the sources. In realistic scenarios, this information is not available and must be estimated from the mixture signal at hand. These limitations motivated the development of statistical BSS methods, usually referred to as ICA-based methods, which estimate the unmixing filters directly from the observed mixtures.

An important class of BSS methods is based on the assumption that the source signals are statistically independent and non-Gaussian. Such algorithms typically employ Independent Component Analysis (ICA) [11], which is a method to find statistically independent sources from mixtures of sources by utilizing higher order statistics. Unlike beamforming methods, which commonly require prior knowledge or estimation of the source directions, ICA estimates the separation filters directly from the observed mixtures by exploiting the statistical properties of the source signals. In its simplest form, the ICA model assumes linear instantaneous mixing without sensor noise and the number of sources being equal to the number of sensors. When trying to solve the problem of separating source signals mixed in a real environment, such assumptions are not valid and model extensions are required. In those cases, observed signals are not instantaneous mixtures of sources, but convolutive mixtures, which means that they are mixed with time delays and convolutions. In order to deal with convolved mixtures, the ICA model formulation has been extended to convolutive mixtures in both the time and the frequency domains.

Many convolutive BSS methods are based on ICA. A widely used extension for convolutive audio mixtures is Frequency-Domain Independent Component Analysis (FD-ICA) [12]. FD-ICA transforms the microphone signals into the Short-Time Fourier Transform (STFT) domain and approximates the convolutive mixture as an instantaneous mixture at each frequency bin. The demixing matrices are estimated directly from the observed mixtures by maximizing the statistical independence of the separated frequency components. Although this formulation simplifies convolutive source separation, each frequency bin is processed independently. Therefore, the estimated source components may appear in a different order across frequencies, leading to the well-known frequency permutation problem.

To avoid the internal permutation problem, statistical coupling between the frequency bins of the demixed signals has been enforced within the class of Independent Vector Analysis (IVA)-based algorithms [13]. IVA is a multivariate-type independent component analysis applied to the frequency domain for convolutive mixtures. In IVA, all frequency components

of each source are modeled as a stochastic vector variable. Thanks to this, IVA is theoretically not affected by the permutation problem, unlike conventional frequency domain ICA.

More recent methods are based on non-negative matrix factorization (NMF). In contrast to ICA-based approaches, NMF techniques represent the spectro-temporal characteristics of the sources. However, although audio recordings are often available in multichannel format, standard NMF is only suited to single-channel data and does not explicitly represent multichannel spatial information. Another interesting extension is complex non-negative matrix factorization (CNMF) [14], which employs a factorization-type model for the magnitude of a time-frequency representation similar to NMF but additionally estimates a phase matrix for each source. The phase information can be used to improve the separation quality of overlapping signals.

Another way of modeling the spatial properties of the multichannel mixture signal is based on spatial covariance matrix (SCM) as signal representation. For each time-frequency point in the STFT, the SCM represents the mixing of the sources by magnitude correlations and phase differences between channels. Contrary to the CNMF approach, SCM is not dependent on the absolute phase of the source signal.

Finally, deep-learning approaches have significantly advanced the performance of audio source separation systems, particularly in speech separation and enhancement. ConvTasNet [15] is a recently proposed model that uses a combination of learned time-domain analysis and synthesis transforms with a time-dilated convolutional network. Hershey et al. [16] proposed deep clustering for multichannel speech separation, combining spectral and spatial features in a deep clustering framework so that the complementary spectral and spatial information can be simultaneously exploited to improve speech separation. Other approaches combine deep learning with classical multichannel statistical models. In [17], DNN-based source spectral models are integrated with a multichannel Gaussian spatial model, allowing both spectral and spatial information to be exploited, while the parameters are estimated in an iterative expectation-maximization (EM) fashion and used to derive a multichannel Wiener filter. Independent deeply learned matrix analysis (IDLMA) [18] is one of the state-of-the-art supervised but spatially blind multichannel audio source separation methods using the source power estimation based on Deep Neural Networks (DNNs). This method combines independence-based multichannel source separation with pretrained DNN source models, where the DNNs estimate the source power spectra while the spatial demixing filters are it-

eratively optimized. However, most of these approaches have not been investigated for separating mixtures of arbitrary sounds of different types, and it is unknown how performance on speech tasks carries over to non-speech tasks.

Audio source separation techniques have been extensively investigated for several application domains. The most known separation problem concerns the cocktail party. This refers to the brain's ability to focus on one voice while tuning out a loud, crowded room full of conversations and background noise. It introduces the challenge of isolating a single audio source (like one person talking) from a complex mixture of background noises, echoes, and overlapping conversations. For that reason, most source separation studies are applied to speech separation [11],[14], where the objective is typically to recover one or more speech signals from interfering speakers. Another well-established application is music source separation, which aims to isolate components such as individual musical instruments from a music mixture. In contrast, the separation of environmental acoustic events represents a more general and difficult problem, since the sources may belong to multiple sound classes and exhibit different temporal and spectral characteristics. Recent research on audio source separation has therefore moved beyond predefined speech or music sources and aims to separate acoustic events from complex mixtures. This broader separation scenario is particularly relevant to the present work, where the mixtures may contain different environmental acoustic events with varying durations, spectral structures, temporal patterns, and energy distributions.

1.2.2 Acoustic Event Classification

Numerous studies have explored the task of AEC. Traditional machine learning models used since the early 2000s include Support Vector Machines (SVM) [19], [20], K-Nearest Neighbors (KNNs) [21], Artificial Neural Networks (ANNs) [22] and hidden Markov models (HMMs) [23]. These methods perform traditional feature extraction from audio recordings and then use the features to classify the audio into different classes. More recently, deep-learning is becoming increasingly popular for AEC. Its ability to learn complex patterns often yields higher accuracy than traditional machine learning methods. The most popular deep-learning architectures for multi-class audio classification are Convolutional Neural Networks (CNNs) [24], Recurrent Neural Networks (RNNs) [25], Autoencoders [26], Transformers [27] and Hybrid Models [28]. CNNs are applied to spectrogram images and can be used to classify audio signals into different categories such as speech, music, and environmental

sounds. They can also be used for speech recognition, speaker identification, and emotion recognition. RNNs are widely used for audio classification and audio segmentation. RNN models can capture temporal patterns of audio signals and be used to classify audio segments into different categories. Another approach is to use autoencoders for learning the features of audio signals and then classifying the signals into different categories. Transformers are also well-suited for audio classification. In particular, temporal and frequency features can be extracted to identify the characteristics of the audio signals. Transformers address the limitations of RNNs and LSTMs by using self-attention mechanisms. Transformer-based models use a Transformer encoder to capture features in the audio spectrogram and have shown superior performance in audio classification tasks by capturing long-range context. Finally, hybrid models for audio classification either combine various deep-learning architectures (i.e. CNN-RNN) or combine deep-learning models with traditional machine learning techniques (i.e. CNN-Support Vector Machine). These hybrid models take advantage of the strengths of different architectures while avoiding their weaknesses. Generally, the deep-learning models are heavily dependent on the amount and the quality of the data they are trained on. If the data are noisy, biased, and incomplete, the results may be significantly impacted.

Since one of the objectives of this Thesis is the classification of the separated audio source signals extracted through the BSS stage, there are a significant number of studies that use BSS as a preprocessing step in order to detect the corresponding acoustic events of the mixture signal. Dai et al. [29] present a BSS model for bird sound recognition in complex acoustic environments. In particular, the model first uses IVA in the frequency domain to separate source signals from the original multichannel mixture signal and then utilizes Xception as the adaptive CNN architecture to identify all foreground bird species. Giannoulis et al. [30] proposed a multichannel separation network that generates K separated signals to be used as a preprocessing stage that passes these signals to a sound-event-classification network. Researchers there adjust a state-of-the-art multichannel sound separation network namely transform-average-concatenate (TAC) in order to exploit, in additional to spectral content, the spatial discrimination of the events present in a mixture clip, while for the sound event classification module they employ a CNN-based architecture. Other studies that relied upon the ability of BSS to be used as a preprocessing step for classification tasks are [31], [32]. In their research, Heittola et al. [31] presented a SED system for natural multi-source environments, using a sound source separation front-end. The audio is preprocessed using NMF

and separated into four individual signals, while each sound event class is represented by a HMM trained using mel frequency cepstral coefficients extracted from the audio. In other work, [32] employed a universal neural sound separation model applied to complex mixtures before a CRNN-based sound event detector and investigated different strategies for combining the predictions obtained from the original mixture and the separated outputs. Their findings showed that source separation can improve the performance of SED, although the benefit depends on the separation quality and the compatibility between the source separation training conditions and the SED test conditions.

The aforementioned methods have been applied to a wide range of audio analysis tasks. Speech-related applications include speaker identification, classification of different foreign-language accents or dialects, and emotion recognition. Music-related applications involve tasks such as music genre and instrument classification. Another major research direction focuses on environmental sound classification, where systems aim to identify everyday acoustic events such as alarms, vehicles, human activities, or other characteristic sounds occurring in indoor and outdoor environments. Moreover, recent studies consider challenging acoustic scene scenarios under noisy conditions, overlapping events, and complicated recording environments. Such scenarios are particularly relevant to the present work, since AEC is performed on separated source estimates that may still contain residual interference and background noise.

1.2.3 Acoustic Source Localization

The estimation of DOA is of special interest for a wide range of applications and has been tackled using different approaches. In the present work, however, the localization problem is considered specifically in the context of BSS, where spatial information recovered during the separation process is subsequently exploited for DOA estimation. Therefore, this subsection reviews related works that investigate the estimation of source direction as a byproduct of multichannel BSS methods.

One of the earliest solutions proposed [33] dates back to 2003, where an ICA method is used for estimating the directions of source signals by computing the inverse of the demixing matrix estimated during the ICA process. Other studies that relied upon the ability of BSS are [34], [35], [36], [14]. Wang et al. [34] proposed an overdetermined source separation approach using an FD-ICA model in each frequency bin directly and solved the permutation

problem by employing a time activity-based clustering approach to cluster the separated frequency components into the existed number of channels. The spatial location of each source was estimated by applying a TF masking-based steered response power algorithm. [35] proposed a source separation and beamforming algorithm that can locate more blind sources than the number of microphones in an array. [36], [14] are both based on multichannel BSS using SCM-CNMF, in which the spatial properties of each source are represented through source-specific SCMs while the source magnitude spectra are modeled within a CNMF framework. Both works produce estimated spatial parameters to be directly related to source direction, so that the direction of each acoustic event can be estimated.

More recent methods are based on deep-learning approaches [37], [38]. Specifically, Chen et al. [37] presented an end-to-end deep learning framework to separate and localize multiple audio sources from the multichannel mixture. The overall framework in [37] consists of three parts: a source separation network, a time difference of arrival (TDOA) estimation network, and DOA estimation network. As source separation network, they used two state-of-the-art speech separation models, SuDoRM-RF [39] and DPTNet [40], while the TDOA estimation network is a 6-layer CNN with four 1D convolution layers followed by two fully connected layers. [38] introduced an unsupervised method to separate and track moving audio sources using microphone array data. Munakata et al. [38] proposed a BSS method called neural full-rank spatial covariance analysis (FCA) that trains a neural separation model based on a nonlinear generative model of multichannel mixtures and can precisely separate unseen mixture signals. In order to estimate the direction of the sources, time-varying spatial covariance matrices and DOA of sources have been introduced into the nonlinear generative model of the neural FCA.

1.3 Our Contribution

As already indicated, this Thesis focuses on two main processing tasks: multichannel ASS and AEC. In addition, spatial information extracted during the separation process is used for ASL. The main contributions of this work are summarized as follows:

1. Formulating the multichannel spatial mixture scenario based on the DCASE2025 Task4 dataset, which contains four-channel spatial mixtures with multiple simultaneous acoustic events, non-target interfering sounds, and background noise.

2. Investigating a challenging and realistic acoustic dataset that includes overlapping environmental acoustic events, variable source activity, reverberation, interference, and spatial information captured by a microphone array.

3. Introducing the general principles of ASS, AEC and ASL.

4. Implementing and analyzing a state-of-the-art BSS framework for multichannel mixture signals namely IVA.

5. Providing an extensive survey of the deep-learning model used for AEC task, namely CNN.

6. Exploiting the source-specific spatial information extracted during the separation stage to characterize how each acoustic source is observed across the microphone channels and to estimate its DOA.

7. Evaluating the performance of the implemented separation and classification methods using appropriate performance metrics, on the same dataset, and against existing scientific works.

1.4 Thesis Structure

The rest of this Thesis is structured as follows:

- Chapter 2 introduces the general framework of ASS as well as the analytical description of the specific BSS model implemented for this Thesis.

- Chapter 3 presents the general principles of AEC as well as the detailed explanation of the CNN model implemented for the classification of the separated source signals.

- Chapter 4 introduces the fundamental principles of ASL, with emphasis on DOA estimation and the exploitation of spatial information associated with the separated signal sources.

- Chapter 5 describes the DCASE 2025 Task4 dataset, presenting its structure and statistics.

- Chapter 6 presents the experimental setup for each approach.

- Chapter 7 includes a detailed description of the evaluation of the proposed methods as well as the evaluation metrics that have been used for both ASS and AEC.
- Chapter 8 presents our experimental results.
- Chapter 9 draws our conclusions and discusses further research directions.

Chapter 2

Audio Source Separation

This chapter presents the theoretical foundations of ASS, with particular emphasis on BSS. It first introduces the definitions of ASS and BSS and formulates the multichannel mixing model on which the separation problem is based.

Building on this theoretical background, the chapter describes the BSS method employed in this work: Independent Vector Analysis (IVA). IVA is a widely used algorithm for BSS tasks since it requires only a few weak assumptions and allows for blindness regarding the original source signals and the acoustic propagation path. Our goal is to describe in detail its main processing steps and mathematical formulation, providing the necessary foundation for the experimental results presented in the following chapters.

2.1 Introduction

ASS refers to the range of techniques aiming to extract the signals of individual sound sources from a given recording. The input recording is referred to as a mixture signal. ASS is closely related to the well-known cocktail party problem, where sounds from different sources mix before arriving at the ear, requiring the brain to estimate individual sources from the received mixture [41]. In this work, we focus on methods that aim at recovering each individual source signal from the mixture without any prior knowledge about the mixture or the existing sources. This is called BSS. The term “blind” is used to emphasize that very little information about the sources or the mixing process is known. BSS has often been proposed as a possible solution to the cocktail party problem, as it carries the promise to recover the sources exactly. BSS is one of the important signal processing techniques for extracting a de-

sired source from mixtures, and it still remains one of the most interesting topics in the signal processing field. BSS can be used as a pre-processing step for classification or recognition tasks.

In the context of ASS and this specific work, the input mixture is referred to as multichannel mixture signal. In multichannel ASS, the acoustic scene is observed through multiple microphones rather than a single recording channel. Since each microphone captures a different version of the same source signal, the resulting signals contain not only spectral information about the sources but also spatial information arising from the different propagation paths between each source and microphone. Spatial information is the manner in which each source is observed across the different microphone channels. This channel-dependent information provides additional cues that can be exploited to distinguish sources that overlap strongly in both time and frequency. Therefore, the multichannel observations improve the procedure of source separation, as we take advantage of both the spectral characteristics of the sources and the spatial diversity introduced by the microphone array.

There are multiple potential applications of multichannel BSS. Typical examples include speech enhancement, automatic speech recognition, hearing-assistance systems, teleconferencing, music signal processing, acoustic monitoring, and robot audition. In addition, source separation, and especially BSS, can serve as a preprocessing stage for subsequent audio analysis. It has already been mentioned in Chapter 1 that the separated source signals will be used as inputs for the audio classification task, and the spatial information of each separated signal will be exploited for the localization task of this Thesis.

2.2 The Mixing Model

The first step to address when considering a source separation problem is to define the mixing model. Specifically, the choice of the mixing model determines which information is available to the separation algorithm, such as spatial diversity, propagation delays, spectral characteristics, or statistical independence between the sources. The BSS problem arises when one or several microphones record a sound that is the mixture of sounds coming from several sources [42].

2.2.1 Multichannel Mixing Model

First, we describe the multichannel mixture model, as it determines the model of this work. We consider a mixture of J source signals, observed by M microphones. Let

$$\mathbf{x}(t) = \Big[x_1(t), \quad x_2(t), \quad \cdots, \quad x_M(t)\Big]^T \in \mathbb{R}^M \tag{2.1}$$

be the multichannel time-domain mixture signal, recorded with an array of microphones, where $(\cdot)^T$ denotes vector transposition. The model introduces the following relation between the m-th mixed signal, the original source signals convolved with the room impulse response (RIR), and some additive sensor noise. The m-th channel can be modeled as

$$x_m(t) = \sum_{j=1}^{J} (h_{mj}(t) * s_j(t)) + n_m(t), \tag{2.2}$$

where $s_j(t)$ is the single-channel dry source signal, $h_{mj}(t)$ is the RIR of the m-th channel at the spatial position of s_j, and n_m is the m-th channel of the multichannel noise signal. The RIR characterizes the acoustic channel between a source and a microphone and represents how the sound propagates in the room [43]. Each RIR between a microphone and a source is called a mixing filter. Because of the convolutional operation between the sources and the mixing filters, the mixing model can be also referred to as convolutive mixture signal.

By concentrating all equations for the M channels, we define the multichannel mixture signal as

$$\mathbf{x}(t) = \mathbf{H}(t) * \mathbf{s}(t) + \mathbf{n}(t) = \begin{bmatrix} \sum_{j=1}^{J} (h_{1j}(t) * s_j(t) + n_1(t) \\ \sum_{j=1}^{J} (h_{2j}(t) * s_j(t) + n_2(t) \\ \vdots \\ \sum_{j=1}^{J} (h_{Mj}(t) * s_j(t) + n_M(t) \end{bmatrix}. \tag{2.3}$$

In detail, if we assume that we have $J = 2$ sources observed by $M = 4$ microphones (a realistic scenario in our work), the time-domain multichannel mixture signal is given by

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} = \begin{bmatrix} h_{11}(t) * s_1(t) + h_{12}(t) * s_2(t) + n_1(t) \\ h_{21}(t) * s_1(t) + h_{22}(t) * s_2(t) + n_2(t) \\ h_{31}(t) * s_1(t) + h_{32}(t) * s_2(t) + n_3(t) \\ h_{41}(t) * s_1(t) + h_{42}(t) * s_2(t) + n_4(t) \end{bmatrix}. \tag{2.4}$$

The following figure represents the modeling of the mixing process for $J = 2$ sound sources and $M = 4$ microphones.

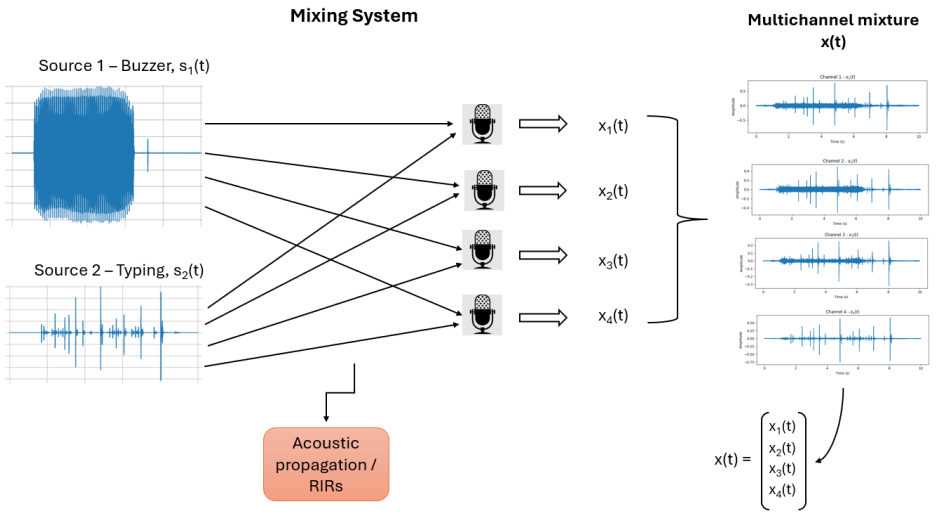


Figure 2.1: Overview of the multichannel mixing model. The following microphone signals $x_1(t), \cdots, x_4(t)$ jointly constitute the multichannel mixture $\mathbf{x}(t)$.

In a multichannel formulation, source separation can exploit the spatial diversity introduced by the microphone array. Since each source reaches the microphones through different acoustic propagation paths, its contribution is characterized by source-dependent time delays and RIRs. Consequently, multichannel separation algorithms can distinguish sources using both their spectro-temporal characteristics and the spatial information provided by the microphone channels.

The multichannel convolutive mixing model provides a realistic representation of an ASS task. This occurs because it perfectly matches how sound is captured and processed in the physical world. The multiple sources captured by multiple microphones, with regard to noise signals and the creation of complex echoes and reflections, makes the separation task highly complex.

2.2.2 Alternative Representations of the Mixing Process

Mixing models can be categorized according to the number of observed channels, the acoustic propagation assumptions, and the signal domain in which the mixture is represented. Although this work considers multichannel convolutive mixtures, the following cases provide useful background for understanding the assumptions and limitations of different separation

approaches.

Single-channel Mixing Model

Single-channel source separation is the task of isolating individual audio sources from a single, combined audio recording. In the time domain, the single-channel mixture is simply the linear summation of J independent source signals $s_j(t)$ and is expressed by

$$x(t) = \sum_{j=1}^{J} h_j(t) * s_j(t) + n(t). \tag{2.5}$$

Since only one observation is available, the algorithm cannot use physical spatial cues (like time delays between multiple microphones). Separation must instead rely entirely on the acoustic characteristics of the sounds and on differences in the spectral and temporal structure of the sources, time-frequency sparsity, predefined source models, or patterns learned from training data. Consequently, single-channel separation is generally more dependent on prior assumptions or supervised learning than multichannel separation.

Instantaneous Mixing Model

In this case we assume that all the signals arrive at the sensors at the same time without being filtered. It is a simplified version of the multichannel mixture. In the instantaneous mixing model, propagation delays and reverberation are neglected, and each microphone signal is modeled as an instantaneous linear combination of the source signals. The mixing model is described by

$$\mathbf{x}(t) = \mathbf{H}\mathbf{s}(t) + \mathbf{n}(t). \tag{2.6}$$

In contrast to convolutive multichannel mixtures, the instantaneous mixing model has no time delays and, therefore, does not exploit the physical spatial information of the microphones.

STFT Representation of the Convolutive Mixing Model

The convolutive mixing process (Eq. 2.3) can be simplified by transforming the mixtures into the frequency domain. Time-frequency (TF) representations are preferred, since they decrease the overlap between the sources and simplify the modeling of their characteristics. The linear convolution in the time domain can be written in the frequency domain as separate multiplications for each frequency:

$$\mathbf{X}(k,n) = \mathbf{H}(k)\mathbf{S}(k,n) + \mathbf{N}(k,n). \tag{2.7}$$

where $\mathbf{H}(k)$ is the $M \times J$ complex matrix called mixing matrix, $\mathbf{S}(k, n)$ is the $J \times 1$ complex vector of source coefficients, and $\mathbf{N}(k, n)$ is a complex $M \times 1$ vector of noise, at each frequency-bin index k and time-frame index n. The frequency transformation is typically computed using STFT. Source separation amounts to estimating the STFT coefficients $\mathbf{S}(k, n)$ of all sources and transforming them back to the time domain using ISTFT.

2.2.3 Mixing Scenarios According to the Number of Sources and Microphones

Another important characterization of a BSS problem concerns the link between the number of observations (sensors) and the number of underlying sources in the mixture [44]. Specifically, BSS is categorized as determined $(M = J)$, overdetermined $(M > J)$ and underdetermined $(M < J)$, where M represents the number of microphones and J represents the number of sources as already mentioned.

The determined scenario constitutes the most straightforward configuration for a BSS problem, particularly for ICA-based approaches [45]. Since the number of observations is equal to the number of sources, the mixing and demixing matrices can be formulated as square matrices, which simplifies the estimation of the separated signals.

In the overdetermined scenario, the additional observations provide spatial information that can improve the robustness of the separation process. A common approach for handling an overdetermined case is to reduce the number of channels from M to J and then apply a determined separation algorithm. This is done for example by selecting the best J channels or by principal component analysis (PCA), a linear dimensionality reduction technique which tends to minimize the information loss in the reduced channels.

In an underdetermined source separation problem, the number of source signals exceeds the number of available microphone observations. Consequently, even if the mixing system is known, the recovery problem does not generally have a unique solution. Additional assumptions about the source signals are therefore required. A common approach is sparse component analysis (SCA), which exploits the sparsity of the sources, particularly in the time-frequency domain, in order to estimate the mixing system and recover the individual signals [46]. Underdetermined configurations arise when several sources are simultaneously active, but only a limited number of sensors are available.

2.3 The Separation Model

The objective of the source separation is to find an estimate, $\mathbf{y}(t)$, which is a model of the original source signal $\mathbf{s}(t)$ [47]. More specifically, the estimated signals should be as close as possible to the original sources, while minimizing interferences from the other sources, background noise, and separation artifacts. However, the recovery of identical copies of the original signals is generally not possible, since BSS is inherently affected by scaling and permutation ambiguities. Therefore, the separated signals are considered estimates of the original sources rather than their exact replicas.

Given the multichannel mixture model in the frequency domain (Eq. 2.7), we need to estimate the separated source signals so that $\mathbf{y} \approx \mathbf{s}$. However, it may not be possible to identify the mixing filters $\mathbf{H}$ explicitly. Instead, it is often sufficient to estimate separation filters $\mathbf{W}$. For the measured signals $\mathbf{x}(t) = [x_1(t), x_2(t), \ldots, x_M(t)]^T$, where $\mathbf{x} = \mathbf{H}\mathbf{s}$, we assume that $\mathbf{W}$ is the inverse system of $\mathbf{H}$, called the demixing matrix, and $\mathbf{y} = \mathbf{W}\mathbf{x}$ provides an estimate of the original sources $\mathbf{s}$.

In the STFT domain, the convolutive separation problem can be approximated as an instantaneous separation problem at each frequency bin, as reported in [48]. The estimated source coefficients are obtained by applying a frequency-dependent demixing matrix to the observed mixture as:

$$\mathbf{Y}(k,n) = \mathbf{W}(k)\mathbf{X}(k,n), \tag{2.8}$$

where k and n denote the frequency-bin and time-frame indices, respectively. The vector

$$\mathbf{Y}(k,n) = \Big[Y_1(k,n), \quad Y_2(k,n), \quad \cdots \quad , Y_J(k,n)\Big]^T \in \mathbb{C}^J \tag{2.9}$$

contains the estimated source coefficients, while

$$\mathbf{W}(k) \in \mathbb{C}^{J\times M} \tag{2.10}$$

denotes the demixing matrix at frequency bin k. Here, $\mathbb{C}$ denotes the set of complex numbers. The demixing matrix $\mathbf{W}(k) \in \mathbb{C}^{J\times M}$ represents a $J \times M$-dimensional complex-valued matrix, where M is the number of microphone channels and J the number of target sources.

The demixing matrix can be expressed in terms of its row vectors as

$$\mathbf{W}(k) = \begin{bmatrix} \mathbf{w}_1^H(k) \\ \mathbf{w}_2^H(k) \\ \vdots \\ \mathbf{w}_J^H(k) \end{bmatrix}, \tag{2.11}$$

where $\mathbf{w}_j(k) \in \mathbb{C}^M$ is the demixing vector associated with the j-th estimated source and $(\cdot)^H$ denotes the Hermitian transpose. Therefore, the separation model and each source coefficient is estimated as

$$Y_j(k,n) = \mathbf{w}_j^H(k)\mathbf{X}(k,n) = \sum_{m=1}^{M} w_{jm}^*(k) X_m(k,n). \tag{2.12}$$

Thus, each separated source is formed as a frequency-dependent complex-valued linear combination of all microphone observations.

By removing the noise, the STFT representation of the mixing model will be transformed to

$$\mathbf{X}(k,n) \approx \mathbf{H}(k)\mathbf{S}(k,n). \tag{2.13}$$

Consequently, the separation model can be written as

$$\mathbf{Y}(k,n) \approx \mathbf{W}(k)\mathbf{X}(k,n) = \mathbf{W}(k)\mathbf{H}(k)\mathbf{S}(k,n). \tag{2.14}$$

In an ideal square case where $M = J$, the estimated sources will be perfectly matched with the original sources ($\mathbf{Y}(\mathbf{k},\mathbf{n}) \approx \mathbf{S}(\mathbf{k},\mathbf{n})$) if

$$\mathbf{W}(k)\mathbf{H}(k) = \mathbf{I}. \tag{2.15}$$

where $\mathbf{I}$ denotes the identity matrix.

However, in BSS, the mixing matrix and the source signals are unknown. Consequently, the demixing matrices must be estimated directly from the observed mixtures by exploiting statistical properties of the source signals. In practice, the condition in Eq. 2.15 cannot be satisfied. The formulation of Eq. 2.15 can, then, be expressed as

$$\mathbf{W}(k)\mathbf{H}(k) = \underbrace{\mathbf{D}(k)\mathbf{\Pi}}_{\mathbf{P}} \tag{2.16}$$

where $\mathbf{P}$ is a scaling, delay, and permutation matrix [48]. That means that the output of the separation equation will be the original sources, arbitrarily scaled, permuted, and delayed. Matrix $\mathbf{P}$ is decomposed into the permutation matrix $\mathbf{\Pi}$ and the complex-valued diagonal

scaling matrix $\mathbf{D}(\mathbf{k})$. The permutation matrix $\mathbf{\Pi}$ indicates that a BSS algorithm cannot inherently determine the correspondence between the estimated outputs and the original source ordering. For instance, if

$$\mathbf{s} = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} \tag{2.17}$$

are the original target sources of the mixing model, a valid solution of the estimated sources is

$$\hat{\mathbf{s}} = \begin{bmatrix} s_2 \\ s_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{s}. \tag{2.18}$$

so that

$$\mathbf{\Pi} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \tag{2.19}$$

The statistical independence between the source components is unaffected by their ordering whether the algorithm returns

$$[s_1, s_2] \tag{2.20}$$

or

$$[s_2, s_1]. \tag{2.21}$$

On the other hand, the diagonal scaling matrix $\mathbf{D}(\mathbf{k})$ represents the arbitrary gain (or attenuation) applied to each separated source and is expressed by

$$\mathbf{D}(k) = \operatorname{diag}\left[d_1(k), d_2(k), \ldots, d_J(k)\right] = \begin{bmatrix} d_1(k) & 0 & \cdots \\ 0 & d_2(k) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}. \tag{2.22}$$

Each non-zero diagonal value represents the unknown scaling factor (gain) for a specific separated source. Since the diagonal coefficients are generally complex-valued, each factor $d_j(k)$ may modify both the magnitude and the phase of the corresponding source component. For example, if only a real-valued gain is considered, source 1 might come out twice as loud if $d_1 = 2$, while source 2 might come out at half-volume if $d_2 = 0.5$. The off-diagonal elements are zero because the scaling ambiguity acts independently on each separated source and does not introduce cross-source mixing.

Finally, the estimated separated source signals in the time domain will be reconstructed as

$$\hat{s}_j(t) = \operatorname{ISTFT}\left\{Y_j(k,n)\right\}. \tag{2.23}$$

2.4 Independent Vector Analysis

As already mentioned, BSS methods aim at decomposing convolutive mixtures into their individual source signals. Therefore, we will focus on multichannel algorithms that require none or only few assumptions about the mixing process.

Many BSS methods use the statistical properties of the source signals for separation [11]. When the main assumption is the statistical independence of the source signals, such algorithms are usually called Independent Component Analysis (ICA) [11]. However, ICA in its original form is suitable only for instantaneous mixtures, which makes direct application to the separation of audio signals inappropriate in most cases, as it does not exploit the spatial information observed by the microphone array [11].

In this work, we focus on convolutive BSS methods based on ICA. Specifically, we consider IVA, a statistical approach based on ICA and applied to the frequency domain for multichannel mixtures. In IVA, all the frequency components of a source are modeled as a stochastic vector variable [13]. The remainder of this section focuses on the IVA algorithm, describes two optimization techniques aiming to estimate the demixing matrix $\mathbf{W}(k)$, and highlights the corresponding steps for its implementation.

The IVA formulation considered in this work assumes a square demixing system, where the number of separated components is equal to the number of observation channels. In the present dataset, however, the mixtures are recorded by $M = 4$ microphones and contain between $J = 1$ and $J = 3$ active sound sources. Therefore, the physical separation problem is generally overdetermined $(M > J)$.

In our implementation, a square $M \times M$ demixing matrix is estimated, resulting in $M = 4$ output components. Since the number of target sources can be smaller than the number of outputs, only J components are subsequently retained as estimates of the active sources, while the remaining components correspond to residual or non-target components. The specific output-selection procedure employed in the experiments is described in the experimental methodology.

As IVA is defined in the STFT domain, we first define the microphone signal vector at time-frequency bin (k, n) as

$$\mathbf{x}(k,n) = [x_1(k,n), \ldots, x_M(k,n)]^{\mathrm{T}} \in \mathbb{C}^{M \times 1}. \tag{2.24}$$

The vector $\mathbf{x}(k,n)$ is a 4×1 vector, where k represents the frequency bin index and n the

time frame index. As already mentioned, $\mathbb{C}$ denotes the set of complex numbers. Therefore, $\mathbf{x}(k,n) \in \mathbb{C}^M$ represents an M-dimensional complex-valued observation vector, where $M = 4$ is the number of microphone channels. The observations are complex-valued because they correspond to STFT coefficients containing magnitude and phase information.

Then, we perform the demixing operation at each frequency bin k and time frame n individually and independently [11] as

$$\underbrace{\begin{bmatrix} y_1(k,n) \\ \vdots \\ y_M(k,n) \end{bmatrix}}_{\mathbf{y}(k,n)} = \underbrace{\begin{bmatrix} w_{11}(k) & \cdots & w_{M1}(k) \\ \vdots & \ddots & \vdots \\ w_{1M}(k) & \cdots & w_{MM}(k) \end{bmatrix}}_{\mathbf{W}(k)} \underbrace{\begin{bmatrix} x_1(k,n) \\ \vdots \\ x_M(k,n) \end{bmatrix}}_{\mathbf{x}(k,n)}. \tag{2.25}$$

A frequency-domain vector is then constructed by capturing all K frequency bins of the STFT-domain output vector for channel m as

$$\mathbf{y}_m(n) = [y_m(1,n), \ldots, y_m(K,n)]^{\mathrm{T}} \in \mathbb{C}^K. \tag{2.26}$$

Once the output vector has been constructed, $\mathbf{y}_m(n)$, we capture the statistical dependencies between all frequency bins to avoid the internal permutation problem by computing the ℓ_2-norm of each source vector. Specifically, for each estimated separated source q and time frame n, we define the Laplacian radius [11] as

$$r_q(n) = \sqrt{\sum_{k=1}^{K} |y_q(k,n)|^2 + \varepsilon}, \tag{2.27}$$

where $\varepsilon > 0$ is a small constant added for numerical stability. The demixing matrices $\mathbf{W}(k)$ are iteratively updated in order to minimize the cost function. In this work, we describe two optimization techniques, namely AuxIVA and GradIVA. For the first iteration, we initialize $\mathbf{W}(k)$ as the identity matrix for each frequency bin k, i.e.,

$$\mathbf{W}(k) = \mathbf{I}, \tag{2.28}$$

The final demixing matrices $\mathbf{W}(k)$ are used to recompute the separated signals $\mathbf{y}_q(n)$. Finally, the time-domain separated signals are reconstructed by applying the ISTFT.

2.4.1 Cost Function

The cost function of the IVA model [11] consists of the Laplacian radius, $\mathbf{r_q(n)}$, and the demixing matrix, $\mathbf{W}(k)$. First, it is defined as:

$$\mathcal{C}_{\text{IVA}}\big(\mathbf{W}(1),\ldots,\mathbf{W}(K)\big) := \sum_{q=1}^{M}\frac{1}{N}\sum_{n=0}^{N-1}\big\{G\big(y_q(n)\big)\big\} - 2\sum_{k=1}^{K}\log\left|\det\mathbf{W}(k)\right|, \tag{2.29}$$

where $G\big(y_q(n)\big)$ is called a contrast function. The cost function consists of two main components: the source-likelihood term and the log-determinant term.

- **source-likelihood term:** The first term models the statistical independence of each source vector using a multivariate Laplacian prior:

$$G\big(y_k(n)\big) = -\log p\big(\mathbf{y}_q(n)\big) \propto \|\mathbf{y}_q(n)\|_2 = r_q(n) \tag{2.30}$$

  where

$$r_q(n) = \|\mathbf{y}_q(n)\|_2 = \sqrt{\sum_{k=1}^{K}|y_q(k,n)|^2 + \varepsilon}, \tag{2.31}$$

  Thus, the source likelihood term becomes:

$$\sum_{q=1}^{M}\frac{1}{N}\sum_{n=0}^{N-1} r_q(n) \tag{2.32}$$

- **Log-determinant term:** In the second term we compute the determinant for each demixing matrix $\mathbf{W}(k)$ and end up to:

$$\sum_{k=1}^{K}\log\left|\det\mathbf{W}(k)\right| \tag{2.33}$$

Combining the above terms, the IVA cost function is written as:

$$\mathcal{C}_{\text{IVA}} = \sum_{q=1}^{M}\frac{1}{N}\sum_{n=0}^{N-1} r_q(n) - 2\sum_{k=1}^{K}\log\left(\left|\det\mathbf{W}(k)\right| + \varepsilon\right) \tag{2.34}$$

2.4.2 Optimization Techniques

The ultimate goal is to estimate the demixing matrices $\mathbf{W}(k)$, in order to minimize the IVA cost function. In this work, we present two widely used optimization approaches for IVA: auxiliary-function-based optimization (AuxIVA) and gradient-based optimization (GradIVA). Both methods aim to minimize the same cost function, but differ significantly in their update rules. We describe both algorithms in detail.

Auxiliary-Function-Based Optimization (AuxIVA)

AuxIVA is a convergence-guaranteed iterative algorithm for IVA with a spherical and super Gaussian source model, and the source model can be characterized by a weighting function [13]. We describe the update rules for the demixing matrices, $\mathbf{W}(k)$, and the conditions that make this approach fast and effective.

In AuxIVA, a spherical and super Gaussian source model is used, and the separation performance should depend on the source model. Moreover, the following two conditions are assumed for the contrast function [13]:

1. **Spherical symmetry:** $G\big(y_q(n)\big)$ is assumed to be a function of only the L_2 norm of $y_q(n)$. This means that $G\big(y_q(n)\big)$ can be represented as:

$$G\big(y_q(n)\big) = G_R\big(r_q(n)\big), \tag{2.35}$$

$$r_q(n) = \|\mathbf{y}_q(n)\|_2, \tag{2.36}$$

   where $G_R(r)$ is a function of a real-valued scalar variable r.

2. **Super Gaussianity:** $G_R(r)$ is assumed to be a continuous and differentiable function of r. The following conditions must be satisfied:

   - $G'_R(r)/r > 0$ and continuous everywhere and is monotonically decreasing. Here, $G'_R(r)$ denotes the first derivative of $G_R(r)$ with respect to r. The ratio $G'_R(r)/r$ corresponds to the weighting function, $\phi(r)$, used in the AuxIVA update. For a super-Gaussian source model, this function is assumed to be positive, continuous, and monotonically decreasing with r.
   - $r \geq 0$.

We now move on to the auxiliary variable updates. First, we update the weighted covariance matrix $\mathbf{V}_q(k)$. The matrix $\mathbf{V}_q(k)$ represents the source-dependent weighted spatial covariance matrix of the multichannel observations for the q-th estimated source at frequency bin k. The weighted covariance matrix describes the relationship between the $M = 4$ microphone signals with regard to the statistical dependence of the sources. The matrix will be updated for all k as follows:

$$r_q(n) = \|\mathbf{y}_q(n)\|_2 = \sqrt{\sum_{k=1}^{K} |y_q(k,n)|^2 + \varepsilon}, \tag{2.37}$$

$$\phi\big(r_q(n)\big) = \frac{G'_R\big(r_q(n)\big)}{r_q(n)}, \tag{2.38}$$

$$\mathbf{V}_q(k) = \frac{1}{N}\sum_{n=1}^{N} \phi\big(r_q(n)\big)\, \mathbf{x}(k,n)\mathbf{x}^H(k,n). \tag{2.39}$$

The weighted covariance matrix $\mathbf{V}_q(k)$ is expressed by two terms:

- the **source-dependent weight function**, $\phi\big(r_q(n)\big)$, which depends on the current estimate of the q-th source and determines the contribution of each time frame to the covariance estimate. We denote $\phi\big(r\big)$ as a weighting function because it works as a weight for calculating a weighted covariance matrix $\mathbf{V}_q(k)$. The weighting function can be defined as:

  $$\phi(r) = \gamma r^{\beta-2}, \tag{2.40}$$

  where γ is a scale parameter and $0 \leq \beta \leq 2$. In case the weighting function becomes very huge, this can lead to numerical divergence or overfitting of $\mathbf{V}_q(k)$. To avoid this, we set the weighting function as:

  $$\phi(r) = \min\left\{\phi_0,\ \gamma r^{\beta-2}\right\}. \tag{2.41}$$

- The **spatial covariance information** of the multichannel mixture signal, $\mathbf{x}(k,n)\mathbf{x}^H(k,n)$, which captures the spatial relationships among the microphone channels. The diagonal entries represent the average power of the signal observed at each microphone, whereas the off-diagonal entries represent the cross-covariance between different microphone channels.

Therefore, $\mathbf{V}_q(k)$ provides the link between the statistical source model used by AuxIVA and the spatial information contained in the multichannel observations.

Then, the demixing matrix updates [13] for all k are:

$$\mathbf{w}_q(k) \leftarrow \left(\mathbf{W}(k)\,\mathbf{V}_q(k)\right)^{-1}\mathbf{e}_q \tag{2.42}$$

$$\mathbf{w}_q(k) \leftarrow \frac{\mathbf{w}_q(k)}{\sqrt{\mathbf{w}_q^H(k)\,\mathbf{V}_q(k)\,\mathbf{w}_q(k)}} \tag{2.43}$$

where $\mathbf{e}_q$ is:

$$(\mathbf{e}_q)_i = \begin{cases} 1, & \text{if } i = q, \\ 0, & \text{otherwise.} \end{cases} \tag{2.44}$$

Gradient-Based Optimization (GradIVA)

Gradient-based optimization is the standard framework for training machine learning models. It uses calculus—specifically derivatives—to iteratively adjust model parameters to minimize a loss function. In the case of IVA, it is used to iteratively estimate the demixing matrices $\mathbf{W}(k)$. In its general form, gradient descent updates the parameters according to:

$$x_{\text{new}} = x_{\text{old}} - \eta \cdot \nabla(f), \tag{2.45}$$

where ∇f is the gradient of the function $F(\mathbf{X})$ and η is the learning rate.

In GradIVA, the update rule is derived from the source-likelihood term of the IVA cost function [49]. First, we define the normalized source coefficients as

$$\phi_q(k,n) = \frac{y_q(k,n)}{r_q(n)} \tag{2.46}$$

Then, for each frequency bin k we create an array [49]

$$\mathbf{A}(k) = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{\phi}(k,n)\,\mathbf{y}(k,n)^H \tag{2.47}$$

Finally, the demixing matrix $\mathbf{W}(k)$ is updated as [49]

$$\mathbf{W}(k) \leftarrow \mathbf{W}(k) + \mu\,(\mathbf{I} - \mathbf{A}(k))\,\mathbf{W}(k) \tag{2.48}$$

where μ is the step size. It is obvious that the performance of GradIVA depends on the choice of the step size μ, which may affect convergence speed and stability and make the selection of μ a difficult task. To overcome these difficulties, [49] proposes a new learning factor iterative approach that can satisfy both the step and accuracy requirements:

$$\mu(k) = \frac{\mu}{\log_\alpha(i\alpha)}, \tag{2.49}$$

where μ denotes the initial learning rate. The parameter $\alpha > 1$ controls the decay of the learning rate throughout the optimization process and $i = 1, 2, \ldots, N$ is the iteration index and N is the total number of iterations.

Finally, an important advantage of AuxIVA over GradIVA method is that AuxIVA does not require the selection of a step size parameter. This makes the implementation of AuxIVA faster and computationally less expensive.

2.4.3 IVA Algorithmic Framework

Having described the IVA signal model, cost function, and the optimization techniques employed by AuxIVA and GradIVA, the complete source separation procedure can now be summarized. Algorithm 1 presents the overall workflow of the IVA framework. The main difference between AuxIVA and GradIVA lies in the optimization step, where the demixing matrices are updated according to their respective update rules.

Algorithm 1 Pseudocode of the overall IVA source separation procedure

1: Prepare the time-domain multichannel audio mixture input signal, $\mathbf{x}(\mathbf{t})$.
2: Define the microphone signal vector, $\mathbf{x}(k, n)$ at time-frequency bin (k, n), by computing the STFT of the mixture signal using Eq. 2.24.
3: Initialize demixing matrices, $\mathbf{W}(k) \in \mathbb{C}^{M \times M}$, using Eq. 2.28.
4: RUN IVA AND OPTIMIZE THE COST FUNCTION LOOP
5: **while** iter $\leq$ num. of iters **do**
6: Compute the separated outputs, $\mathbf{y}(k, n)$, at each frequency bin k and time frame n using Eq. 2.25. Then, construct the separated output as a vector across all frequency bins using Eq. 2.26.
7: Compute Laplacian radius, $\mathbf{r_q}(n)$, using Eq. 2.27.
8: Compute the cost function, $\mathcal{C}_{\text{IVA}}$, using Eq. 2.34.
9: Update $\mathbf{W}(k)$ using Eq. 2.43 (AuxIVA) or 2.48 (GradIVA).
10: **end while**
11: Recompute the separated output vector, $\mathbf{y}$, using Eq. 2.25.
12: Reconstruct the separated spectrograms by running the ISTFT for each source and obtain the separated time-domain signals.

Chapter 3

Acoustic Event Classification

This Chapter presents the fundamental concepts of AEC. It introduces the concept of classification within the context of machine learning and defines AEC. Afterwards, it describes the general workflow of an AEC task and includes a detailed explanation on the signal representation techniques for audio classification. Then, it presents a detailed explanation of the structure of the CNN algorithm built for the classification of the separated sources. As discussed in Chapter 2, BSS was used as a pre-processing step for audio recognition tasks. Therefore, the CNN model developed for this work is applied to the separated source signals obtained by the BSS stage and belong to the 18 classes of the DCASE2025 Task 4 dataset, allowing the identification of the acoustic event corresponding to each extracted source.

3.1 Introduction

Classification is a supervised machine learning method [50] where the model tries to predict the correct label of a given input data. Supervised learning is the most dominant type of machine learning [51]. In this approach, the model is trained on a labeled dataset. In other words, the data are accompanied by labels that the model is trying to predict. In classification, the model is fully trained using the training labeled data, and then it is evaluated on test data before being used to perform prediction on new unseen data. In binary classification problems, a model predicts whether data fits into one of two classes. The learning techniques that are applied during training have models assess the features in the training data and predict which of two possible labels apply to each data point: positive or negative, true or false, and yes or no. Multiclass classification problems classify data with more than two class labels,

all of which are mutually exclusive. In this way, multiclass challenges are similar to binary classification tasks, except with more classes. In practical terms, this could look like an emotional speech recognition process, wherein a dataset of speech signals where each signal is labeled as “happy”, “angry”, or “sad”, a supervised model can recognize and categorize new emotional speech signals accurately. Multi-label classification allows a single piece of data to belong to multiple categories at the same time. Unlike multiclass classification, where each data point is assigned only one class, this approach allows the model to assign multiple labels to the same input. Fig. 3.1 illustrates these types of classification problems.

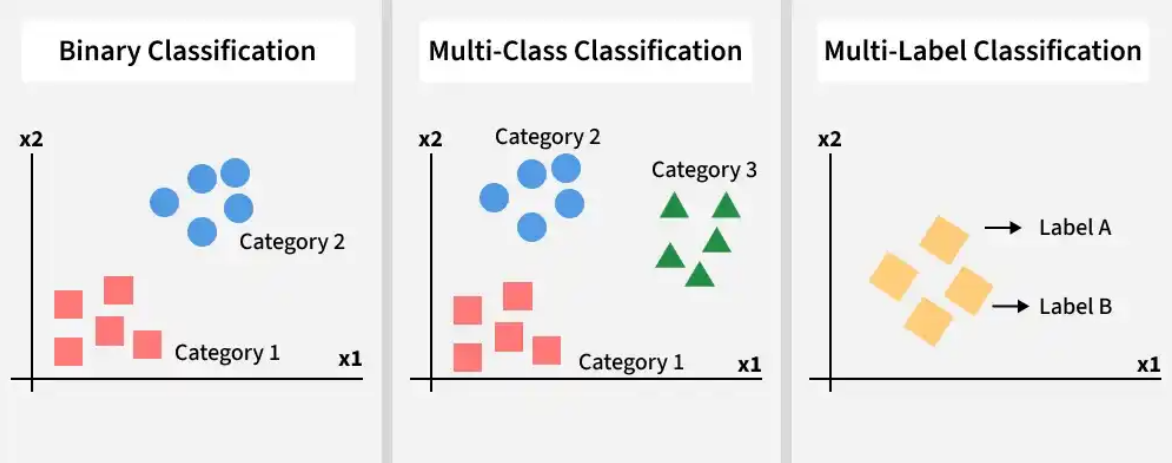


Figure 3.1: Types of classification.

AEC is the analysis and classification of audio signals into various categories. Depending on the task, the assigned label may describe the entire recording, the acoustic environment, or a particular acoustic event contained in the signal. It is an important tool in audio signal processing as it helps to organize, analyze and understand audio signals.

There are many AEC applications such as environmental sound classification applications, music genre identification, and speech recognition. In general, AEC can be broadly grouped into acoustic data classification, speech classification, music classification, and environmental sound classification. Given the diversity of the eighteen audio classes included in the DCASE2025 Task4 Dataset, the problem addressed in this work is more specifically formulated as an acoustic event classification task. In particular, the objective is to assign an acoustic event label to each separated sound source signal.

Audio classification technology can help make our everyday life easier. Specifically, it can provide assistance for people with disabilities to control devices in their home, like turn on lights or open the door without having to physically move. In security and surveillance

systems, audio classification can support the automatic recognition of events such as glass breaking, alarms, gunshots, screams, or unauthorized activity.

Sound event detection (SED) [52] aims to detect what sound events happen in an audio recording and when they occur. SED has many applications in everyday life. For instance, SED can be used to monitor “baby cry” sound at home or to detect “gunshot” and “scream” sounds at public places.

This work does not aim to estimate when an acoustic event occurs. Instead, its objective is to assign an acoustic class label to each source signal obtained by the preceding BSS stage. Therefore, the considered problem is formulated as an AEC task rather than a SED task.

3.2 General Workflow of an Audio Classification System

In this section we describe the different steps of an audio classification task. In general, the main steps involved in a classification task are:

- **Data Collection**: Start with a dataset where each data point has a correct label.
- **Feature Extraction**: Identify important features that help distinguish classes.
- **Model Selection**: Choose the model that best suits the case of the problem.
- **Model Training**: The algorithm learns patterns that connects feature to the correct class.
- **Model Evaluation**: The trained model is tested on unseen data to measure its accuracy.
- **Prediction**: The model predicts the class of new data based on learned patterns.
- **Model Improvement**: If performance is not satisfactory, the model or its parameters are adjusted and retrained.

Based on these stages, we define the general workflow of an AEC system as follows.

1. **Definition of the Classification Problem**: First, we define the classification problem that need to be solved. For example, the present study focuses on the single-label classification task, where each separated sound recorded is assigned to one of the predicted acoustic event classes. In addition, the type of classification problem depends on how many classes exist and how the categories are structured.

2. **Dataset Acquisition and Annotation**: As long as the definition of the classification problem has been defined, we select an appropriate audio dataset. The quality of the dataset affects significantly the performance of the model. The audio classification dataset should include the set of classes that will allow us to collect labeled data for training and evaluation [53], the number of samples at each class, the duration and the sampling rate of the audio signals, noise and reverberation, and the subsets of training, validation and test data.

3. **Audio Signal Preprocessing**: The primary objective of data preprocessing is to transform the raw data into a format that is more easily and effectively processed by the models before being used by the classification system. Data preprocessing techniques are generally used at the earliest stages of the machine learning development pipeline to ensure that the final results are accurate, reliable, and effective. This process involves several techniques such as denoising techniques used to remove noise from data and normalization techniques used to organize data for more efficient access and processing.

4. **Feature Extraction**: Feature extraction in audio classification turns raw sound waves into useful numbers. This process changes time-based sound signals into clear properties like frequency and energy. The preprocessed audio signals are transformed into representations that describe their temporal and spectral characteristics. These numerical values help the models to recognize speech, music, or environmental acoustic events. Audio features break down into different groups based on what part of the sound they measure. The most used feature extraction techniques are spectral features which analyze the frequency content using tools like the STFT to show how energy spreads across frequencies and Mel-Frequency Cepstral Coefficients (MFCCs) which capture the power spectrum while matching how human ears perceive pitch.

5. **Model Selection**: A suitable classification model is selected according to the type of problem, the size of the dataset and the extracted feature representation. Traditional machine-learning classifiers, such as SVMs, and KNNs are commonly applied to handcrafted feature vectors. Deep learning architectures, such as CNNs, RNNs, and transformers, are usually applied to spectrogram-based representations.

6. **Model Training**: During the training stage, the model learns the relationship between

the input audio features and their corresponding acoustic event labels. The model parameters are iteratively adjusted by minimizing a loss function using the training dataset. A validation subset is typically used for hyperparameter selection and to prevent overfitting.

7. **Model Evaluation**: After training, the performance of the model is assessed using unseen audio samples. Common evaluation metrics include classification accuracy, precision, recall, F1-score, and the confusion matrix. These metrics indicate both the overall classification performance and the ability of the system to distinguish between individual acoustic event classes.

8. **Prediction**: Finally, the trained model receives a new audio signal, applies the same preprocessing and feature extraction procedure, and produces a probability for each predefined acoustic event class. The class with the highest probability is selected as the final predicted acoustic event.

Fig. 3.2 is a representation of this process.

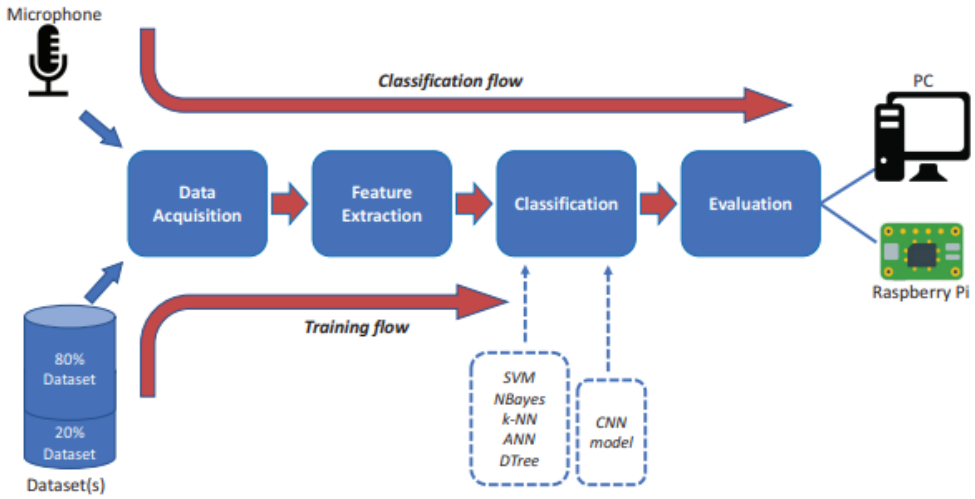


Figure 3.2: Different steps of the AEC [1].

3.3 Signal Representation Techniques for Audio Classification

Audio signals must be represented in an appropriate form before they can be effectively processed by most machine learning and deep learning classification models. Although some

end-to-end architectures can operate directly on raw audio waveforms, audio classification systems commonly rely on TF representations. These representations reveal relevant temporal and spectral characteristics of the signal and facilitate the extraction of important features. Common representations include sound waveforms, Fourier spectra, spectrograms, Mel-frequency Cepstral coefficients (MFCCs), and Mel filter-bank coefficients. Once the audio signal is represented in a suitable form, it can be fed into a deep learning model.

3.3.1 Sound Waveform

A sound waveform represents an audio signal in the time domain, illustrating how its amplitude varies over time. In digital audio, the waveform consists of a sequence of discrete amplitude samples recorded at a specific sampling rate. This representation preserves the temporal structure of the signal and provides the most direct form of the recorded acoustic information.

Raw waveforms can be used directly as input to deep learning models [54] [55], particularly to one-dimensional CNNs. Their high temporal resolution can be beneficial for identifying short-duration or transient acoustic events. However, the high dimensionality of raw waveform inputs may make model training more challenging, since the network must learn suitable low-level representations directly from the audio samples [55]. Consequently, they may require larger datasets and greater computational resources than models operating on predefined time-frequency representations. Moreover, variations in noise, amplitude, temporal alignment, and phase can make the direct analysis of raw waveforms more challenging and harder for deep learning models to process directly.

Fig. 3.3 presents the time-domain waveforms of representative acoustic events from the employed dataset. Each event exhibits a different temporal structure. For example, impulsive events are characterized by short and abrupt amplitude variations, whereas continuous or periodic sounds exhibit more sustained waveform patterns.

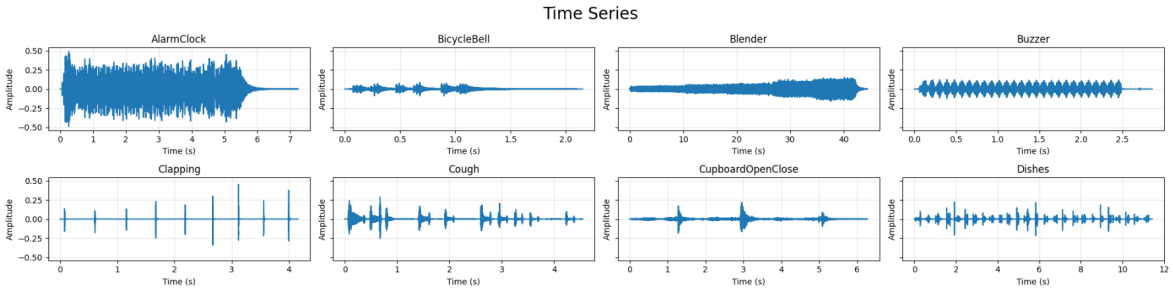


Figure 3.3: Time-domain representations of multiple acoustic event classes, illustrating differences in their amplitude and temporal structure.

3.3.2 Fourier Transform

Although the waveform of a sound illustrates how the signal amplitude varies over time, it provides only limited information about the frequency content of the sound. The Fourier Transform (FT) allows the signal waveform to be decomposed into its frequency components and to go from the time-domain to the frequency domain. For each frequency component, the FT produces a complex coefficient characterized by a magnitude and a phase. A high magnitude at a particular frequency indicates that the corresponding sinusoidal component contributes strongly to the signal. In simple words, the Fourier Transform converts a signal into individual spectral components and thereby provides frequency information about the signal.

Fig. 3.4 presents the magnitude spectra corresponding to the sound waveforms shown in Fig. 3.3. The horizontal axis represents frequency, while the vertical axis indicates the magnitude of each frequency component. Periodic sounds are generally characterized by distinct spectral peaks, whereas transient or broadband acoustic events tend to distribute their energy across a wider frequency range. However, the Fourier Transform does not preserve the temporal location of the frequency components. This limitation is important for non-stationary signals, such as acoustic events whose spectral content changes over time.

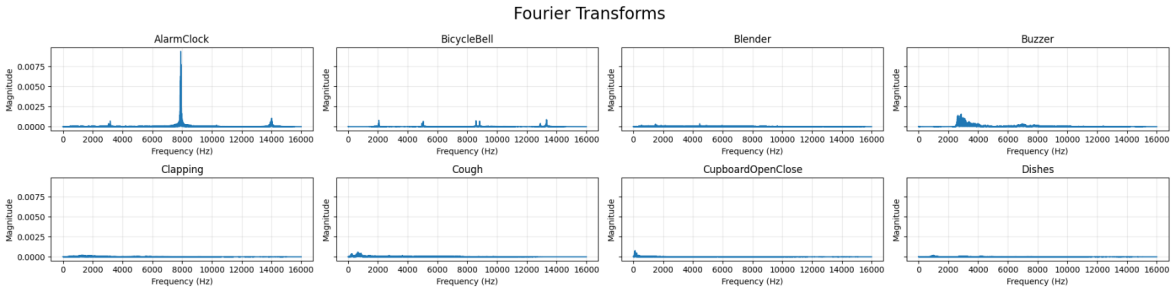


Figure 3.4: Fourier transform of the acoustic event waveforms presented in Fig. 3.3.

3.3.3 Spectrograms

Although Fourier Transform (FT) describes which frequencies are present, it does not explore when they occur during the recording, leading to a lose of useful information about the signal. A spectrogram is a 2D visual representation of audio signals in the frequency domain that displays how the frequencies within a sound evolve over time by breaking down an audio signal into small segments and computing the intensity of different frequency components within each segment. In simple terms, spectrograms capture the frequency content of an audio signal since the audio signal varies with time. The spectrogram, or time-frequency representation of an audio signal, provides valuable insights about the audio content, like distinguishing between various sounds, patterns, or characteristics. The efficient creation of spectrograms is a key step in audio classification as they are used as inputs to the deep learning models.

In general, spectrograms are generated using the STFT. STFT is an improved form of Fourier Transform (FT) to provide temporal details of signals in both the time and frequency domains. Using a window in STFT, the signal is divided into fixed-sized time-domain segments. Each segment is then subjected to FT to reveal different features of the signal. In essence, STFT uses equally spaced, identical, and symmetrical bandpass filters in the frequency domain to analyze the signal. The mathematical formulation of any signal $s(t)$, can be written as

$$S(f,t) = \int_{-T}^{T} s(\tau)\, w(\tau - t)\, e^{-j2\pi f\tau}\, d\tau. \tag{3.1}$$

To obtain this representation, signal s(t) is partitioned into segments, over a windowing function w(t) in Eq. 3.1. The window length must be the same as the length of the signal segments, and it is assumed that the signal does not change (stationary) inside a window duration. The spectrogram is obtained by using STFT by taking the magnitude squared value

of the TF representation expressed as

$$\text{Spectrogram} = |S(f,t)|^2 . \tag{3.2}$$

Fig. 3.5 presents the spectrograms of the representative acoustic events. In each spectrogram, the horizontal axis represents time, while the vertical axis represents frequency, and the color intensity indicates the magnitude of the corresponding time-frequency components. Brighter regions correspond to components with higher spectral magnitude. The acoustic events exhibit distinct spectro-temporal patterns. Continuous sounds, such as the "Blender" and "AlarmClock", maintain spectral energy over extended time intervals, whereas events that contain high energy peaks and silent intervals between these peaks, such as "Clapping", appear as short vertical structures in terms of frequency range. These differences demonstrate how spectrograms provide more useful information than time-domain waveforms by simultaneously representing the temporal and frequency characteristics of an acoustic event.

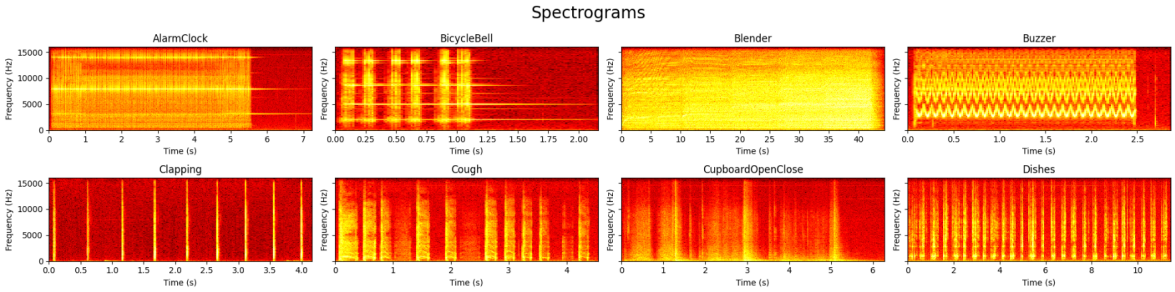


Figure 3.5: Time–Frequency representations (spectrograms) of acoustic events.

3.3.4 Filter Bank Coefficients

Filter banks are very popular features, especially for deep learning models and audio classification tasks. In general, the procedure to obtain filter banks [2] begins by passing the signal through a pre-emphasis filter; then gets sliced into (overlapping) frames and a window function is applied to each frame; afterwards, a Fourier Transform is applied on each frame (or more specifically a Short-Time Fourier Transform) and the power spectrum is calculated; and the filter-bank coefficients are subsequently computed.

The final step in computing filter bank coefficients involves applying triangular filters, typically 40 filters, (nfilt = 40) on a Mel-scale to the power spectrum to extract frequency bands. The Mel-scale aims to mimic the non-linear human ear perception of sound, by being more discriminative at lower frequencies and less discriminative at higher frequencies. The

conversions between Hertz (f) and Mel (m) are described, respectively, as:

$$M(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right), \tag{3.3}$$

$$f = 700\left(10^{m/2595} - 1\right). \tag{3.4}$$

where $M(f)$ is the Mel frequency for a given frequency (f). It is derived from a logarithmic scale and related to the way humans perceive sound. The formula is based on the assumption that frequencies have a logarithmic relationship to pitch, so it is used to convert frequencies into a Mel-frequency scale, which better represents how humans perceive pitch.

Each filter in the filter bank is triangular and has a response of 1 at the center frequency and decrease linearly to zero until reaching the center frequencies of the two adjacent filters where the response is 0, as shown in Fig 3.6:

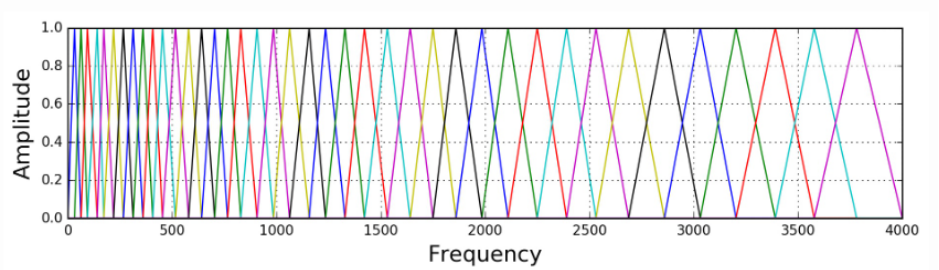


Figure 3.6: Filter bank on a Mel-Scale [2].

After applying the filter bank to the power spectrum of the signal, we obtain the spectrograms in Fig. 3.7. Fig. 3.7 illustrates the Mel filter-bank representations of the same acoustic events previously illustrated using spectrograms. Unlike the spectrogram, whose frequency axis consists of the individual frequency bins produced by the STFT, the filter-bank representation groups the spectral energy into a smaller number of perceptually motivated frequency bands. In the present case, 26 Mel filters are employed. Therefore, each time frame is represented by a 26-dimensional feature vector, where each coefficient indicates the energy contained within the corresponding Mel-frequency band. This transformation reduces the dimensionality of the original spectrogram and smooths fine spectral variations, while maintaining the main temporal and spectral patterns of each acoustic event.

As illustrated in Fig. 3.7, transient sounds, such as “Clapping”, remain visible as short vertical energy patterns, whereas periodic and tonal events, such as “Buzzer” and “Bicycle-Bell”, produce repeated or sustained structures across specific Mel bands. Consequently, Mel

filter-bank features provide a compact representation that can be effectively used as input to deep learning audio classification models.

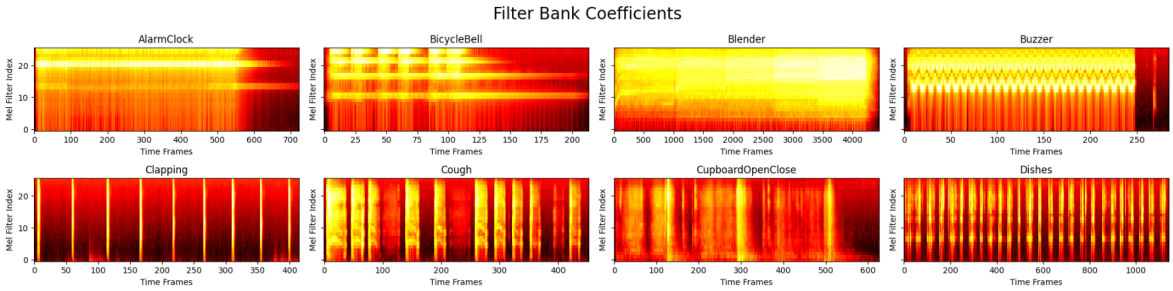


Figure 3.7: Mel-scaled filter banks representations of acoustic events.

3.3.5 Mel Frequency Cepstral Coefficients

Mel Frequency Cepstral Coefficients (MFCCs) are another type of TF feature representation that can be used to convert 1D audio signals to a 2D representation. Computing filter bank coefficients and MFCCs involve somewhat the same procedure, where in both cases filter banks are computed and with a few more extra steps MFCCs can be obtained.

To obtain MFCCs [2], a Discrete Cosine Transform (DCT) is applied to the filter banks to decorrelate the filter bank coefficients and yield a compressed representation of the filter banks, retaining a number of the resulting coefficients while the rest are discarded. The reason for discarding the other coefficients is that they represent fast changes in the filter bank coefficients.

Fig. 3.8 presents the MFCCs of the same acoustic events previously represented using Mel filter-bank energies. As already mentioned, the MFCCs are obtained by applying the DCT to the logarithmic Mel filter-bank energies. In the present implementation, the spectral information contained in 26 Mel-frequency bands is represented using the first 13 cepstral coefficients. Unlike the Mel filter-bank representation, the vertical axis of the MFCC representation does not correspond to individual frequency bands. Each cepstral coefficient is calculated as a weighted combination of all Mel filter-bank energies. Therefore, the MFCCs provide a compact and less correlated representation of the spectral energy, while some fine spectral details are discarded.

The temporal characteristics of the acoustic events remain visible in the MFCC representations. Periodic sounds, such as “Buzzer”, produce repeated cepstral patterns, whereas transient events, such as “Clapping” and “Coughing”, produce rapid variations across multi-

ple coefficients. Continuous acoustic events generally exhibit more stable cepstral structures over time.

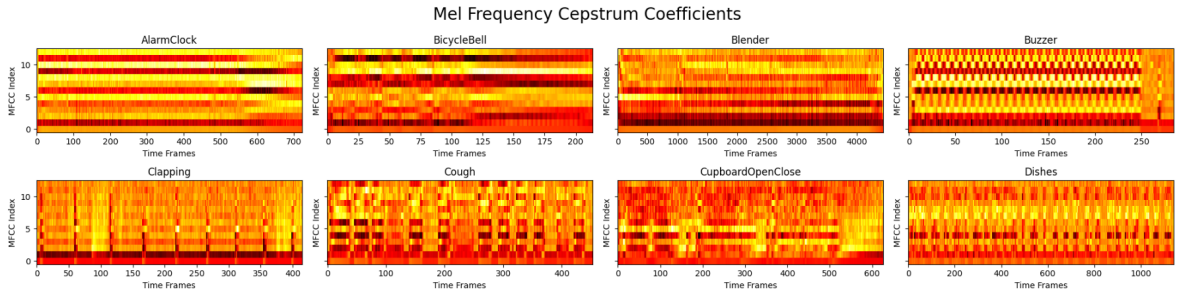


Figure 3.8: MFCC representations of acoustic events.

3.4 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are neural network architectures inspired by the human visual system, designed to process image data by capturing spatial relationships between pixels [56]. A CNN is a type of feedforward neural network that learns features via filter (or kernel) optimization. This type of deep learning network has been applied to process and make predictions from many different types of data, including text, images and audio. In the context of AEC, CNNs are commonly used for speech recognition, environmental sound classification, music recommendation, audio source separation (i.e. separating audio sources from a single-channel recording), and many more application areas.

For AEC tasks, inputs to CNNs are audio signals. However, CNNs make the explicit assumption that inputs are images. Therefore, CNN-based methods do not use the raw one-dimensional (1D) signals as input, but preprocess the input audio signal from a 1D signal into a 2D representation. The 2D representation of the audio signal is then input to a CNN model. This 1D to 2D conversion is generally executed to generate spectrograms. Spectrograms capture spectrum frequencies of an audio signal since the audio signal varies with time. Other types of spectrograms that can be used to convert 1D audio signals to a 2D representation are Fast Fourier Transform (FFT), STFT, Mel-Frequency, log-Mel frequency. Fig. 3.9 illustrates the several types of spectrograms of an audio signal.

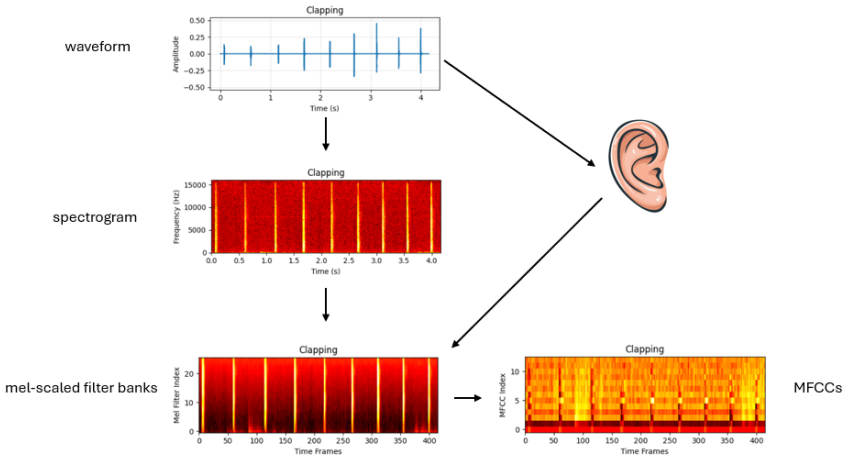


Figure 3.9: Sample spectrogram representation of an audio signal.

The general structure of a CNN [56] consists of multiple convolutional layers, a Rectified Linear Unit (ReLU), pooling layers, fully connected layers, and a softmax output layer. The convolutional layer is responsible for extracting important features from the input data. It is a layer that applies a convolutional filter operation to the input image to produce feature maps, and then passes the obtained feature maps to the next layer. It uses small filters (i.e., 2×2, 3×3, 5×5) to scan the input image. The generated feature maps capture patterns such as edges, textures and shapes. Fig. 3.10 illustrates the general structure of a CNN for a classification task.

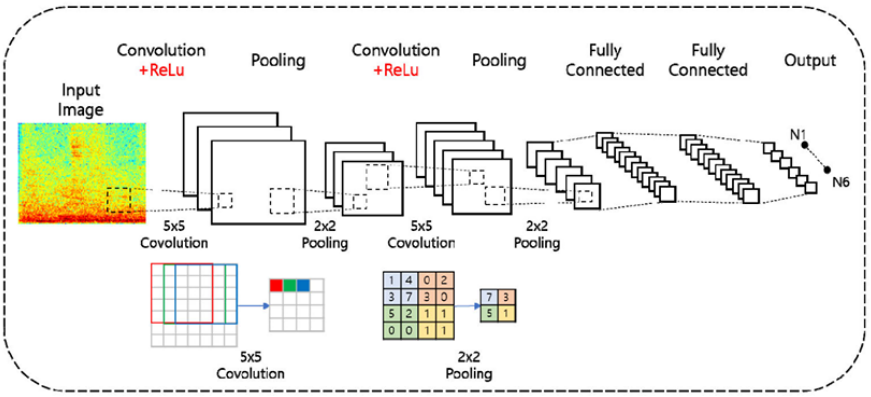


Figure 3.10: Structure of CNN [3].

In order to extract the significant features from the obtained feature maps in a convolution layer, the ReLU activation function [4] is usually applied. The activation function plays a major role in the learning process of a neural network and enables the model to learn complex patterns beyond linear relationships. In a ReLU, negative input values become 0 and

the same input value remains for the non-negative numbers. Therefore, a ReLU enhances the non-linearity of the model and prevents overfitting to the trained input samples and does not activate all neurons at the same time. A ReLU is the most commonly used activation function in CNNs, although there are other activation functions such as sigmoid activation, tanh activation and others. The ReLU is illustrated in Fig 3.11 and can be described mathematically as

$$f(x) = max(0, x). \tag{3.5}$$

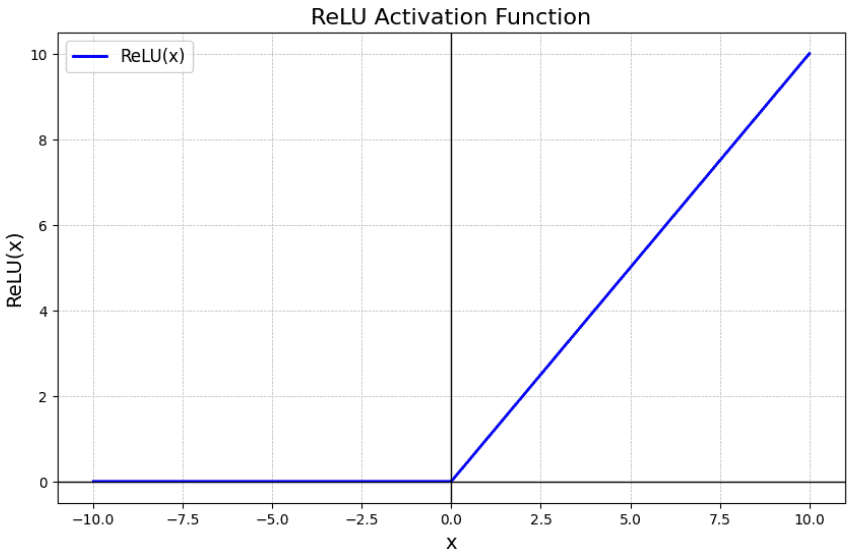
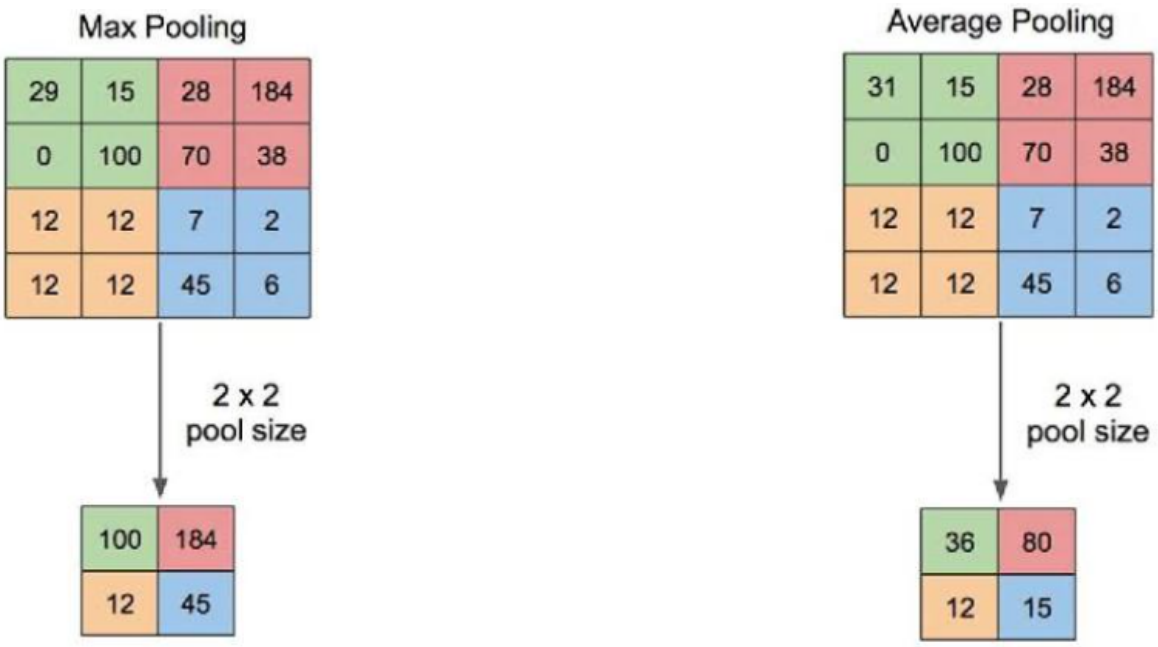


Figure 3.11: ReLU activation function [4].

A pooling layer [57] is a layer in a CNN to reduce the size of the input while still retaining important information, making computation faster, reducing memory usage and helping to prevent overfitting. This can be achieved by the pooling layers reducing the dimensions of the input feature map. This enables the model to focus on more important features, thus improving accuracy. The most commonly used pooling layers are max and average pooling. Max pooling selects the maximum value from each region of the feature map, capturing the most prominent features. Average pooling computes the mean value of elements within each region of the feature map, capturing overall feature information. Fig. 3.12 illustrates how max and average pooling work.



(a) Max Pooling Layer (b) Average Pooling Layer.

Figure 3.12: Max pooling layer and average pooling layer.

One of the essential operations in CNNs, especially when transitioning from convolutional layers to fully connected layers, is flattening. Flattening transforms a multi-dimensional tensor into a one-dimensional tensor, making it compatible with linear layers. This operation is represented in the following figure.

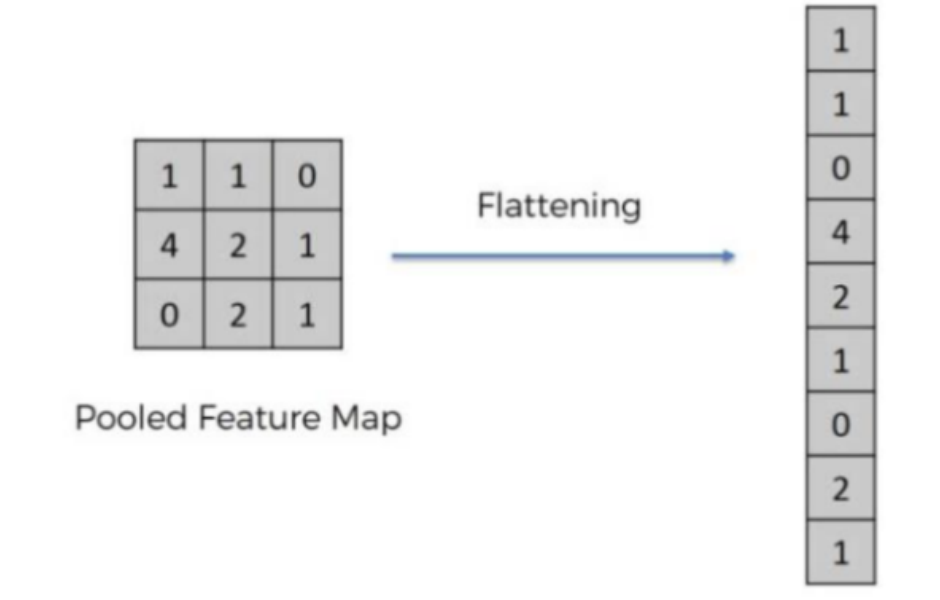


Figure 3.13: Flattening operation in CNNs [5].

A fully connected layer [58] in a CNN is used to connect all neurons in a previous layer to all neurons in the next layer. Fully connected layers generally come after the convolutional layer. A fully connected layer is a neural network layer with a fully connected configuration, and it is used to process the output of the convolutional layer after the flattening operation. The fully connected layers convert extracted feature maps into final predictions or output classes. The structure of a fully connected layer is based on complete connectivity, where every neuron in one layer is connected to every neuron in the next layer. Fully connected layers play a major role in combining learned features and generating final predictions in neural networks. The visualization of this layer is presented in Fig. 3.14.

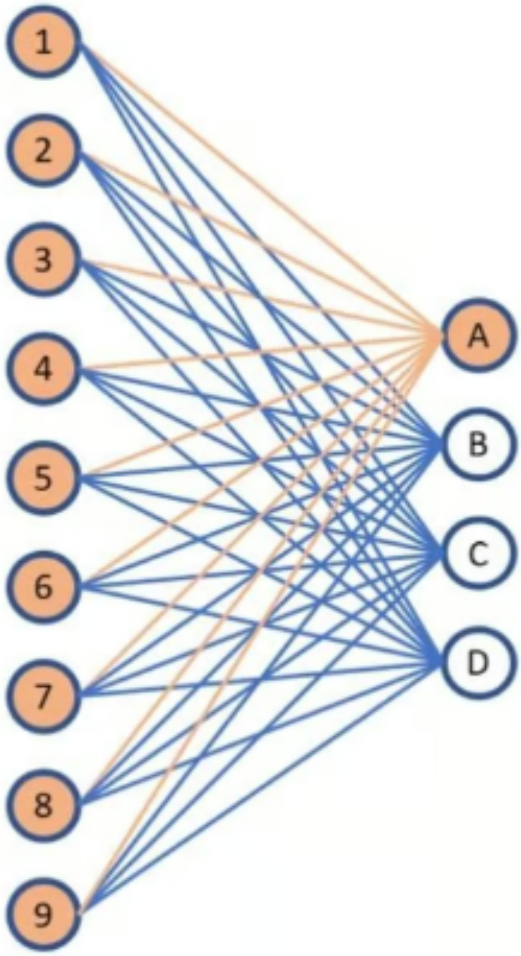


Figure 3.14: A fully connected layer [6].

A softmax layer in a CNN is a type of output convolutional layer that uses a softmax activation function [59], which is a type of logistic sigmoid function used to turn arbitrary real-valued scores into a probability distribution. In the softmax layer, the output layer produces a probability distribution over the possible classes for a given input. This enables the

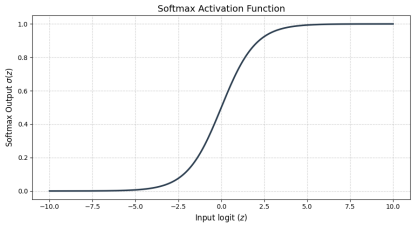
model to predict which class a particular input signal belongs to on the basis of the probabilities it produces. It is also a good choice for multi-class classification problems because it produces probabilities that sum to 1, making it ideal for assigning a class label to an input. The softmax activation function transforms the output vector $\mathbf{z} = [z_1, z_2, \ldots, z_n]$ into a probability distribution which can be defined mathematically as

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_j}}. \tag{3.6}$$

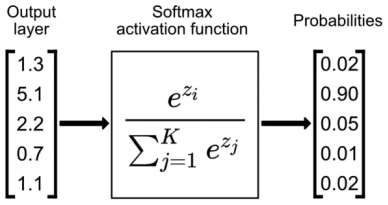
where:

- e^{z_i}: Exponentiation of the input value.
- $\sum_{j=1}^{n} e^{z_j}$: Sum of all exponentiated values used to normalize the outputs.

Each output $\sigma(z_i)$ represents the probability that the input belongs to class i. The processing of the softmax function is illustrated in Fig. 3.15.



(a) Softmax activation function.



(b) Example of the softmax function [60].

Figure 3.15: Softmax function.

The final steps of a CNN are represented in Fig. 3.16. The data is flattened into a one-dimensional vector, which is fed into the fully connected layer. The final layer produces the network's output. For multi-class audio classification problems, the output layer has as many neurons as there are classes (i.e. 18 classes for this work), with a suitable activation function such as softmax for multi-class classification.

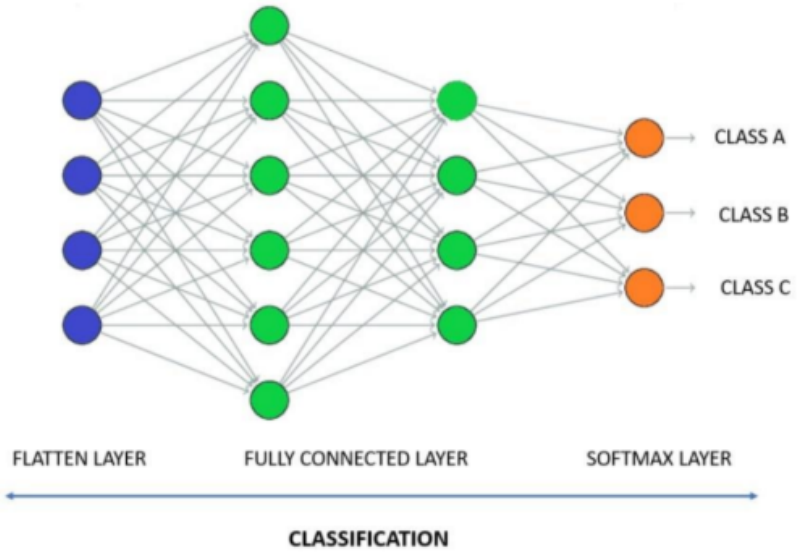


Figure 3.16: Final stage of a CNN [5].

Finally, a CNN is very useful to analyze and classify images, and has produced promising results for a variety of acoustic event analysis applications since 1D audio signals are generally converted into 2D patterns (spectrograms) for the analysis using a CNN.

Chapter 4

Acoustic Source Localization

Acoustic source localization (ASL) has been a topic of significant research activity in the field of signal processing and it has been serving the purpose of estimating the source of sound with respect to an array of microphones. This Chapter focuses on ASL in multi-channel scenarios, with particular emphasis on the spatial information already contained in the multichannel mixing and separation process. Rather than treating localization as an entirely independent processing task, the chapter focuses on the multichannel mixing model presented in Chapter 2 but this time from a spatial perspective and examines how source-specific spatial information can be recovered from the separated outputs. In the experimental part of this Thesis, the localization task focuses primarily on measuring the average power of each source at each microphone channel and on estimating the DOA of each separated source by exploiting spatial information obtained during the preceding source separation stage.

4.1 Introduction

ASL refers to the range of techniques aiming to estimate the spatial position or direction of an acoustic source with respect to an array of microphones. The ability to distinguish the direction of particular sounds coming from the surrounding noise environment is an important aspect of the human auditory system and has been a growing area of study by researchers from the field of signal processing, robotics, and medicine.

ASL has a wide range of applications in modern technologies. One the most important applications is hearing aid for people with hearing loss who suffer from the disability of being unable to recognize and distinguish speech between multiple speakers or environmental

acoustic events in the presence of background noise. ASL is also used for navigation, with Sonar systems [61] using this technique to find the location of a target. Furthermore, an ASL technology can be integrated into a robot to automatically move towards the direction of the speaker or a desired sound, or rotate its head to the desired direction in a conference room to improve communication and make the interaction with the speakers feel more human [62].

As we are concerned with a multichannel scenario consisting of $M = 4$ microphones, each propagation path from the acoustic source to the microphone m may be represented by an attenuation and a propagation delay of the source signal [7]. Since the microphones occupy different spatial positions, the same source is observed differently across the channels of the array. These differences can be characterized in terms of propagation delay, phase, and magnitude and therefore contain information about the spatial relationship between the source and the microphone array. Such inter-channel relationships constitute spatial cues that can be exploited to estimate the spatial characteristics of the source.

Localizing a sound source can be formulated in many possible ways, ranging from the estimation of a complete three-dimensional source physical position to the estimation of its direction relative to a microphone array. The present work primarily considers localization in terms of the DOA estimation for each separated acoustic event, which determines only the angular direction from which the acoustic wave reaches the microphone array, typically in terms of azimuth and elevation, without estimating the source distance. Moreover, the spatial response of each separated source across the microphone array is examined through source-specific spatial covariance information derived from the separation stage. In particular, the average power of each source at each microphone channel is used to determine how strongly the specific source is observed across the array.

Many algorithms have been proposed for ASL using microphone arrays. In many approaches, localization is treated as an independent processing task, where the spatial position or direction of an incoming sound source is estimated directly from the multichannel observations. However, ASL does not necessarily need to be treated as an entirely independent processing problem. As reported in [7], spatial localization information may also emerge as a byproduct of multichannel BSS algorithms. This statement is particularly relevant to the present work, where DOA estimation is performed by exploiting the source-specific spatial information recovered during the preceding separation stage.

4.2 Spatial Information in the Multichannel Mixing Model

As we are concerned with multichannel approaches, the sound scene must be recorded with multiple microphones. The following figure [7, Fig. 6.1] illustrates how the same source s is observed differently by each microphone channel, since each microphone occupies a different spatial position. Fig. 4.1 represents M microphones at positions r_m capturing a source at position r_s. The direct path represents the shortest propagation path between the source and a microphone, whereas the reflected paths contribute to the RIR associated with the corresponding source–microphone pair.

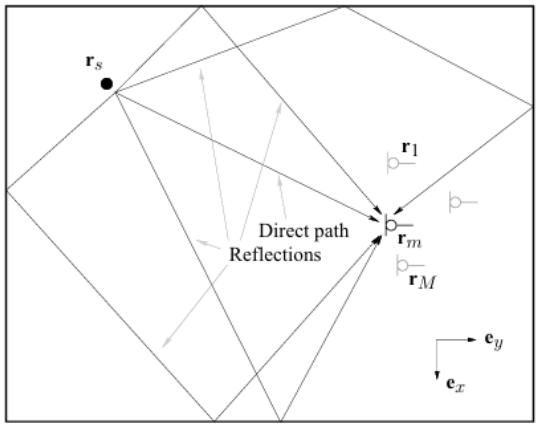


Figure 4.1: Acoustic signal paths for a particular microphone in the x–y plane. r_s and r_m denote the locations of the source and the m-th microphone, respectively. Each path from the source to the microphone m may be represented by an attenuation and a delay of the source signal. The direct path possesses the least delay. The sum of all the paths constitutes the impulse response of the room for the particular source and microphone position [7].

Rather than introducing a separate signal model for localization, this section revisits the multichannel mixing model defined in Section 2.2 and examines the spatial information encoded in its mixing parameters. We consider again the frequency domain mixing model as in Eq. 2.7:

$$\mathbf{X}(k,n) = \mathbf{H}(k)\mathbf{S}(k,n) + \mathbf{N}(k,n).$$

As already mentioned, $\mathbf{S}(k,n)$ contains the source coefficients and $\mathbf{N}(k,n)$ represents the multichannel noise, in each frequency-bin index k and time-frame index n. For the rest of this section, the quantity of primary interest is in the complex mixing matrix $\mathbf{H}(k) \in \mathbb{C}^{M\times J}$, since its elements characterize how the sound of source j is transmitted to each microphone

channel m.

In the multiple source case, the mixing matrix $\mathbf{H}(k)$ can be written as

$$\mathbf{H}(k) \approx \begin{pmatrix} H_{11}(k) & \cdots & H_{1J}(k) \\ \vdots & \ddots & \vdots \\ H_{M1}(k) & \cdots & H_{MJ}(k) \end{pmatrix}. \tag{4.1}$$

Each column of the mixing matrix corresponds to a particular source. More specifically, the j-th column

$$\mathbf{h}_j(k) = \Big[H_{1j}(k), H_{2j}(k), \ldots, H_{Mj}(k)\Big]^T \tag{4.2}$$

characterizes how source j is observed across the M microphone channels at frequency bin k. Consequently, the column vector $\mathbf{h}_j(k)$ contains source-specific spatial information determined by the acoustic propagation between the source and the microphone array.

Assuming that the direct propagation path is dominant, the mixing coefficient $\mathbf{H}_{mj}(k)$ may be decomposed into a direct-path component and an indirect propagation component as

$$\mathbf{H}_{mj}(k) = h'_{mj}(k)e^{-j\omega_k\tau_{mj}} + h''_{mj}(k) = \mathbf{H}'(k) + \mathbf{H}''(k), \tag{4.3}$$

where $|h'_{mj}| \gg |h''_{mj}|$. The elements of $\mathbf{H}_{mj}(k)$ represent the spatial response from source j to microphone m. $h'_{mj} \in \mathbb{R}$ represents the attenuation along the direct path and $h''_{mj} \in \mathbb{C}$ constitutes the disturbances caused by the reflections along the indirect paths. τ_{mj} represents the absolute time delay of the signal from source j to the microphone m along the direct path. Then, the direct-path component, $H'(k)$, of the mixing matrix takes the form

$$\mathbf{H}'(k) = \begin{pmatrix} h'_{11}(k)e^{-j\omega_k\tau_{11}} & \cdots & h'_{1J}(k)e^{-j\omega_k\tau_{1J}} \\ \vdots & \ddots & \vdots \\ h'_{M1}(k)e^{-j\omega_k\tau_{M1}} & \cdots & h'_{MJ}(k)e^{-j\omega_k\tau_{MJ}} \end{pmatrix} \tag{4.4}$$

The propagation matrix $\mathbf{H}'(k)$ is directly related to the geometric arrangement of the sources and the sensors and thus is the key to solving the localization problem.

For convenience, to express the channel-dependent propagation relative to a common reference, a reference point r_0 is introduced. The source spectrum at this reference point is defined as

$$\mathbf{S}_j(k) = h'_{0j}(k)e^{-j\omega_k\tau_{0j}}S_{0j}(k). \tag{4.5}$$

Then, the propagation matrix $\mathbf{H}'(k)$ multiplied with the source signal spectra $\mathbf{S}_j(k)$ can be

written as

$$\mathbf{H}'(k) = \begin{pmatrix} \frac{h'_{11}(k)}{h'_{01}(k)} e^{j\omega_k \tau_{01} - j\omega_k \tau_{11}} & \dots & \frac{h'_{1J}(k)}{h'_{0J}(k)} e^{j\omega_k \tau_{0J} - j\omega_k \tau_{1J}} \\ \vdots & \ddots & \vdots \\ \frac{h'_{M1}(k)}{h'_{01}(k)} e^{j\omega_k \tau_{01} - j\omega_k \tau_{M1}} & \dots & \frac{h'_{MJ}(k)}{h'_{0J}(k)} e^{j\omega_k \tau_{0J} - j\omega_k \tau_{MJ}} \end{pmatrix} \tag{4.6}$$

where $h'_{mj}(k)$ defines the attenuation of the direct path from source j to microphone m while $h'_{0j}(\Omega)$ defines the attenuation of the direct path to a reference point. What needs to be mentioned here is the time delay of arrival (TDOA) defined as

$$\mathbf{\Delta\tau_{mj}} = \tau_{0j} - \tau_{mj}. \tag{4.7}$$

TDOA measures the difference in time it takes for a specific source signal to reach multiple microphones at known locations.

Assuming that the reference point is close to the microphone array ($\frac{h'_{mj}(k)}{h'_{0j}(k)} \approx 1.$), Eq. 4.6 may be simplified to

$$\mathbf{H}'(k) = \begin{pmatrix} e^{j\omega_k \Delta\tau_{11}} & \dots & e^{j\omega_k \Delta\tau_{1J}} \\ \vdots & \ddots & \vdots \\ e^{j\omega_k \Delta\tau_{M1}} & \dots & e^{j\omega_k \Delta\tau_{MJ}} \end{pmatrix}. \tag{4.8}$$

The direct path contributions encode the spatial positions of the j sources in terms of TDOA. The indirect paths and noise contributions constitute a disturbance. Finally, the propagation matrix $\mathbf{H}'(\mathbf{k})$ constitutes the propagation vector that can be parametrized by the source location of the j-th source as

$$\mathbf{H}'(r_j, k) = \left(e^{j\omega_k \Delta\tau_{1j}(r_j)}, \dots, e^{j\omega_k \Delta\tau_{Mj}(r_j)}\right)^T. \tag{4.9}$$

The propagation vector in Eq. 4.9 is of particular relevance to the framework adopted in the separation stage. For a given source j, Eq. 4.9 describes how the source is observed across the microphone array. Respectively, each separated source extracted from the IVA method can be represented not only by its spectral content, but also by the relative way in which it is observed across the microphone channels, as mentioned in section 2.4.

The aforementioned observations are perfectly congruent with the spatial multichannel mixing model defined in section 2.2. Under an idealized anechoic or direct-path-dominant assumption, the contribution of the indirect path will be neglected, allowing the source-specific propagation vector to be described mainly through its relative delay and thus $\mathbf{H}(k) = \mathbf{H}'(k)$

(see Eq. 4.3). Consequently, the propagation-vector interpretation is directly relevant to the separation framework, since the spatial mixing parameters associated with each source characterize its relative observation across the microphone channels. After separation, this source-specific spatial information can be recovered or reconstructed from the estimated separation/mixing parameters and, subsequently, analyzed for localization, an important attribute of multichannel BSS methods.

4.3 Spatial Analysis of Separated Sources

In spatial audio BSS, the objective is not only to decompose a multichannel mixture into individual source estimates, but also to represent information about their spatial position or response from their originating location to each receiving microphone [36]. Many BSS methods are based on the spatial diversity of the channels and the way each source is observed across the microphone channels. Therefore, once the individual sources have been separated, their associated spatial characteristics can be analyzed independently. In the present work, this source-specific spatial information is exploited to characterize how each separated source is observed across the microphone array and to estimate its DOA.

4.3.1 Channel-Wise Power Analysis

Within the IVA framework, spatial relationships among the microphone observations are involved in the covariance matrices used during the estimation of the demixing matrix. As discussed in Section 2.4, the outer product $\mathbf{x}(k,n)\mathbf{x}^H(k,n)$ in Eq. 2.39 denotes the mixing of the sources by magnitude and phase differences between the recorded channels, and is not dependent on the absolute phase of the source signal. Particularly, spatial covariance calculations translate the absolute phase of the mixture to a phase difference between each microphone pair. The corresponding SCM in Eq. 2.39 ends up representing the spatial information of all the sources in the mixture at different spatial locations. Once again, the diagonal entries of the SCM represent the average power of the source signals observed at each microphone.

The covariance representation discussed above describes the spatial relationships contained in the original multichannel mixture. However, the objective of this section is to characterize the spatial information associated with each individual source after separation. Once

IVA has been applied to the observed mixture, the separated source estimates are obtained as

$$\mathbf{y}(k,n) = \mathbf{W}(k)\mathbf{x}(k,n), \tag{4.10}$$

where each output vector can be represented as

$$\mathbf{y}_q(k,n) = \begin{bmatrix} y_1(k,n) \\ y_2(k,n) \\ y_3(k,n) \\ y_4(k,n) \end{bmatrix}. \tag{4.11}$$

These are the four estimated separated outputs of the IVA algorithm. For localization, we need to know how each estimated source is captured at each spatial channel. First, an estimate of the corresponding mixing matrix can be obtained as

$$\hat{\mathbf{A}}(k) = \mathbf{W}^{-1}(k), \tag{4.12}$$

where

$$\hat{\mathbf{A}}(k) = \begin{bmatrix} \hat{a}_1(k) \\ \hat{a}_2(k) \\ \hat{a}_3(k) \\ \hat{a}_4(k) \end{bmatrix} \tag{4.13}$$

describes the spatial signature of the q-th source. Then, the spatial reconstructed images of the sources are denoted as

$$\hat{\mathbf{y}}_q^{\text{img}}(k,n) = \hat{\mathbf{a}}_q(k) y_q(k,n) \tag{4.14}$$

which are the source signals as seen by the array. For each source q, the source-specific SCM at each microphone channel m is presented as

$$\hat{\mathbf{R}}_q(k) = \frac{1}{N} \sum_{n=1}^{N} \hat{\mathbf{y}}_q^{\text{img}}(k,n) \hat{\mathbf{y}}_q^{\text{img}^{\text{H}}}(k,n). \tag{4.15}$$

Unlike the covariance representation of the original mixture in Eq. 2.39, $\hat{\mathbf{R}}_q(k)$ describes the spatial covariance associated with a single separated source q.

Finally, the source-specific SCM can also be used to characterize the average contribution of each separated source across the microphone channels. In particular, the diagonal elements of $\hat{\mathbf{R}}_q(k)$ correspond to the average power of source q at each microphone channel m for frequency bin k. The average power of source q at microphone m is defined as

$$E_{q,m} = \frac{1}{K} \sum_{k=1}^{K} \left[\hat{\mathbf{R}}_q(k) \right]_{mm}. \tag{4.16}$$

Since the diagonal entries of the covariance matrix is given by

$$\left[\hat{\mathbf{R}}_q(k)\right]_{mm} = \frac{1}{N}\sum_n \left|\hat{y}^{img}_{q,m}(k,n)\right|^2, \tag{4.17}$$

the channel-wise average power can equivalently be expressed as

$$E_{q,m} = \frac{1}{KN}\sum_{k=1}^{K}\sum_{n=1}^{N} \left|\hat{y}^{img}_{q,m}(k,n)\right|^2. \tag{4.18}$$

$\mathbf{E}_{q,m}$ will be 1×4 vector (i.e. $[0.8, 0.1, 0.03, 0.07]$), where $M = 4$ represents the number of channels, and thus

$$m^*_q = \arg\max_m \mathbf{E}_{q,m} \tag{4.19}$$

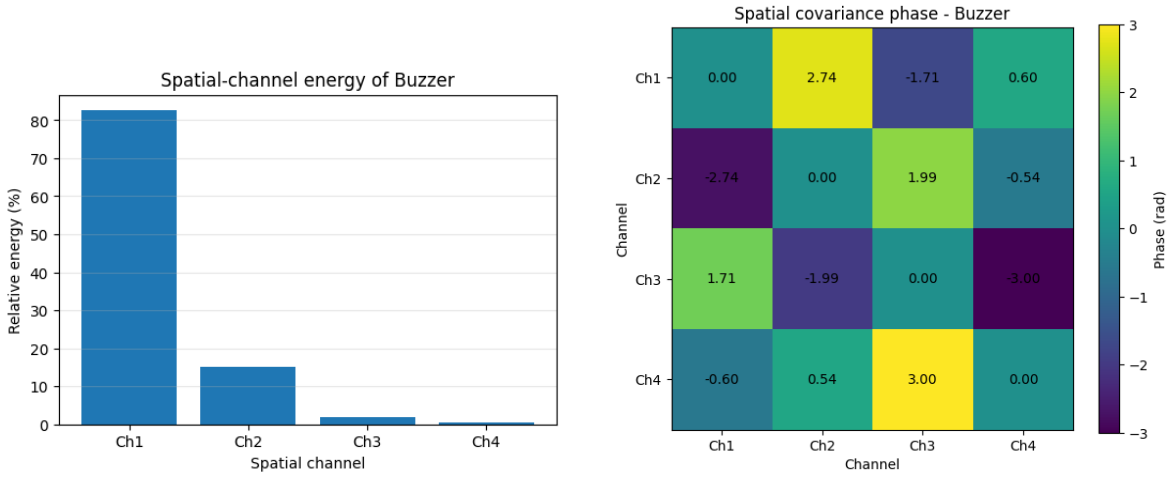
describes which of the four spatial channels m has the biggest contribution for the specific estimated source q (i.e. channel 1).

To illustrate the above spatial representation, we consider the first multichannel mixture from the DCASE2025 evaluation dataset (namely eval0000.wav) which consists of two sources: “Buzzer” and “Typing”. After applying IVA as in section 2.4, each separated source is reconstructed in the original four-channel spatial domain as in Eq. 4.14. Then, for a given source q, the source-specific spatial covariance matrix $\hat{\mathbf{R}}_q(k)$ is computed, and its diagonal entries are used to obtain the average power observed in each spatial channel as in Eq. 4.16 and 4.18 equivalently. If we outrun to the experimental setup for ASS methods (section 6.1), we do know which output correspond to each active source of the mixture. Therefore, the first and last separated outputs correspond to classes “Buzzer” and “Typing” respectively as

$$y_1 \leftrightarrow \text{Buzzer} \qquad y_4 \leftrightarrow \text{Typing}.$$

Fig. 4.2.a presents the normalized channel-wise power distribution for the separated “Buzzer” source. The reconstructed source image is strongly dominated by the first spatial channel, which contains approximately 82.5% of the total reconstructed energy. The second channel contributes approximately 15%, whereas the remaining two channels exhibit 1.9% and 0.6% respectively. According to Eq. 4.19 the dominant channel for this source is channel 1. While the diagonal entries characterize the channel-wise power distribution, the off-diagonal entries contain information about the relationships between different spatial channels. Fig. 4.2.b visualizes the phase of these cross-channel covariance terms for the same separated “Buzzer” source. As expected for an Hermitian covariance matrix (see Eq. 4.15), the phase matrix is

antisymmetric around its main diagonal: the phase associated with channel pair (m, ℓ) has the opposite sign of the corresponding (ℓ, m) entry, while all diagonal phases are zero.

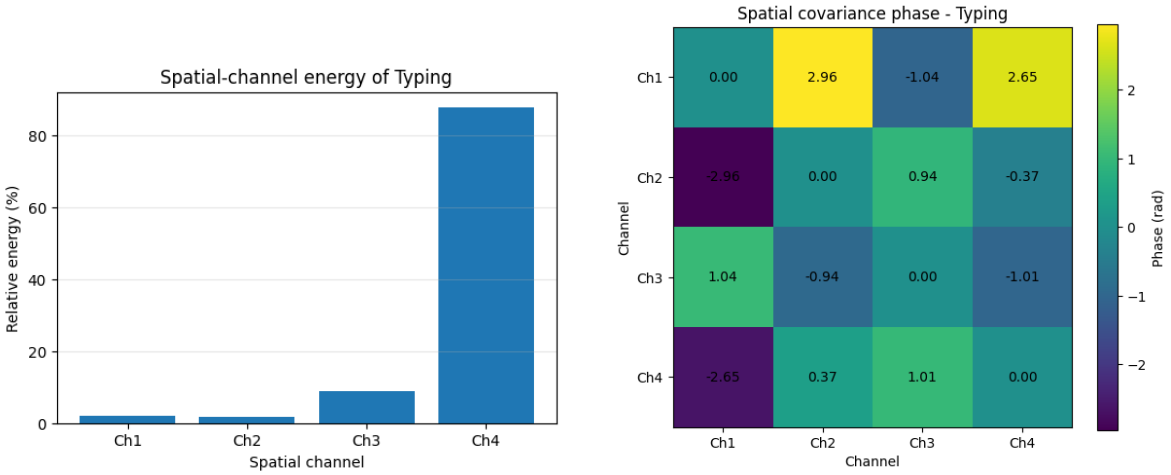


(a) Average spatial-channel energy of the reconstructed source "Buzzer".

(b) Relative phase relationship between the microphone channels of the reconstructed source "Buzzer".

Figure 4.2: Spatial analysis of the separated "Buzzer" source from the first mixture in the DCASE2025 Task4 evaluation dataset, showing the relative channel-wise power distribution and the corresponding inter-channel covariance phase relationships.

Following the same direction, Fig. 4.3.a illustrates the normalized channel-wise energy for the separated "Typing" source. This source is dominated by the fourth channel which contains 87.6% of the total energy, while channels 1,2, and 3 contain 1.9%, 1.7%, and 8.8% respectively. Consequently, the dominant channel for source "Typing" is channel 4.



(a) Average spatial-channel energy of the reconstructed source “Typing”.

(b) Relative phase relationship between the microphone channels of the reconstructed source “Typing”.

Figure 4.3: Spatial analysis of the separated “Typing” source from the first mixture in the DCASE2025 Task4 evaluation dataset, showing the relative channel-wise power distribution and the corresponding inter-channel covariance phase relationships.

Last but not least, the above example also represents the connection between the separation and spatial-analysis stages of the proposed framework. IVA exploits the differences in the spatial mixing characteristics of the simultaneously active sources in order to separate them. In the presented example, the separated “Buzzer” and “Typing” events exhibit clearly different channel-wise power distributions, with their maximum contributions occurring in channel 1 and channel 4, respectively. This is in complete agreement with the correspondence of the separated sources with respect to the microphone array, since $y_1 \leftrightarrow$ Buzzer and $y_4 \leftrightarrow$ Typing.

4.3.2 Direction-of-Arrival Estimation

The channel-wise power analysis presented in the previous subsection provides a useful description of how strongly each separated source is observed across the microphone array, an important information that many BSS methods rely on. Methods that explore the areas with higher energy reveal possible source positions and provide good DOA estimates for a single source. However, identifying the microphone channel with the highest source power for multiple sources does not by itself constitute source localization, since it does not pro-

vide an explicit estimate of the direction from which the acoustic wave arrives. In this work, we mainly rely on the estimation of the acoustic intensity vector, which represents the flow of energy in each frequency band and provides an estimate of the DOAs of each source. Specifically, localization is performed by estimating the 2D coordinates of the DOA for each separated sound event using azimuth and elevation. Therefore, the subsection focuses on estimating the DOA of each acoustic source from its source-specific spatial information.

In order to capture the spatial information, the sound scene must be recorded using multiple microphones. In this work, the multi-source mixtures have been synthesized by arranging the microphones as a spherical array, ensuring than no direction in space is favored. The recordings can then be stored in Ambisonics format, a full-sphere surround sound format for describing a spatial 3D sound field, indicating a suitable case for DOA estimation as it directly encodes the spatial properties of the sound field. Before we describe the DOA estimation system used in this work, we provide a general background on the Ambisonics format.

The Ambisonics format relies on the spatial decomposition of the sound field in the orthogonal basis of spherical harmonic functions. In practice, the sound field is recorded by a spherical microphone array and converted into Ambisonics with an encoding matrix [63]. The spatial recordings employed in the DCASE2025 Task 4 dataset are based on first-order Ambisonics (FOA). FOA uses 4 channels and is useful for basic recording, simple spatial work, compatibility, and many entry-level microphone workflows [64]. In addition, the RIRs were captured using a Sennheiser AMBEO VR microphone and are provided in B-format using the AmbiX convention. B-Format [64] is the Ambisonics signal representation used for production, exchange, decoding, and archiving. It describes a sound field around a listening point. First-order B-Format has four channels. This indicates that the relevant configuration of the channels does not correspond to four independent physical directions. In contrast, these channels can be seen as the recordings obtained from an omnidirectional microphone W and three cardioid microphones X, Y, Z, all four being coincident in space [8, Fig. 1a].

According to the AMBEO VR microphone specification [65, p. 18], the order and the numeric output format of the four channels when AmbiX is selected (B-format) is defined as $W - Y - Z - X$, rather than the traditional FuMa ordering $W - X - Y - Z$. Since the source-specific spatial image $\hat{\mathbf{y}}_q^{\mathrm{img}}(k, n)$, presented in Eq. 4.14, is reconstructed in the same multichannel spatial domain as the original mixture, its four components follow the same

Ambisonic channel convention as the input recording. Particularly, the reconstructed spatial image of source q can be written as

$$\hat{\mathbf{y}}_q^{\text{img}}(k,n) = \begin{bmatrix} \hat{\mathbf{y}}_1^{\text{img}}(k,n) \\ \hat{\mathbf{y}}_2^{\text{img}}(k,n) \\ \hat{\mathbf{y}}_3^{\text{img}}(k,n) \\ \hat{\mathbf{y}}_4^{\text{img}}(k,n) \end{bmatrix} = \begin{bmatrix} W_q(k,n) \\ Y_q(k,n) \\ Z_q(k,n) \\ X_q(k,n) \end{bmatrix}. \tag{4.20}$$

Here, W_q corresponds to the order-0 spherical harmonic function of the reconstructed source image, namely what an omnidirectional microphone would record, whereas the three other channels, X_q, Y_q, and Z_q, correspond to the order-1 functions of the basis, and to what three cardioid microphones aligned to the three spatial axes would record.

Once the source-specific Ambisonic image has been reconstructed (Eq. 4.20), its directional components can be exploited to estimate the direction from which the corresponding acoustic event arrives. For complex sound fields involving multiple sources or reverberation, advanced techniques need to be used to recover that spatial information each reconstructed spatial source image contains. This is achieved through the acoustic-intensity representation commonly employed for FOA signals. The general acoustic intensity vector is defined as

$$\mathbf{I}_q(k,n) = \mathcal{R}\left\{\mathbf{p}_q(k,n)\mathbf{v}_q^*(k,n)\right\}. \tag{4.21}$$

and represents the flow of sound energy in a point of space, with $\mathbf{p}_q(k,n) = \mathbf{W}_q(k,n)$ and $\mathbf{v}^*(k,n)$ the particle velocity. This vector can be expressed in the Ambisonics formalism as

$$\mathbf{I}_q(k,n) = \begin{bmatrix} \mathcal{R}\left\{W_q(k,n)X_q^*(k,n)\right\} \\ \mathcal{R}\left\{W_q(k,n)Y_q^*(k,n)\right\} \\ \mathcal{R}\left\{W_q(k,n)Z_q^*(k,n)\right\} \end{bmatrix}. \tag{4.22}$$

The relative values of the three entries in Eq. 4.22 provide a Cartesian description of the dominant spatial direction associated with source q. For instance, if $\left|I_x^{(q)}\right| \gg \left|I_y^{(q)}\right|, \left|I_z^{(q)}\right|.$, the spatial direction of source q is basically coming from x-axes. Therefore, the following conversion

$$(W_q, X_q, Y_q, Z_q) \rightarrow \left(I_x^{(q)}, I_y^{(q)}, I_z^{(q)}\right). \tag{4.23}$$

is basically the transition from Ambisonic representation to Cartesian direction representation.

Since the instantaneous intensity estimates may vary across TF bins due to the spectral content of the acoustic event, residual interference, and reverberation, a single source-direction vector is obtained by averaging the source-specific intensity components across time and frequency as

$$\bar{\mathbf{I}}_q = \frac{1}{KN} \sum_{k=1}^{K} \sum_{n=1}^{N} \mathbf{I}_q(k, n). \tag{4.24}$$

and

$$\bar{\mathbf{I}}_q = \begin{bmatrix} \bar{I}_x^{(q)} \\ \bar{I}_y^{(q)} \\ \bar{I}_z^{(q)} \end{bmatrix}. \tag{4.25}$$

The averaging is performed over the same TF representation used in the IVA algorithm, where K denotes the number of frequency bins and N the number of time frames.

Finally, the Cartesian components of $\bar{\mathbf{I}}_q$ in Eq. 4.25 are converted into the corresponding azimuth and elevation angles using the standard Cartesian-to-spherical coordinate transformation, in order to estimate the 2D coordinates of the DOA for each source signal. The azimuth and elevation angles are defined, respectively, as

$$\hat{\theta}_q = \operatorname{atan2}\left(\bar{I}_y^{(q)}, \bar{I}_x^{(q)}\right). \tag{4.26}$$

and

$$\hat{\phi}_q = \operatorname{atan2}\left(\bar{I}_z^{(q)}, \sqrt{\left(\bar{I}_x^{(q)}\right)^2 + \left(\bar{I}_y^{(q)}\right)^2}\right) \tag{4.27}$$

The azimuth $\hat{\theta}_q$ is obtained from the projection of the source-direction vector onto the horizontal xy-plane, whereas the elevation $\hat{\phi}_q$ describes the angle between this horizontal projection and the vertical z-component.

Considering the same example from the above subsection, we move forward to the computation of the DOAs of the first mixture in the DCASE2025 Task4 evaluation dataset. Fig. 4.4 illustrates that the estimated "Buzzer" direction is $(-86.3°, -2.9°)$, while the "Typing" acoustic event is estimated at $(-174.5°, 5.1°)$. The estimates are visualized in the azimuth–elevation plane to provide a realistic representation of the spatial direction of the two acoustic events.

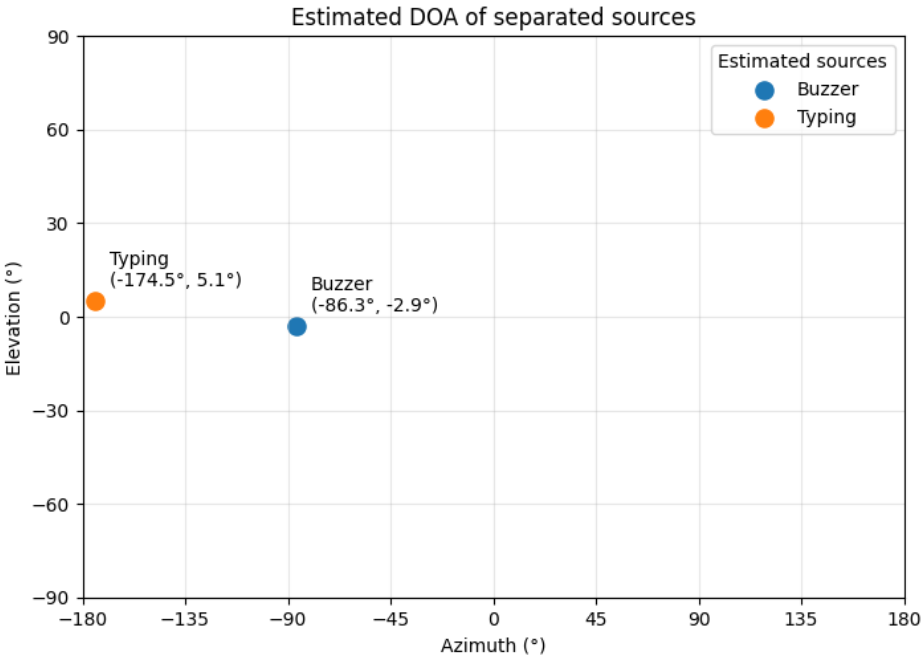


Figure 4.4: Example results in a two-source scenario from the first mixture in the DCASE2025 Task4 evaluation dataset: “Buzzer” and “Typing”. The estimated DOAs are marked by a blue circle for source “Buzzer” and an orange circle for source “Typing”.

Chapter 5

Dataset

In this Thesis we employ the DCASE2025Task4Dataset [66], [67], [68]. It is divided into a development set, an evaluation set, and a ground truth evaluation set. The dataset is available at Zenodo. The size of the dataset is about 15.1 GB. The DCASE2025Task4Dataset is composed of the following main components:

- isolated target sound events s_k
- Room Impulse Responses (RIRs) $h_k^{(m)}$
- Environmental noise $n^{(m)}$
- Interference sounds

The dataset is particularly suitable for the objectives of this Thesis, since it provides multi-channel spatial mixtures for audio source separation, isolated target audio events for acoustic event classification, and reference source signals for objective performance evaluation.

5.1 Development Dataset

The development set of the DCASE2025 Task4 Dataset [66] was constructed from various datasets, including both existing and newly recorded data specifically for this task. The isolated target sound events for the development set consist of newly recorded data and curated samples from FSD50K [69] and EARS [70]. RIR dataset was constructed by merging newly recorded RIRs for this task with curated material from the publicly released FOA-MEIR dataset [71]. All environmental-noise signals were curated from the publicly available

FOA-MEIR dataset [71]. All interference sounds were curated from the publicly available dataset.

The development dataset is divided into three subsets: training, validation, and test. All the subsets include isolated target sound events s_k belonging to 18 classes, namely: “Alarm-Clock”, “Bicycle Bell”, “Blender”, “Buzzer”, “Clapping”, “Cough”, “CupboardOpenClose”, “Dishes”, “Doorbell”, “FootSteps”, “HairDryer”, “MechanicalFans”, “MusicalKeyboard”, “Percussion”, “Pour”, “Speech”, “Typing”, and “VacuumCleaner”. The RIR assigned to each sound event is selected randomly. Mixtures $y^{(m)}$ for the test subset were pre-synthesized, while for the training and validation subsets, mixtures can be generated using the provided data. The SNR for each target sound event ranges from 5 to 20 dB.

Fig. 5.1 presents the distribution of the isolated training target sound events included in the DCASE2025 Task4 Development Dataset. Respectively, Fig. 5.2 illustrates the distribution of the isolated validation and test target samples. It can be observed that the dataset is not perfectly balanced in training and validation subsets, as some classes are represented by more recordings than others.

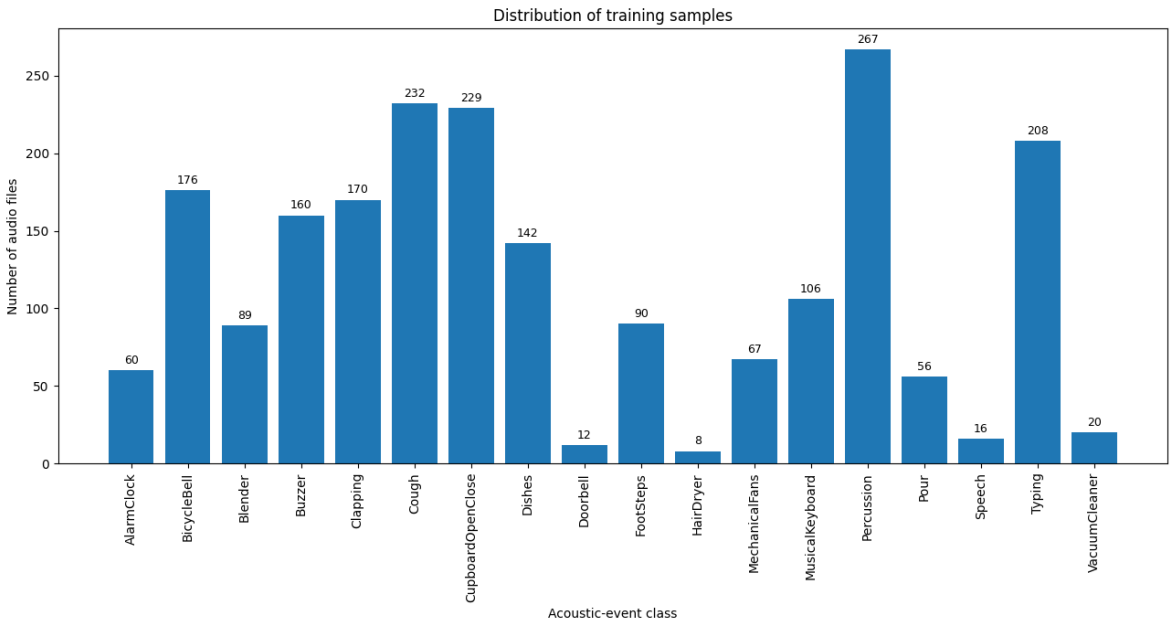
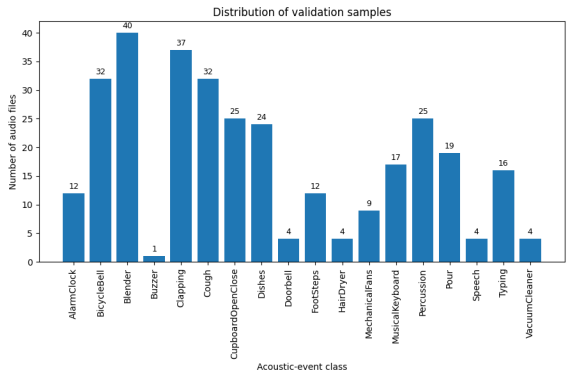
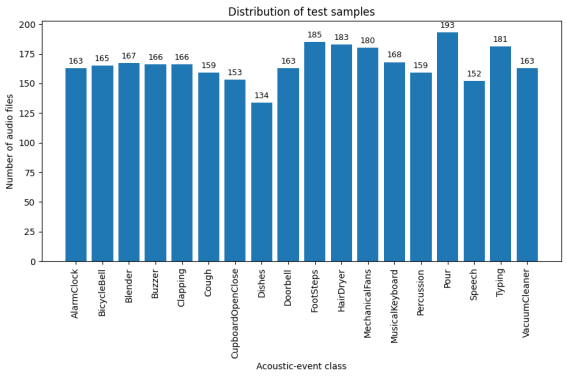


Figure 5.1: Distribution of isolated target training samples across the 18 sound event classes of the DCASE2025 Task4 Development Dataset.



(a) Distribution of isolated target validation samples across the 18 sound event classes of the DCASE2025 Task4 Development Dataset.



(b) Distribution of isolated target test samples across the 18 sound event classes of the DCASE2025 Task4 Development Dataset.

Figure 5.2: Distribution of audio samples across the 18 sound event classes of the DCASE2025 Task4 Development Dataset on validation and test subset.

The list of the 18 target sound event classes consists of multiple acoustic event classes, which exhibit different acoustic characteristics. Fig. 5.3 represents audio classes that contain high energy peaks. These high-energy transient sources, such as “Clapping”, “CupboardOpenClose”, “Percussion”, “FootSteps”, and “Typing”, are characterized by short-duration acoustic events with abrupt amplitude peaks. Their energy is concentrated within relatively small temporal intervals, while the regions between these intervals typically exhibit considerably lower energy.

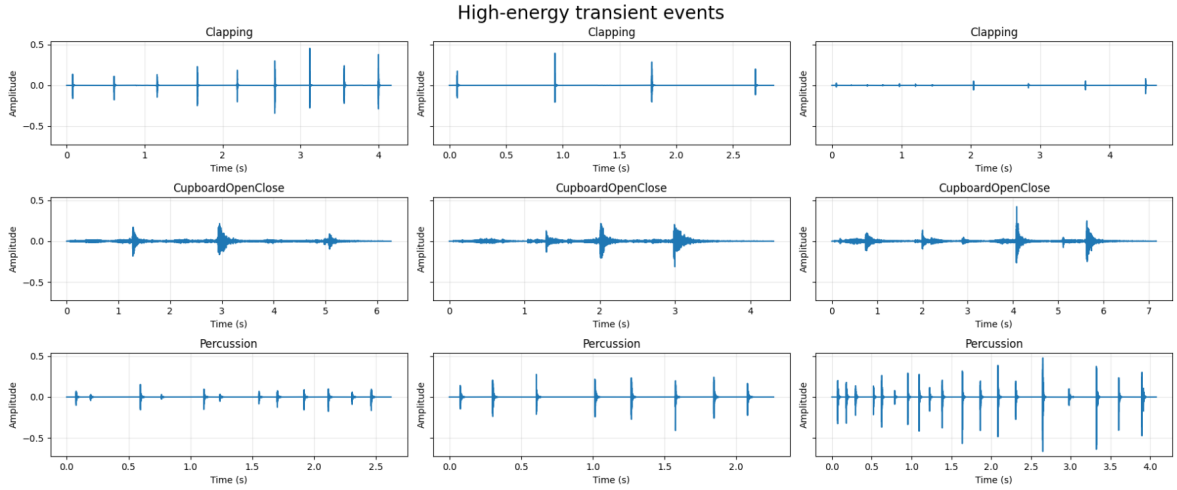


Figure 5.3: High-energy transient sound event classes of the development training subset.

In contrast, Fig. 5.4 illustrates continuous noise-like sources, such as “HairDryer”, “Alarm-

Clock", and "VacuumCleaner". These audio classes are usually non-speech, and non-musical acoustic signals. Their waveforms exhibit sustained and dense temporal activity, with signal energy distributed over extended time intervals rather than concentrated around isolated transient events.

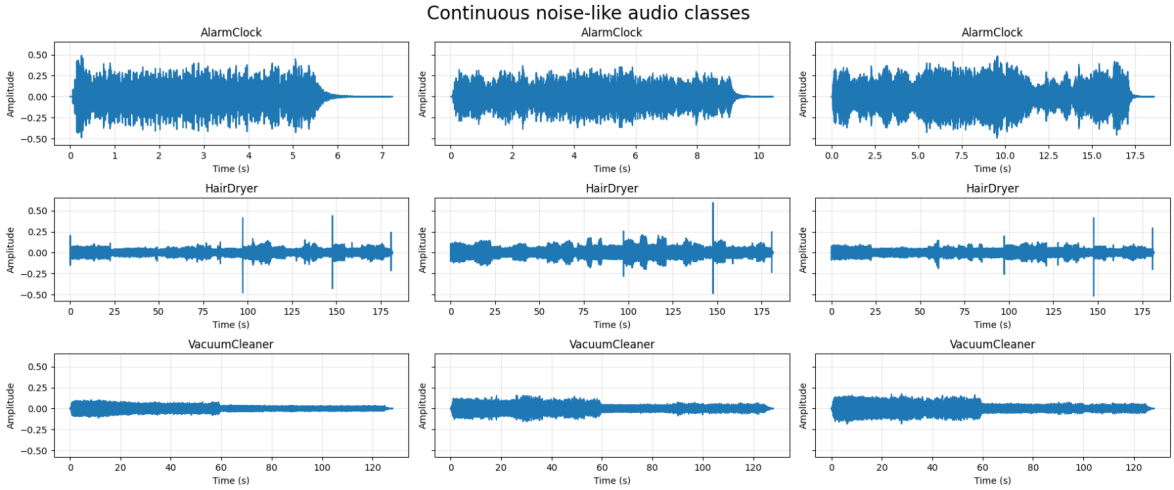


Figure 5.4: Continuous noise-like sound event classes of the development training subset.

The following figure (Fig. 5.5) illustrates periodic sources. Classes such as "BicycleBell", "MusicalKeyboard", and "Doorbell" exhibit repeated acoustic activations with a relatively organized temporal structure. Their waveforms contain repetitive amplitude patterns that reflect the periodic nature of the generated sounds.

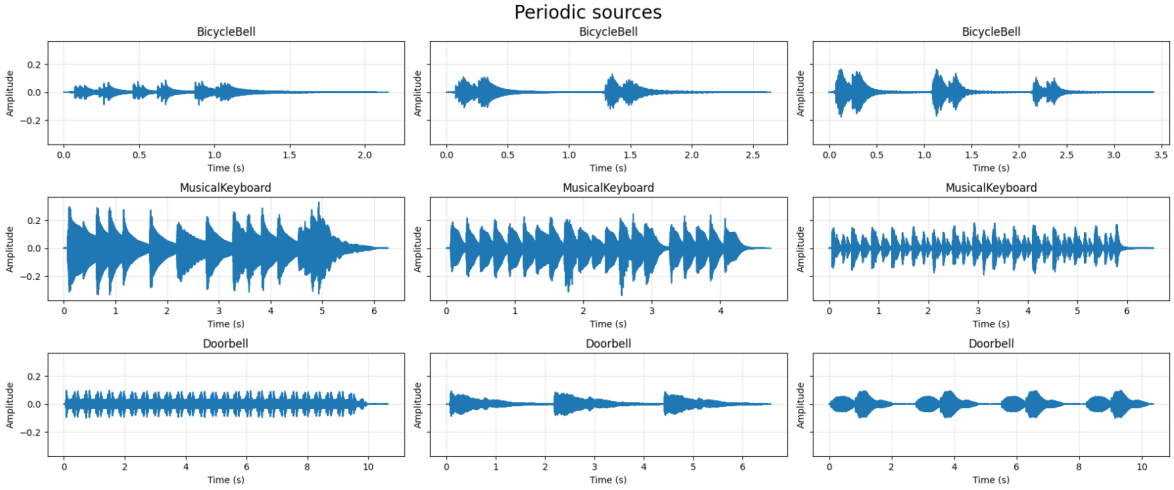


Figure 5.5: Periodic sound event classes of the development training subset.

The variability of the acoustic event classes in terms of temporal and spectral characteristics along with the spatial information observed by the different microphone channels

suggests a more realistic and challenging benchmark for audio source separation and acoustic event analysis.

Finally, each isolated target sound event is stored as a four-channel signal. The four channels represent simultaneous recordings of the same acoustic event obtained from different microphone positions. Fig. 5.6 illustrates an example from the "BicycleBell" class, where the waveforms share a common temporal structure, since they originate from the same source, but they exhibit channel-dependent variations in amplitude and waveform shape. These differences result from the spatial arrangement and directional response of the microphones used during the recording process.

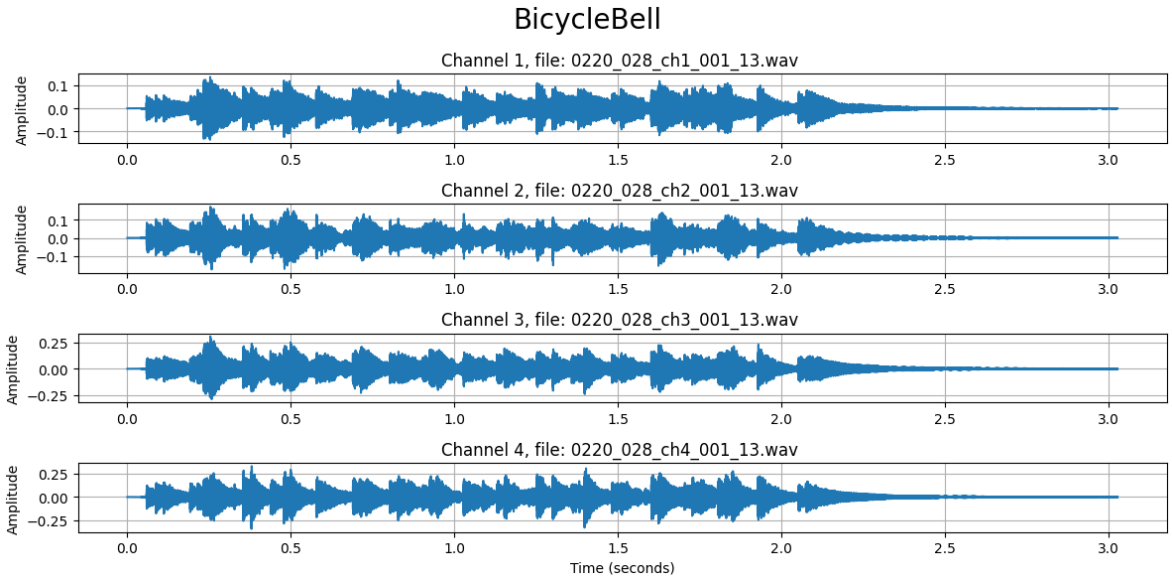


Figure 5.6: Four-channel waveform representation of an isolated "BicycleBell" target sound event from the DCASE2025 Task4 Development Dataset.

5.2 Evaluation Dataset

The evaluation set [67] includes all the multichannel spatial mixture signals that are generated by mixing 1 to 3 samples from the 18 sound event classes recorded in an anechoic chamber convolved with recorded RIRs. Each mixture also includes 1 or 2 interfering sound events, with SNRs ranging from 0 to 15 dB. Each mixture is represented by four channels in first-order Ambisonics B-format (AmbiX).

Since this dataset is designed for evaluation, it does not include the isolated sound events, RIR, or noise of the sound event itself, but only the mixtures after synthesis. The evaluation

data does not include any publicly available data.

Evaluation mixtures $y^{(m)}$ are synthesized using Spatial Scaper [72]. Spatial Scaper is a library used to simulate and augment soundscapes for sound event localization and detection (SELD) in realistic rooms. The evaluation set consists of 2290 files with the following characteristics:

- The duration of all mixture clips is fixed at 10 seconds.
- Each mixture clip contains between one and three simultaneous acoustic events.
- All files are available in 16-bit wav format at 32 kHz.

Fig. 5.7 illustrates the first 6 files (eval_0000.wav , ... , eval_0005.wav) of the evaluation set. Each soundscape of Fig. 5.7 has been generated by mixing 1 to 3 samples from the 18 sound event classes. The evaluation set does not provide any information about the labels of the acoustic events existed in each mixture.

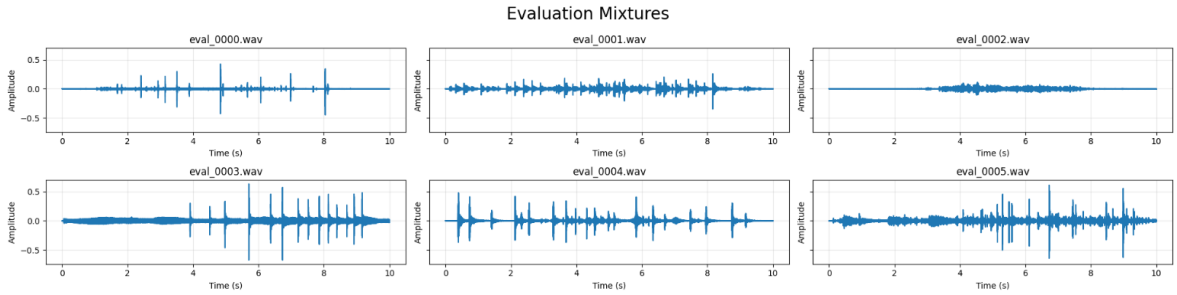


Figure 5.7: Waveforms of the first 6 evaluation mixtures from the evaluation set.

The observed multichannel mixture signals of the evaluation set can be modeled as

$$y^{(m)}(t) = \sum_{k=1}^{K} h_k^{(m)} * s_k \; + \; n^{(m)}. \tag{5.1}$$

Specifically, $h_k^{(m)}$ represent multichannel RIRs measured in various rooms, s_k stand for the one-shot recordings of diverse sound event classes, and $n^{(m)}$ presents the environmental noise recorded with the same multichannel microphone array used for the RIRs.

5.3 Evaluation Ground Truth

This dataset provides the ground truth data [68] for the DCASE2025 Task4 Evaluation Dataset. It contains the isolated ground truth signals for the target sound events in the 2290

mixtures of evaluation set. These reference isolated target sources are used to objectively evaluate the performance of the separation algorithms by comparing the estimated sources with their corresponding ground truth signals. Fig. 5.8 illustrates the first three evaluation mixtures together with their corresponding ground truth target sources.

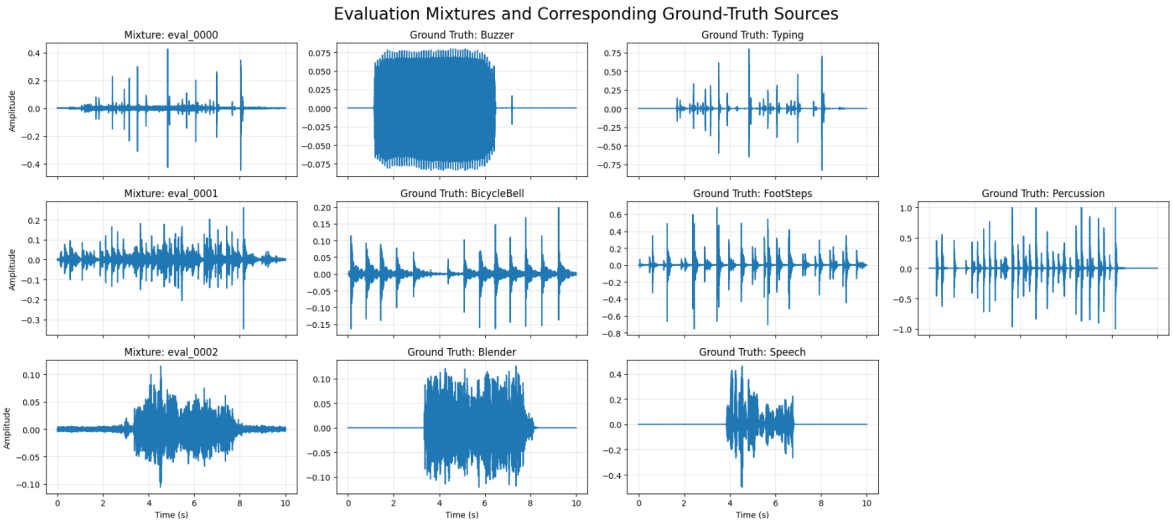


Figure 5.8: Relationship between evaluation mixtures and their corresponding isolated ground truth target source signals.

5.4 Recording Details

Isolated target sound events s_k were recorded in an anechoic chamber using microphones: three (3) cardioid microphones to capture the sound events from the left, front, right and one (1) omnidirectional microphone to capture the sound from above.

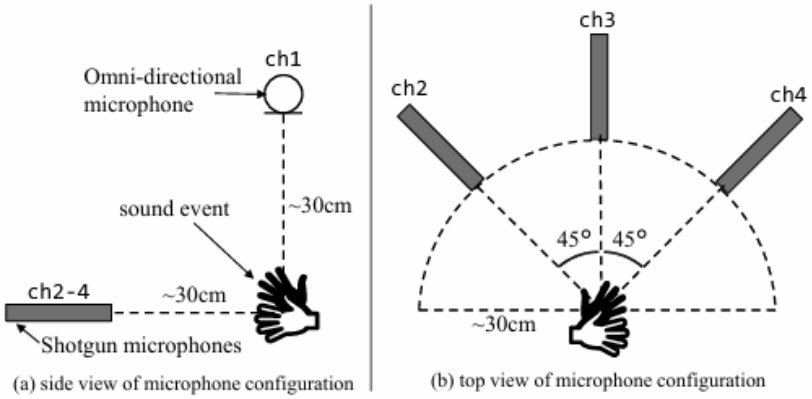


Figure 5.9: The microphone configuration used to record isolated sound events in an anechoic chamber [8].

All RIRs were captured with the same first-order Ambisonics microphone (Sennheiser AMBEO VR Mic) and are provided in B-format (AmbiX) [8]. The new recordings comprise 540 RIRs measured at five positions distributed across three acoustically distinct rooms. At each position, 108 RIRs were gathered while systematically varying the source-microphone geometry:

- the azimuth was swept in 20° steps to cover the full 360°,
- the elevation was set to -20°, 0°, or 20°,
- the source-microphone distance was chosen from the range 0.75-1.50 m.

Chapter 6

Experimental Setup

This Chapter describes the experimental setup used to evaluate the BSS and AEC stages of this work. Since the two tasks involve different processing pipelines, the chapter is divided into two main parts. The first part presents the configuration of the IVA method and the two optimization techniques implemented for this Thesis, including the STFT parameters, the number of iterations and the core parameters. The second part describes the classification experiments, including the input representation, model configuration, and training procedure. The proposed algorithms and experiments were implemented in Python and executed on the Leonardo EuroHPC supercomputer (CINECA, Italy) using compute nodes equipped with 32 Intel Ice Lake CPU cores (2.60 GHz) and 512 GB RAM.

6.1 Experimental Setup for Independent Vector Analysis

The goal of the experiments is to analyze the separation performance of the IVA approach described in Chapter 2. We mainly focus on AuxIVA and GradIVA, in a determined scenario of 1-3 sources and four microphones, considering only the performance after convergence. Each multichannel mixture signal has a 10s duration and $f_s = 32kHz$ sampling rate. In order to determine the best performance of our models, we have executed multiple experiments by trying different STFT configurations and core parameters.

6.1.1 AuxIVA

For AuxIVA, we have chosen a Hamming window, as it has showed the best performance in our experiments. The algorithm has been tested in three STFT configurations, as shown in

Table 6.1:

Table 6.1: STFT parameters for AuxIVA

FFT length	Hop length	Iterations
2048	1024	20
4096	1024	15
4096	512	15

By changing the STFT parameters, the experiments showed no significant difference in terms of the evaluation metrics, while increasing the FFT length to 4096 required more processing time. Consequently, a Hamming window with FFT length 2048, 50% overlap and 20 iterations has been selected as the final AuxIVA configuration.

As discussed in Section 2.4, the core parameters that describe the AuxIVA are β, γ, and ϕ_0. The AuxIVA has been applied to different $\beta = 0, 0.2, 0.4, 0.6, 0.8, 1$. The following figure (Fig. 6.1) represents the performance of the cost function across all different βs. The cost function decreases rapidly during the first iterations for all tested β values and converges to approximately the same final value after 4–5 iterations. Although AuxIVA almost converged at 4 or 5 iterations, we applied 20 iterations to evaluate the best separation performance in this experiment. As mentioned in [13], the best choice of β ranges between 0.2 and 0.4. Therefore, we choose $\beta = 0.2$ for source separation. Moreover, $\gamma = 1$ and $\phi_0 = 1000$ were adopted following the parameter settings reported in [13].

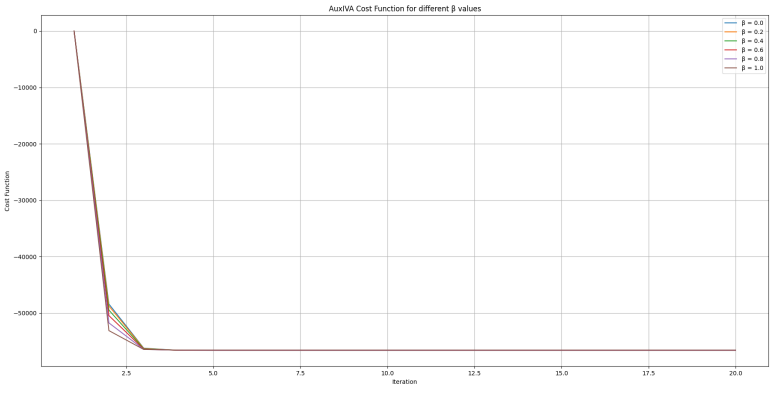


Figure 6.1: Convergence behavior of the AuxIVA cost function for different values of β.

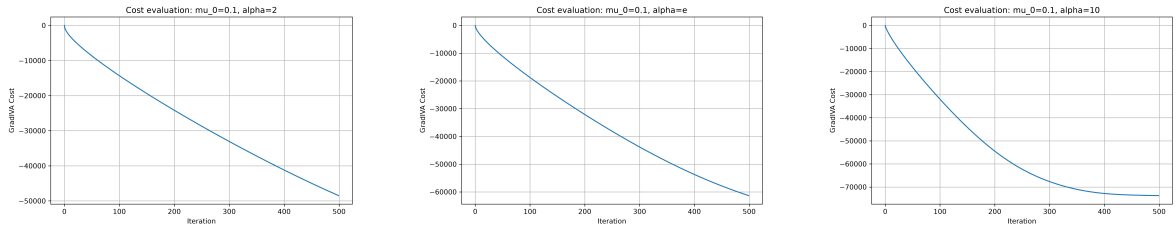
As mentioned in Section 2.4, the IVA implementation considered in this work operates on $M = 4$ microphone channels and consequently produces four estimated output signals.

However, each mixture contains, at most, $J = 3$ active sources, resulting in more estimated outputs than target sources. For evaluation purposes, the reference source signals provided by the ground truth dataset are used to identify the estimated output corresponding to each target source. Specifically, the separation metrics are computed between the estimated outputs and the corresponding reference signals, allowing the relevant outputs to be retained while the remaining non-target output(s) are discarded.

6.1.2 GradIVA

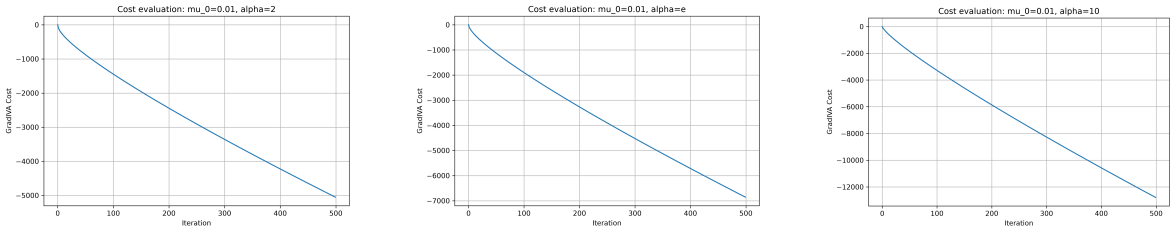
For GradIVA, the FFT of the STFT has been also set to 2048, with 50% overlap and a Hamming window as well. Moreover, in order to simplify the complexity of the algorithm, demean and whitening data preprocessing methods have been applied to the original mixtures in order to reduce the amount of computation during iteration, as [49] reports. Finally, the GradIVA requires the specification of the learning rate μ, the parameter α that controls the decay of the learning rate, and the number of iterations N.

To investigate the influence of the GradIVA optimization parameters, different learning rates and decay parameters were evaluated. The learning rate μ was set to 0.1, 0.01 and 0.001. The parameter α will be tested in three scenarios: 2, e and 10. The number of iterations, N, was initially fixed to 500, resulting in nine parameter configurations. Fig. 6.2–6.4 illustrate the corresponding evolution of the IVA cost function.



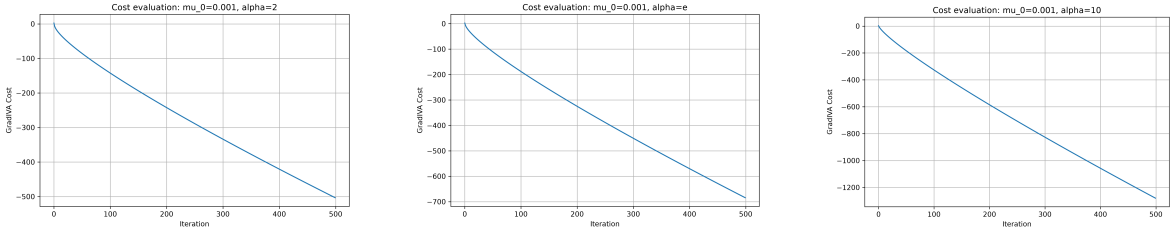
(a) $\mu = 0.1, \alpha = 2$ (b) $\mu = 0.1, \alpha = e$ (c) $\mu = 0.1, \alpha = 10$

Figure 6.2: GradIVA Cost Function Evolution for $\mu = 0.1$ and Different Values of α



(a) $\mu = 0.01, \alpha = 2$ (b) $\mu = 0.01, \alpha = e$ (c) $\mu = 0.01, \alpha = 10$

Figure 6.3: GradIVA Cost Function Evolution for $\mu = 0.01$ and Different Values of α



(a) $\mu = 0.001, \alpha = 2$ (b) $\mu = 0.001, \alpha = e$ (c) $\mu = 0.001, \alpha = 10$

Figure 6.4: GradIVA Cost Function Evolution for $\mu = 0.001$ and Different Values of α

As illustrated in Fig. 6.2–6.4, the GradIVA cost function exhibits a considerably slower stabilization compared with AuxIVA. For most of the examined parameter combinations, the cost function continues to decrease throughout the 500 iterations without reaching into convergence, as occurs in AuxIVA, indicating that a more stable cost function does not necessarily correspond to improved source separation performance. Since the optimization behavior is similar to all the possible combinations of μ and α, the separation performance have been checked primarily in terms of evaluation metrics. The most balanced performance along the aforementioned experiments is considered to be the configuration $\mu = 0.01$, $\alpha = 2$, and $N = 500$, as described in 7.1.3.

In contrast to AuxIVA, GradIVA is considerably more difficult to optimize and typically requires a larger number of iterations to achieve satisfactory separation performance. While AuxIVA usually converges within a relatively small number of iterations due to its auxiliary-function-based update rule, GradIVA may require several hundred or even thousands of iterations, as observed in our experiments and also reported in [11], [49]. One of the main challenges in GradIVA is the selection of an appropriate learning rate. As discussed in [49], the learning rate determines the convergence speed and accuracy of algorithm. If the factor is too small, the computations will increase greatly, and at the same time have no obvious

improvement on the performance. If the factor is too big, it can make the algorithm have no convergence or have big performance error after convergence, so the selection of iterative factor should make the algorithm convergent fast and its performance errors should be as small as possible at the guarantee convergence conditions.

6.2 Experimental Setup for Convolutional Neural Network

In this Section we analyze the experimental setup of the CNN model implemented to classify the separated signals extracted through the separation stage. We mainly focus on the feature extraction of the training audio signals, the structure of the model and the training procedure.

Feature Extraction

For each training audio signal, a spectrogram is computed using the STFT with an FFT size of 1024 samples and a hop length of 512 samples, corresponding to 50% overlap between consecutive analysis frames. Then mel filter banks with 128 bands are applied on the spectrogram followed by logarithm operation to obtain log-Mel spectrogram as input to the CNN architecture.

Neural Network Architecture

In the proposed CNN model, the input to the network is a log-Mel spectrogram with dimensions $1 \times 626 \times 128$, where 1 represents the input feature map, 626 corresponds to the number of time steps, and 128 represents the Mel-frequency bands. Then, four two-dimensional convolutional layers are used, where each of them is followed by a 2×2 max-pooling operation which reduce the temporal dimension from 626 to 313, 156, and 78 respectively and the Mel-frequency dimension from 128 to 64, 32, and 16 while retaining the number of feature maps, before moving to the next convolutional layer. The convolutional layers employ 3×3 kernels with 32, 64, 128, and 256 filters, respectively. Each convolutional layer uses a ReLU activation function and Batch Normalization [73].

Through the convolution operations, the network learns local spectro-temporal patterns from the input log-Mel spectrogram. After the final convolutional layer, a Global Average Pooling [57] is then applied across the temporal and frequency dimensions, transforming the $256 \times 78 \times 16$ feature representation into a fixed 256-dimensional feature vector. A dropout rate [74] of 0.4 is then applied to reduce the effect of overfitting. This vector is further processed a fully connected layer containing 128 neurons with ReLU activation function and Batch Normalization. Again, a dropout rate of 0.3 is applied to reduce the effect of overfitting. Finally, the output feed-forward neural network layer has 18 output nodes where each

of which corresponds to an acoustic event class and employs a softmax activation function to obtain the class probabilities and assign a class label to each estimated separated input source. The integrated structure of the implemented CNN model is described in Table 6.2.

Table 6.2: Configuration of CNN.

Layers	**Output size** (feature maps × time steps × mel bins)
Input log mel spectrogram	$1 \times 626 \times 128$
Conv2D $\{3 \times 3, 32, \mathrm{BN}, \mathrm{ReLU}\}$	$32 \times 626 \times 128$
Max Pooling $\{2 \times 2\}$	$32 \times 313 \times 64$
Conv2D $\{3 \times 3, 64, \mathrm{BN}, \mathrm{ReLU}\}$	$64 \times 313 \times 128$
Max Pooling $\{2 \times 2\}$	$64 \times 156 \times 32$
Conv2D $\{3 \times 3, 128, \mathrm{BN}, \mathrm{ReLU}\}$	$128 \times 156 \times 32$
Max Pooling $\{2 \times 2\}$	$128 \times 78 \times 16$
Conv2D $\{3 \times 3, 256, \mathrm{BN}, \mathrm{ReLU}\}$	$256 \times 78 \times 16$
Global Average Pooling	256
Dropout (0.4)	256
Dense $\{\mathrm{ReLU}, \mathrm{BN}\}$	128
Dropout (0.3)	128
Dense $\{\mathrm{Softmax}\}$	18

Training Procedure

The CNN model was trained using the Adam optimizer [75] with an initial learning rate of 10^{-3} and the sparse categorical cross-entropy loss function [76], which is suitable for multi-class classification. Training was performed for a maximum of 50 epochs. Several callback mechanisms were employed to improve training stability and reduce overfitting. The learning rate was reduced by a factor of 0.5 when the validation loss did not improve for 3 consecutive epochs, with a minimum learning rate of 10^{-6}. Early stopping [77] was applied by monitoring the validation loss, with a patience of 10 epochs, while restoring the model weights corresponding to the best validation performance. In addition, the model achieving the lowest validation loss was also saved using a model checkpoint mechanism.

Chapter 7

Evaluation

In this Chapter we report the experimental results of the separation and classification algorithms using the DCASE2025 Task4 Dataset. First, we introduce the evaluation metrics employed for each task. Then, we present the performance of the implemented methods.

7.1 Evaluation in Blind Source Separation

In this Section we discuss the evaluation performance of the IVA algorithm [42]. The separate performance measures are computed by comparing each estimated separated source to its original wanted true source. The only assumption we make is that the true sources are known from the ground truth evaluation dataset.

7.1.1 Performance Metrics

We present the measures that have been used to evaluate the quality of separated signals with respect to the true source signals. First, separate performance measures are computed for each estimated source $\hat{s_j}$ by comparing it to a given true source s_j. If necessary, $\hat{s_j}$ may be compared with all the target sources of the mixture signal and the "true source" may be selected as the one that gives the best results.

The computation of the evaluation involves two successive steps. In a first step, we decompose $\hat{s_j}$ as

$$\hat{s_j} = s_{target} + e_{interf} + e_{noise} + e_{artif} \tag{7.1}$$

where $s_{target} = f(s_j)$ is a version of the true signal s_j modified by an allowed distortion, and where e_{interf}, e_{noise}, and e_{artif} are, respectively, the interferences, noise, and artifacts error

terms. These four terms should represent the part of the estimated separated target source $\hat{s_j}$ perceived as coming from the wanted source s_j and the unwanted sources $(s_{j'})_{j' \neq j}$, from sensor noises $(n_i)_{1 \leq i \leq M}$, and from other causes (like forbidden distortions of the sources and/or "burbling" artifacts). In a second step, we compute energy ratios to evaluate the relative amount of each of these four terms either on the whole signal duration or on local frames.

Before we define the evaluation metrics that have been used for this task, we show how to decompose $\hat{s_j}$ in four terms as in Eq. 7.1. Consider $\mathbf{A}$ is a time-invariant instantaneous matrix and the mixture is separated by applying a time-invariant instantaneous matrix $\mathbf{W}$, $\hat{s}_j$ can be decomposed as

$$\hat{s}_j = (\mathbf{WA})_{jj} s_j + \sum_{j' \neq j} (\mathbf{WA})_{jj'} s_{j'} + \sum_{i=1}^{m} W_{ji} n_i. \tag{7.2}$$

Since $(\mathbf{WA})_{jj}$ is a time-invariant gain, it seems natural to identify the three terms of this sum with s_{target}, e_{interf}, and e_{noise}, respectively, ($e_{\text{artif}} = 0$ here). However, this decomposition cannot be used as a definition of s_{target}, e_{interf}, e_{noise}, and e_{artif}, since the mixing and demixing systems are unknown. Moreover, the first two terms may not be perceived as separate sound objects when an undesired source $s_{j'}$ is strongly correlated with the desired source s_j. Instead, the decomposition that [42] proposes is based on orthogonal projections. Let $\Pi\{y_1, \ldots, y_k\}$ denote the orthogonal projector onto the subspace spanned by the vectors $y_1, \ldots, y_k$. The projector is a $T \times T$ matrix, where T is the length of these vectors. We consider the following three orthogonal projectors:

$$P_{s_j} := \Pi\{s_j\}. \tag{7.3}$$

$$P_s := \Pi\left\{(s_{j'})_{1 \leq j' \leq n}\right\}. \tag{7.4}$$

$$P_{s,n} := \Pi\left\{(s_{j'})_{1 \leq j' \leq n}, (n_i)_{1 \leq i \leq m}\right\}. \tag{7.5}$$

The estimated source $\hat{s}_j$ is then decomposed into four components [42]:

$$s_{\text{target}} := P_{s_j} \hat{s}_j. \tag{7.6}$$

$$e_{\text{interf}} := P_s \hat{s}_j - P_{s_j} \hat{s}_j. \tag{7.7}$$

$$e_{\text{noise}} := P_{s,n}\hat{s}_j - P_s\hat{s}_j. \tag{7.8}$$

$$e_{\text{artif}} := \hat{s}_j - P_{s,n}\hat{s}_j. \tag{7.9}$$

Starting from the decomposition of $\hat{s_j}$ in (7.3)-(7.9), we now define numerical performance criteria used for this task by computing energy ratios expressed in decibels. We define the signal-to-distortion ratio as

$$\text{SDR} = 10\log_{10}\frac{\|\mathbf{s}_{\text{target}}\|^2}{\|\mathbf{e}_{\text{interf}} + \mathbf{e}_{\text{noise}} + \mathbf{e}_{\text{artif}}\|^2}. \tag{7.10}$$

The SDR serves as a global performance measure for the overall separation quality. Low SDR indicates bad separation. Then, we define the signal-to-noise ratio as

$$\text{SNR} = 10\log_{10}\frac{\|\mathbf{s}_{\text{target}} + \mathbf{e}_{\text{interf}}\|^2}{\|\mathbf{e}_{\text{noise}}\|^2}. \tag{7.11}$$

The SNR quantifies the amount of noise in the estimated signal. Low SNR indicates too much noise in the estimated signal. Finally, we define the signal-to-artifacts ratio as

$$\text{SAR} = 10\log_{10}\frac{\|\mathbf{s}_{\text{target}} + \mathbf{e}_{\text{interf}} + \mathbf{e}_{\text{noise}}\|^2}{\|\mathbf{e}_{\text{artif}}\|^2}. \tag{7.12}$$

The SAR assesses the absence of forbidden distortions and “bubbling” artifacts. Low SAR indicates artificial distortion / musical noise.

In general, higher values of SDR, SNR, and SAR indicate better separation performance. Positive values suggest that the desired source component dominates the corresponding distortion, noise, or artifact component, while larger values reflect a more accurate and perceptually cleaner source estimate. In other words, a high and positive SDR value indicates a good separation performance.

7.1.2 Experimental Evaluations of AuxIVA

Now that we have discussed the role of each evaluation metric, we present the results of performance measurement for the IVA algorithm as introduced in Chapter 2. We first introduce the performance of the AuxIVA method.

The separation performance of AuxIVA was evaluated using different STFT configurations, including variations in the FFT size and hop length (see Table 6.1). These parameters

determine the TF representation available to the separation algorithm. A larger FFT size provides a better frequency resolution, which can facilitate the discrimination of sources with overlapping spectral components. However, increasing the analysis window length reduces the ability to localize rapid temporal variations. Conversely, a shorter analysis window provides better temporal localization at the expense of frequency resolution. The hop length determines the temporal spacing between consecutive STFT frames; thus, a smaller hop size, corresponding to a larger overlap, provides a denser representation of the temporal evolution of the signals, although at a higher computational cost. The acoustic event mixtures considered in this work contain both complex spectral characteristics and highly non-stationary components. Therefore, an appropriate STFT configuration should maintain sufficient frequency resolution while also capturing the temporal variations of short-duration and transient events.

Each STFT configuration was evaluated using the first 100 multichannel mixtures of the evaluation dataset. As illustrated in Fig. 7.1, only minor differences in the average separation metrics are observed among the three evaluated configurations. Nevertheless, the configuration with an FFT size of 2048 and 50% overlap provides the best overall performance while also requiring the lowest processing time among the tested settings. This configuration was therefore selected for the remaining AuxIVA experiments as the most favorable trade-off between frequency resolution, temporal representation, separation performance, and computational efficiency.

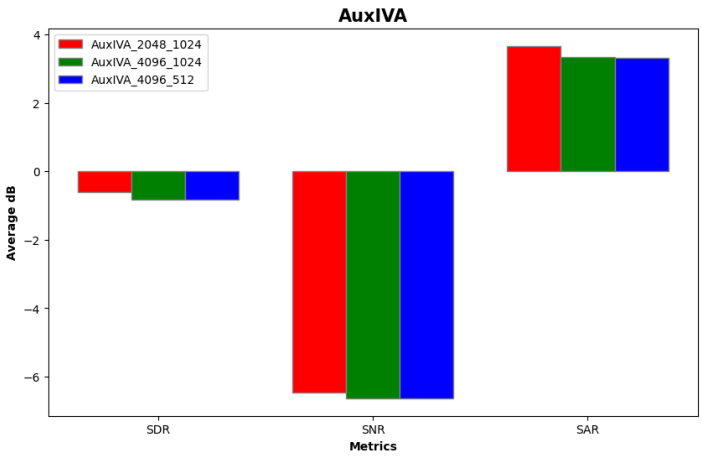


Figure 7.1: Mean SDR, SNR, and SAR values obtained by AuxIVA for the three evaluated STFT configurations. The left bars correspond to SDR, the middle bars to SNR and the right bars to SAR.

Table 7.1 presents the average values of each evaluation metric for the AuxIVA method that are presented in Fig. 7.1. The SDR, which describes the overall separation performance of the estimated signal with respect to its original target source, is slightly below zero. Specifically, the highest average SDR among the evaluated STFT configurations is −0.60 dB. This suggests that AuxIVA does not achieve consistently high separation performance across the entire evaluation dataset. At first glance, this occurs because of the complexity of the acoustic scene considered in this Thesis. Specifically, the mixtures of the dataset consist of multiple acoustic events that might differ in spectral content and temporal structure, and also occur simultaneously causing their corresponding signals to overlap within the recorded mixture. In addition, the IVA algorithm is based on the assumption that the source signals are statistically independent, meaning that the coexistence of acoustic events exhibiting similar acoustic characteristics in the same mixture make the separation performance more difficult. Moreover, the separation performance of AuxIVA depends on the number of the existing sources and their spatial diversity across the microphone array.

Table 7.1: Mean values (dB) of SDR, SNR, SAR for AuxIVA, evaluated on the same 100 multichannel mixture signals.

Method	**Filter length**	**Hop length**	**mean SDR**	**mean SNR**	**mean SAR**
AuxIVA	2048	1024	-0.60	-6.47	3.67
AuxIVA	4096	1024	-0.83	-6.63	3.34
AuxIVA	4096	512	-0.82	-6.63	3.31

Since AuxIVA achieved the best performance for FFT length 2048 and 50% overlap, the algorithm has been applied to 100 more mixtures in order to secure more reliable evaluation results. Therefore, the aforementioned STFT configuration has been tested on the first 200 mixtures of the DCASE2025 Task4 evaluation dataset. The resulting average SDR, SNR, and SAR values were changed to: -0.79 dB, -7.08 dB, and 3.31 dB respectively. Although the average SDR is close to 0, several individual mixtures achieved SDR values greater than 5 dB, demonstrating that AuxIVA can produce reliable source separation under favorable acoustic conditions. Fig. 7.2 illustrates the distribution of the SDR values obtained by AuxIVA for all separated target sources using the best evaluated STFT configuration. The red horizontal line corresponds to the average SDR value (-0.79 dB), while the orange horizontal line corresponds to 5 dB SDR value. It can be observed that a considerable number of the sepa-

rated sources produced an SDR value above these two lines, indicating a superior separation performance.

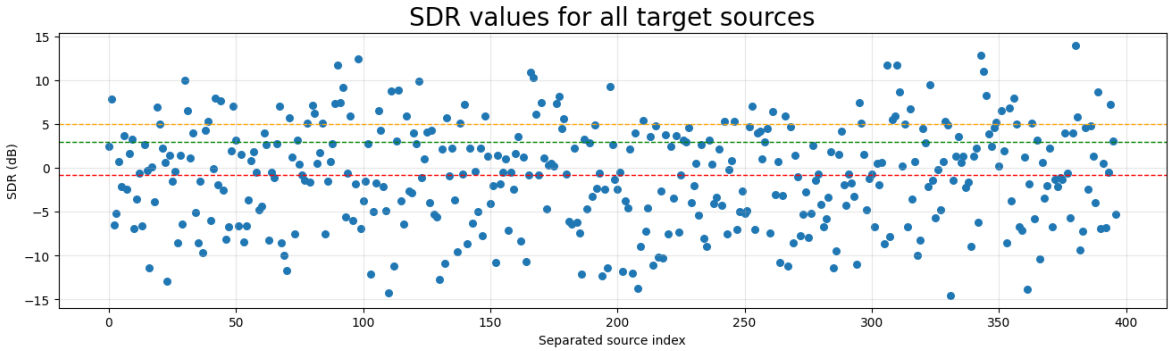


Figure 7.2: Distribution of the SDR values obtained by AuxIVA for all the separated target sources (398 sources). The plot corresponds to the best evaluated STFT configuration (FFT = 2048, 50% overlap, Hamming window). The red dashed horizontal line indicates the mean SDR value, highlighting the variability of the separation performance across different target sources.

Figure 7.3 illustrates the distribution of SDR values across the different acoustic event classes of the DCASE dataset. The line inside each box represents the median SDR of the corresponding acoustic event class and therefore indicates its central tendency. Although some differences in the median SDR and variability can be observed among the classes, no acoustic event class consistently exhibits either high or low separation performance. Most classes contain both positive and negative SDR values. This indicates that the separation performance is not strongly determined by the acoustic event class alone, but is likely influenced by the characteristics of each individual mixture, including the number of active sources and their spatial and spectro-temporal relationships.

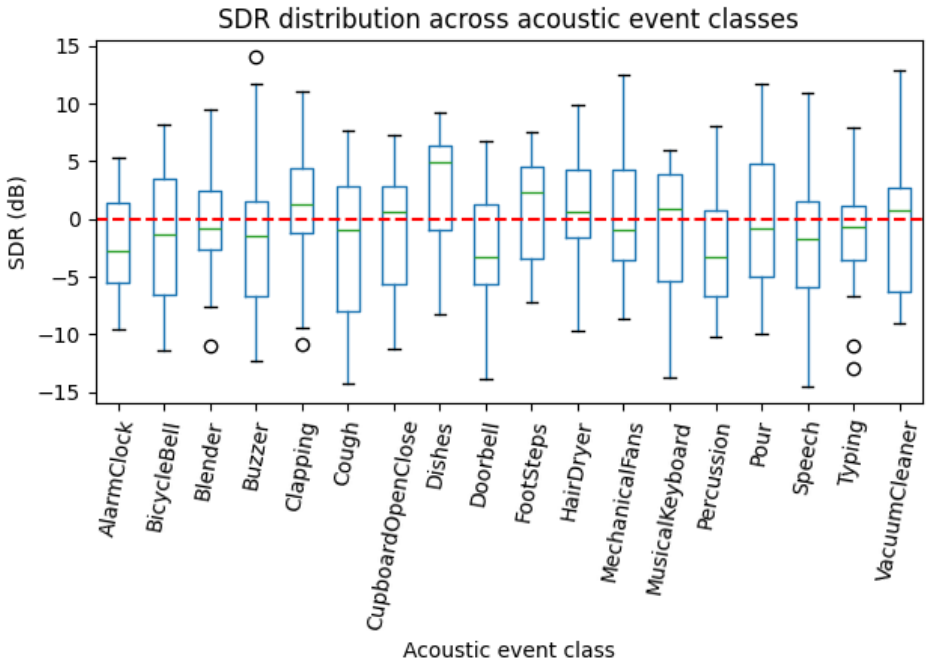


Figure 7.3: Distribution of AuxIVA SDR values across acoustic event classes.

Although Fig. 7.2 and 7.3 illustrate the variability of the SDR values across all separated target sources and acoustic event classes, they do not directly quantify the number of successful and unsuccessful separations. Since the 200 evaluated mixtures contain a total of 397 separated target sources, Table 7.2 summarizes the distribution of the separated target sources across different SDR ranges for the best-performing STFT configuration (FFT length = 2048, 50% overlap).

Table 7.2: Number of separated target sources across different SDR performance ranges for AuxIVA.

SDR range	**Number of sources**	**Separation performance**
SDR < 0 dB	212	Poor separation
0 dB < SDR < 3 dB	74	Marginal separation
3 dB < SDR < 8 dB	89	Good separation
SDR > 8 dB	22	High-quality separation

The previous analysis provided an overall view of the SDR distribution across the mixtures and the acoustic event classes, resulting in the observation of a substantial variability in

separation performance among the individual sources. However, this analysis does not reveal any pattern that may influence the separation performance of the sources. A more detailed examination of specific mixtures is required in order to identify potential factors associated with successful or unsuccessful separation. The following analysis focuses on selected best- and worst-performing cases in order to identify recurring patterns related to the spectral characteristics of the acoustic events, the number of active sources and their spatial configuration.

Table 7.3 summarizes some of the best-performing AuxIVA mixtures. A mixture was considered as a best-performing case if it contained more than one active source and all active sources achieved a positive SDR value. A first observation is that all selected cases contain two simultaneously active sources, suggesting that the separation task becomes more favorable when fewer sources are present compared with the more challenging three-source mixtures and the number of microphones. Moreover, the two sources within most of these mixtures exhibit different dominant microphone channels. In particular, spatial channels 1 and 4 constitute the dominant pair of channels in the majority of the selected cases, indicating a stronger differentiation in the spatial observations of the sources. This suggests that stronger spatial differentiation between simultaneously active sources may facilitate their separation by the multichannel IVA framework. The dominant microphone channel for each source was determined according to the procedure described in Section 4.3.1. On the other hand, no clear pattern can be identified with respect to the acoustic characteristics of the acoustic event classes, since the best-performing mixtures contain sources with substantially different acoustic characteristics.

Table 7.3: Selected Best-Performing AuxIVA Mixtures.

Mixture	Source	SDR (dB)	Dominant Spatial Channel
eval_0000	Typing	7.88	4
	Buzzer	2.50	1
eval_0016	Cough	6.60	1
	Typing	1.12	4
eval_0042	Dishes	7.32	2
	Cough	2.80	3
eval_0043	Buzzer	11.70	1
	Dishes	7.49	4
eval_0060	HairDryer	9.93	1
	Cough	2.83	4
eval_0091	Dishes	5.67	4
	BicycleBell	4.47	1
eval_0112	Dishes	3.19	4
	FootSteps	3.03	2
eval_0123	MusicalKeyboard	5.35	1
	Doorbell	0.89	4
eval_0128	Buzzer	4.26	4
	CupboardOpenClose	1.08	1
eval_0165	BicycleBell	3.55	1
	Buzzer	0.63	4
eval_0174	Doorbell	4.63	1
	CupboardOpenClose	2.47	2
eval_0175	Doorbell	5.22	4
	MusicalKeyboard	0.27	1

In contrast, Table 7.4 presents some of the most challenging AuxIVA cases involving the maximum number of simultaneously active sources ($J = 3$). Again, a mixture was considered a worst-performing case if it contained more than two active sources and at least two active sources achieved a negative SDR value. A common pattern can be observed across these mixtures, since AuxIVA is generally able to separate only one source with a clearly positive SDR

value, while the remaining two sources exhibit substantially lower, mostly negative, SDR values. This suggests that, for mixtures containing the maximum number of active sources ($J = 3$), AuxIVA tends to recover at most one source reliably, while the remaining sources are separated less effectively.

Table 7.4: Selected Worst-Performing AuxIVA Mixtures.

Mixture	**Source**	**SDR (dB)**
eval_0017	MusicalKeyboard	+3.99
	Pour	-5.02
	VacuumCleaner	-8.53
eval_0018	Typing	+4.34
	CupboardOpenClose	-1.53
	HairDryer	-9.66
eval_0020	Percussion	+8.01
	HairDryer	-0.08
	Blender	-6.23
eval_0021	Cough	+7.65
	Buzzer	-1.94
	VacuumCleaner	-2.54
eval_0035	MusicalKeyboard	+5.16
	Dishes	-1.43
	Speech	-1.57
eval_0054	VacuumCleaner	+8.74
	Buzzer	-4.81
	Cough	-14.27
eval_0065	Dishes	+5.71
	Buzzer	-0.87
	Cough	-10.88

The observations derived from Tables 7.3 and 7.4 suggest that the number of simultaneously active sources is strongly related with the separation performance of AuxIVA. To further investigate this relationship, Table 7.5 reports the average SDR obtained for mixtures containing one, two, and three active sources. It can be observed that as the number

of sources increases, the mean SDR decreases. Single-source mixtures achieve the highest average SDR of 4.29 dB, whereas the average SDR decreases to 0.28 dB for two-source mixtures and becomes negative (-2.92 dB) when three sources are simultaneously active. These results support the previous observations from the selected best- and worst-performing cases, indicating that the separation task becomes increasingly challenging when the acoustic scene contains more simultaneously active sources. However, the number of sources should not be considered the only factor affecting the separation performance of AuxIVA, since the spatial configuration and acoustic characteristics of each individual mixture may also contribute to the observed SDR variability.

Table 7.5: Average AuxIVA SDR According to the Number of Active Sources.

Active Sources	Number of Mixtures	Average SDR (dB)
1	63	4.29
2	75	0.28
3	32	-2.92

Finally, the negative mean SNR value of -7.08 dB indicates that an amount of residual noise remains in the estimated source signals. This is expected, since AuxIVA primarily aims at separating statistically independent sources rather than explicitly suppressing environmental noise, and no additional denoising stage was applied to the mixtures. On the other hand, the positive mean SAR value of 3.31 dB indicates that the energy of the useful reconstructed signal components remains higher than the energy of the artifacts introduced by the separation process. Therefore, although the separated outputs remain affected by residual noise, the distortions introduced by AuxIVA itself do not dominate the reconstructed signals.

7.1.3 Experimental Evaluation of GradIVA

As already discussed in the experimental setup, GradIVA proved considerably more difficult to optimize and at the same time more computationally demanding than AuxIVA in our experiments. While AuxIVA typically reached a stable solution within approximately 20 iterations, GradIVA was evaluated using between 500 and 1000 iterations, resulting in a significantly higher computational cost. In addition, GradIVA experiments generally produced lower separation performance than AuxIVA. For these reasons, GradIVA was not evaluated

systematically over the full set of mixtures considered for AuxIVA. Instead, the analysis was limited to a small number of experiments in order to investigate the effect of its main optimization parameters on separation performance.

Despite its overall worse performance, the experiments revealed several observations regarding the influence of the learning rate μ, the decay parameter α, and the number of iterations N. First, we compared the performance of GradIVA for a fixed learning rate of $\mu = 0.1$, while varying $\alpha = 2, e, 10$, with the number of iterations fixed at $N = 500$. The results indicate a slight degradation in separation performance as α increases, reflected by the decreasing SDR values of the same sources. This can be illustrated in Fig. 7.4 where all the three configurations of α were applied to the target sources contained in the first five mixtures of the evaluation dataset. The blue, orange, and green bars correspond to $\alpha = 2$, $\alpha = e$, and $\alpha = 10$ respectively. Overall, increasing α either decreases the SDR values or leaves them nearly unchanged for most target sources, indicating that larger values of α do not improve the separation performance. The only exception is the "BicycleBell" target source in mixture "eval_0001", for which a larger α results in a higher SDR. Therefore, $\alpha = 2$ is considered to be the most appropriate among the tested values.

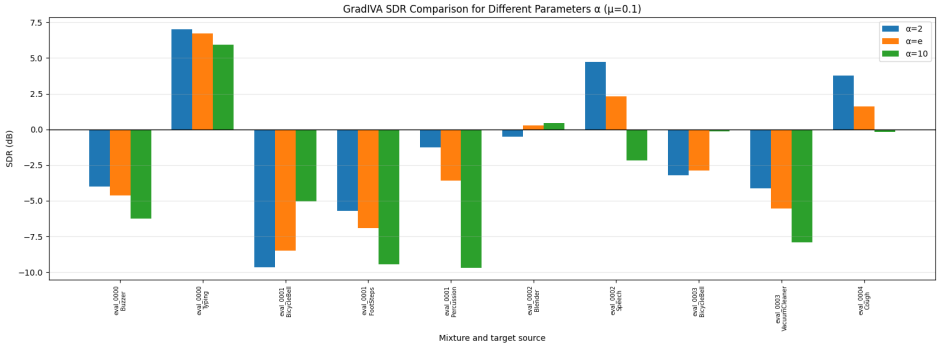


Figure 7.4: Comparison of the SDR values obtained by GradIVA for $\alpha = 2, e, 10$, with a fixed value of $\mu = 0.1$ and $N = 500$.

Since increasing α did not improve the separation performance of GradIVA, the rest of the experiments mainly focused on a fixed value of $\alpha = 2$. For $N = 500$, the learning rates $\mu = 0.01$ and $\mu = 0.001$ produced very similar SDR values across the evaluated target sources. This comparison is illustrated in Fig. 7.5, where both configurations were applied to the target sources contained in the first ten mixtures of the evaluation dataset. The blue bars correspond to $\mu = 0.01$, while the orange bars correspond to $\mu = 0.001$. The results show only minor differences between the two learning rates, with the SDR values remain-

ing nearly identical in most cases. Nevertheless, as discussed in the experimental setup, the configuration $\mu = 0.01$ and $\alpha = 2$ was selected as the most balanced configuration considering the overall separation performance. In addition, by increasing the number of iterations to $N = 1000$ as [11] reports, the SDR values remained approximately unchanged. Consequently, the additional computational cost introduced by the larger number of iterations does not provide a meaningful benefit in terms of separation performance.

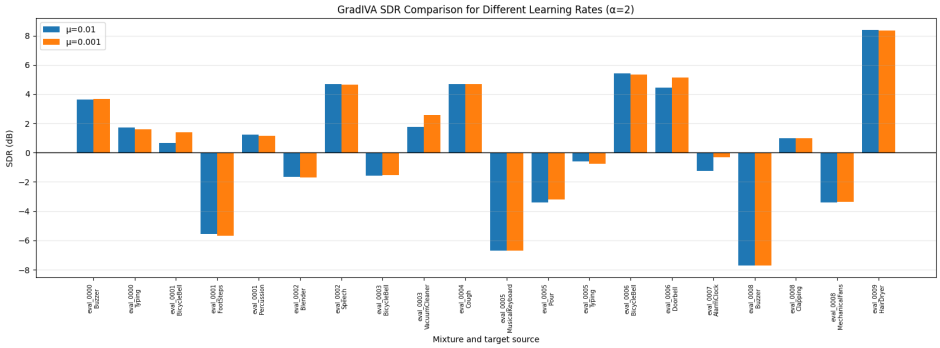


Figure 7.5: Comparison of the SDR values obtained by GradIVA for $\mu = 0.01$ and $\mu = 0.001$, with a fixed value of $\alpha = 2$ and $N = 500$.

7.2 Evaluation in Acoustic Event Classification

In this Section we discuss the evaluation performance of the AEC methods along with the evaluation metrics that have been used.

7.2.1 Performance Metrics

Metrics are a set of measures [3] used to evaluate the performance of machine learning models and to determine how well a model can learn from the training data and to identify areas of improvement in order to optimize the model's performance. The most common of these metrics for classification tasks are accuracy, precision, recall, F1-score, ROC curve (receiver operating characteristic), and AUC (area under the ROC curve) score [78]. Other metrics, such as confusion matrix, sensitivity, and specificity, may also be used to evaluate the performance of deep-learning models.

First, we need to define the confusion matrix, because the calculation of the key measures such as accuracy, precision, and recall is based on this matrix. Confusion matrix [79] is a simple table used to measure how well a classification model is performing. It compares the predictions made by the model with the actual results and shows where the model was right or wrong. This helps you understand where the model is making mistakes so you can improve it. It breaks down the predictions into four categories:

- **True Positive (TP)**: The model correctly predicted a positive outcome.
- **True Negative (TN)**: The model correctly predicted a negative outcome.
- **False Positive (FP)**: The model incorrectly predicted a positive outcome.
- **False Negative (FN)**: The model incorrectly predicted a negative outcome.

Table 7.6: Confusion Matrix.

	Predicted Positive	**Predictive Negative**
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Accuracy is the percentage of the correctly classified data points by an algorithm compared with all data points, described by

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{7.13}$$

Precision is the ratio of correctly predicted data points to the total predicted data points. It actually measures how many of the positive predictions made by the model are actually correct and is defined as

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{7.14}$$

Recall or Sensitivity measures how many of the actual positive cases were correctly identified by the model and is calculated as

$$\text{Recall/Sensitivity} = \frac{TP}{TP + FN}. \tag{7.15}$$

F1-score measures the overall performance of a model by taking the harmonic mean of precision and recall and calculated as

$$\text{F1-score} = \frac{2\,(\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} = \frac{TP}{TP + \frac{1}{2}(FP + FN)}. \tag{7.16}$$

AUC and ROC, on the other hand, are graphs that illustrate TP and FP rates for a given classification model. These graphs are used to evaluate the performance of a model regarding its ability to correctly classify true positives and true negatives.

All the metrics described above range from 0 to 1 with higher values and closer to 1 showing better model performance. The performance of the classification methods will be measured in terms of accuracy, F1-score, precision and recall.

7.2.2 Experimental Evaluation of CNN

Having introduced the evaluation metrics, we now present the performance of the CNN model implemented in this Thesis. Table 7.7 summarizes the classification performance of the trained CNN on the held-out test split in terms of accuracy, F1-score, precision, and recall. The model achieved an overall classification accuracy of 97.39%, indicating that the majority of the test samples were assigned to the correct acoustic event class. The precision of 95.32% indicates that the model produced relatively few FP predictions across the different classes, while recall of 94.86% indicates that a high proportion of the samples belonging to each acoustic event class were correctly identified. The F1-score of 94.34% demonstrates a strong

balance between precision and recall across the 18 classes. However, the CNN has been trained and evaluated on clear target sources, while the final classification task considered in this Thesis involves estimated separated signals, which may contain residual noise and artifacts from other sources. Therefore, the evaluation metrics of Table 7.6 indicate that the model achieves high classification performance on isolated acoustic events, but the ultimate goal of this work is to evaluate the performance of the model on the estimated separated signals.

Table 7.7: Classification performance of the CNN model on the held-out test set.

Model	**Accuracy**	**F1-score**	**Precision**	**Recall**
CNN	0.9739	0.9434	0.9532	0.9486

Fig. 7.6 illustrates the confusion matrix of the CNN classifier on the unseen audio samples of the test set. The strong concentration of predictions along the main diagonal confirms the high overall classification performance reported in Table 7.6. Several classes, including “AlarmClock”, “BicycleBell”, “Buzzer”, “Clapping”, “Cough”, “CupboardOpenClose”, and “FootSteps”, were correctly identified for all test samples. However, specific classes were not classified correctly. Most classification errors are concentrated in only a few classes. In particular, “Dishes” exhibits several misclassifications, with 23 out of 28 samples correctly identified. “MechanicalFans” also shows a small number of errors, with 11 out of 13 samples correctly classified. In addition, it should be mentioned that the test set is considerably imbalanced across the different acoustic event classes. While classes such as “BicycleBell”, “Buzzer”, “Clapping”, “Cough”, contain more than 30 test samples, other classes, including “HairDryer”, “Doorbell”, “Speech” and “VacuumCleaner”, are represented by only a small number of test recordings. The class imbalance across the acoustic event classes is also illustrated in Fig. 5.1 and 5.2(a) of Chapter 5. This imbalance can affect not only the reliability of the evaluation metrics per class, but also the learning process itself. Classes with a limited number of training samples provide the network with a smaller variety of acoustic patterns, which may make it more difficult for the model to learn how to classify these events. Consequently, even if these classes achieve only correct predictions, the result should be interpreted with caution due to their very limited support.

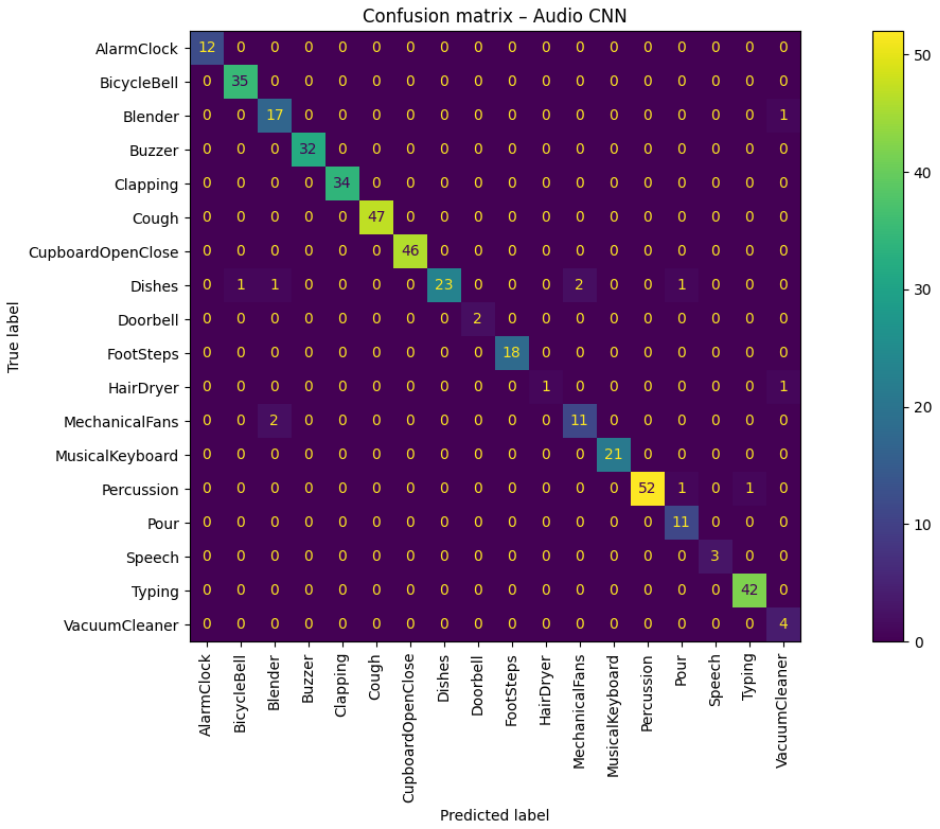


Figure 7.6: Confusion Matrix

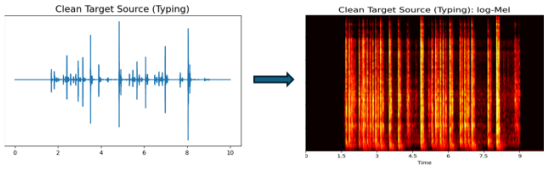
Having evaluated the CNN classifier on clean target source signals, the next and most important experiment investigates its ability to classify the estimated sources produced by the source separation stage. As described in the experimental evaluation of AuxIVA, the evaluated mixtures contain a total of 397 target sources for which a corresponding separated estimate was obtained. Each estimated source was processed using the same preprocessing and feature-extraction procedure employed during CNN training and was subsequently provided to the trained classifier.

Using the highest-probability class as the final prediction, only 135 of them were classified correctly, leading to a 34.0% accuracy. In comparison, the same classifier correctly identified 2,136 out of 3,000 clean target sources of the test set from the DCASE2025 Task4 development set, corresponding to an accuracy of 71.2%. These results indicate a substantial decrease in classification performance when moving from clean target sources to BSS estimated sources, suggesting that residual interference and separation artifacts strongly affect the discriminative acoustic features used by the classifier. Table 7.8 summarizes the classification performance of the CNN model on the estimated separated sources and clean target sources, revealing a substantial difference in classification accuracy for the two cases.

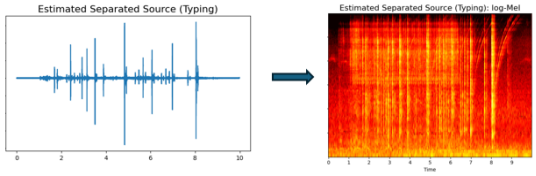
Table 7.8: Classification performance on clean and estimated separated sources.

Input to classifier	**Correct**	**Accuracy**
Estimated separated sources (AuxIVA)	135 / 397	34.0%
Clean target sources	2136 / 3000	71.2%

To sum up, Fig. 7.7.1 and 7.7.2 illustrate the differences between the clean and estimated "Typing" sources. Although their waveforms retain similar transient events, their log-Mel spectrograms differ substantially. The clean source exhibits clearly defined vertical patterns against a low-energy background, whereas the separated source displays more diffuse energy across time and frequency, reducing the contrast of individual typing events. This example illustrates how residual interference and separation artifacts can alter the acoustic features provided to the CNN, potentially contributing to the decrease in classification accuracy.



(a) Waveform and log-Mel spectrogram of the clean "Typing" source, showing distinct transient events and low background energy.



(b) Waveform and log-Mel spectrogram of the "Typing" source estimated by AuxIVA, showing increased background energy and less distinct transient patterns.

Figure 7.7: Waveforms and log-Mel spectrograms of the clean "Typing" source (7.7.1) and the estimated separated source by AuxIVA (7.7.2), illustrating changes in the TF representation associated with residual interference and separation artifacts.

Chapter 8

Experimental Results

This chapter presents the experimental results obtained from the BSS and AEC stages. For each stage, the selected algorithmic configurations are briefly summarized, followed by the corresponding evaluation results.

As a particular method to address the BSS problem of this work, we have used the IVA algorithm. To iteratively optimize the IVA cost function, AuxIVA and GradIVA have been considered. For the STFT transform, both methods used an FFT size of 2048, 50% overlap, and a Hamming window. For AuxIVA we used $\beta = 0.2$, $\gamma = 1$, $\phi_0 = 1000$ and 20 iterations. For GradIVA, the selected configuration was a learning rate $\mu = 0.01$, $\alpha = 2$, and 500 iterations. As discussed in Section 7.1.3, GradIVA was not evaluated over the complete dataset due to its considerably higher computational cost and lower separation performance compared with AuxIVA. These parameters have been identified experimentally and the separation performance of the algorithms has been measured in terms of SDR, SNR and SAR metrics.

The separation performance of the aforementioned configuration of the AuxIVA method was evaluated on the first 200 mixtures of the DCASE2025 Task4 evaluation dataset. The multichannel mixtures consist of one to three simultaneously active acoustic events, observed by four microphone channels. The average SDR value was -0.79 dB, indicating that AuxIVA does not achieve consistently high separation performance across the complete evaluation dataset. However, the individual SDR values of the estimated sources exhibited substantial variability, with several sources achieving clearly positive and high SDR values. The detailed evaluation showed that mixtures containing fewer simultaneously active sources generally yielded better separation performance. In the selected challenging three-source mix-

tures, AuxIVA typically recovered at most one source with a clearly positive SDR, while the remaining sources exhibited substantially lower values. Spatial differentiation between the sources also appeared to contribute positively to separation, while no consistent dependence on the acoustic event class was observed. Overall, the number of simultaneously active sources and the spatial differentiation between them appear to be important factors influencing the separation performance. The average SNR value was -7.08 dB indicating that residual noise remained in the separated signals, whereas the positive mean SAR of 3.31 dB suggests that the artifacts introduced by the separation process did not dominate the reconstructed sources.

GradIVA was not evaluated as extensively as AuxIVA due to its higher computational requirements and lower separation performance in the experiments. While AuxIVA typically reached a stable solution within approximately 20 iterations, GradIVA required several hundred iterations and, in most tested configurations, its cost function continued to decrease without converging. The parameter study indicated that $\alpha = 2$ provided the most consistent performance, while $\mu = 0.01$ and $\mu = 0.001$ produced very similar SDR values. Moreover, increasing the number of iterations from $N = 500$ to $N = 1000$ did not result in a meaningful improvement in separation performance. Therefore, the configuration $\mu = 0.01$, $\alpha = 2$, and $N = 500$ was retained as the most balanced GradIVA setting, although its overall performance remained inferior to that of AuxIVA in the experiments conducted in this work.

The next stage of the proposed framework performs AEC on the estimated separated sources. The CNN classifier introduced in Chapter 6 is employed for this purpose. Features were first extracted from the audio signals using log-Mel spectrograms. The generated spectrograms were then input to the trained CNN, which produces a probability distribution over the 18 acoustic event classes. Before evaluating the separated signals, the model was applied to clean target sources in order to establish its classification capability. The model achieved an accuracy of 97.39% on the unseen audio samples of the held-out test split and correctly classified 2136 out of the 3000 clean target sources on the test set of the DCASE2025 Task4 development dataset, corresponding to an accuracy of 71.2%. The trained CNN was then applied to the estimated sources obtained from the BSS stage. The corresponding separated source signal was also transformed into a log-Mel representation and classified using the same CNN without retraining. Among the 397 evaluated separated target sources obtained using AuxIVA, only 135 were correctly classified according to the highest-probability prediction,

corresponding to an accuracy of 34%. This degradation in the classification performance of the CNN model may be attributed to residual interference, separation artifacts, and distortions introduced by the BSS stage, which may affect the spectro-temporal characteristics of the acoustic event classes the CNN has been trained on.

The final stage of the proposed framework investigates the spatial characteristics of the estimated separated sources in order to estimate the direction of the sound. Since the separation stage produces a source-specific representation for each recovered acoustic event, the spatial analysis can be exploited directly from the corresponding separated output. Two approaches are considered for this work: first, the spatial covariance representation is used to examine how strongly each reconstructed source is represented across the Ambisonic channels; second, the source-specific Ambisonic information is exploited to estimate its DOA in terms of azimuth and elevation. However, these analyses are presented only for representative mixtures and separated sources rather than for the complete evaluation set used for BSS, since no corresponding localization ground-truth information was available to perform a systematic evaluation of the estimated DOAs.

Chapter 9

Conclusions

In this Thesis we tackled the problem of separation, classification, and localization of acoustic events from multichannel mixture signals. The proposed workflow was structured into three consecutive stages. First, we focused on decomposing these mixtures into their individual sources. For that purpose, BSS was performed using IVA-based algorithms capable of separating the existing active sources with little information about the sources or the mixing process, relying entirely on the exploitation of the statistical dependencies and the spatial information captured by the microphone array. With the separation module acting as a preprocessing step, the resulting separated signals were then used as inputs to the CNN classifier, where the objective was to identify the acoustic event represented by each estimated source. Finally, the spatial information associated with the separated sources was exploited to estimate their DOAs relative to the microphone array. In particular, each source was spatially characterized in terms of its azimuth and elevation, together with its source-specific response across the microphones. Overall, the developed framework investigated how BSS can support both the semantic identification and the spatial characterization of multiple acoustic events occurring within the same acoustic scene.

For each stage of the proposed framework, both a theoretical foundation and an experimental investigation were provided. The theoretical part introduced the fundamental principles of each task and presented a detailed description of the employed IVA-based separation algorithms and the CNN classification model, together with the mathematical formulation used to exploit the spatial information captured by the microphone array for source localization. The experimental part examined their implementation and performance using data derived from the DCASE2025 Task4 dataset, comprising multichannel mixtures of overlap-

ping acoustic events from 18 classes, recorded using four microphones and accompanied by their corresponding clean reference signals.

Two optimization algorithms were used for IVA to separate acoustic sources: AuxIVA and GradIVA. Across a total of 200 mixtures from the evaluation dataset, the average separation performance of AuxIVA in terms of SDR was -0.79 dB. We attribute this score to the complexity of the dataset with regard to the variability of the acoustic event characteristics, residual interference, background noise, and processing artifacts. The GradIVA algorithm was not applied on the full set of mixtures because of its computational complexity and its poorer separation performance with respect to AuxIVA. Unfortunately, we did not have time to develop other BSS methods to compare them with the IVA approach. The method nevertheless achieved satisfactory performance under certain conditions, although we would have liked to achieve an even higher average SDR value. For the classification of the estimated separated source signals, the CNN model classified correctly 135 out of the 397 individual separated sources from the 200 mixtures, an accuracy of 34.0%. The performance of the CNN model on clean target sources was much better, classifying correctly 2136 out of the 3000 clean target sources of the test set, an accuracy of 71.2%. The huge gap between the classification accuracies obtained for the separated and clean sources indicates that the quality of the BSS output strongly affected the performance of the CNN model. This degradation can primarily be associated with residual interference from competing sources, background noise, and separation distortions, which affect the discriminative acoustic patterns learned by the CNN from clean training examples. The relatively low average SDR obtained by AuxIVA along with the negative SNR value (-7.08 dB) further supports this interpretation, as most of the separated signals remained considerably different from their corresponding clean references.

Finally, this Thesis developed an integrated framework inspired by the ability of the human auditory system to perceive and interpret complex acoustic environments. By combining BSS, AEC, and ASL, the proposed approach aimed not only to identify the multiple simultaneously active sources, but also to isolate their signals and estimate their spatial directions relative to the microphone array. Fig. 9.1 illustrates a practical example of the complete processing pipeline, from a multichannel mixture to separated source signals, predicted acoustic event classes, and estimated DOAs. The experimental findings demonstrate the feasibility of this processing pipeline while also revealing the strong interdependence among its individual

stages. In particular, the separation performance of the BSS directly affects the classification and localization stages, establishing the BSS stage as the most important step of the complete framework. Nevertheless, the study provides a structured foundation for the analysis of overlapping acoustic events and identifies clear directions for improving the separation performance, classification accuracy, and DOA estimation of future multichannel acoustic scene analysis systems.

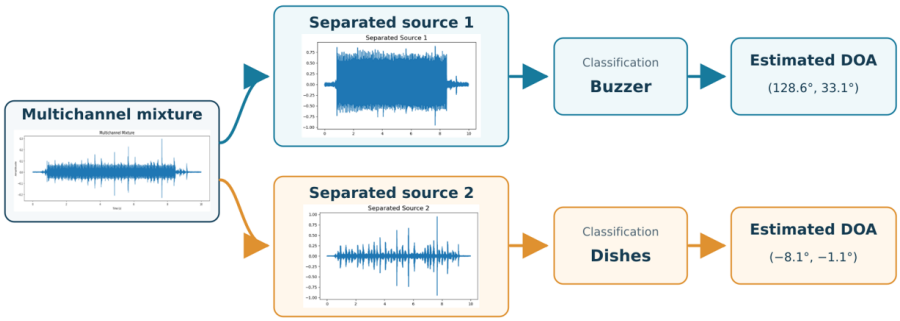


Figure 9.1: Complete processing pipeline of the Thesis in practice. The two separated sources are classified correctly as “Buzzer” and “Dishes”, and a DOA is estimated for each.

9.1 Future Work

A number of promising research directions and system improvement opportunities are noted in conclusion to this Thesis contribution and its implementation work. The proposed system extensions are aimed not only at improving the performance and reliability of the implemented algorithms in the separation and classification stages, but also supporting more recent methods for BSS and AEC in order to significantly increase the system’s capabilities.

First, we conclude that future work should investigate more powerful ways for the optimization of the GradIVA cost function and GradIVA should be evaluated on the same complete set of mixtures used for AuxIVA, allowing a more direct comparison between the two optimization approaches.

A second direction concerns the generalization of the AuxIVA results. In this work, AuxIVA was evaluated on complex multichannel acoustic scenes containing multiple simultaneously active acoustic events. Applying the same implementation to additional benchmark datasets or to datasets previously used in related source separation studies would provide a useful reference for assessing how strongly the complexity of the acoustic scenario affects

separation performance. Such experiments could help distinguish limitations associated with the AuxIVA method itself from those introduced by the specific characteristics of the mixtures considered in this Thesis.

Another critical enhancement involves the integration of IVA with deep-learning approaches for source separation. Recent BSS approaches increasingly combine traditional separation methods, such as ICA and NMF, with trainable neural architectures. Following this direction, the iterative optimization procedure of AuxIVA could be incorporated into a deep-learning framework, where a fixed number of AuxIVA iterations could be used for the optimization of the training procedure. Such an approach could maintain the spatial modeling capabilities of AuxIVA while allowing some of its parameters to be learned directly from data, potentially improving convergence speed and separation performance in complex acoustic environments.

In addition, future work should investigate whether the differences observed between AuxIVA and GradIVA at the separation stage are also reflected in the localization performance. In particular, the spatial information exploited by the separated outputs of both algorithms should be compared in order to determine whether they lead to similar or significantly different source localization results. Such an analysis would clarify whether a method that achieves better performance in terms of separation metrics also provides more reliable spatial information for localization. A further extension would be to evaluate the complete localization framework on datasets for which the ground truth DOAs of the sources are known. This would allow the estimated spatial information and localization outputs to be compared directly with the true source directions using localization metrics. In this way, the effectiveness of the proposed spatial information exploitation techniques could be evaluated.

Finally, a major direction for future work concerns the improvement of the performance of the AEC stage. The classification model used in this Thesis is trained primarily on clean target source signals, whereas during the final evaluation it receives estimated separated signals that may contain residual interference, artifacts, and spectral distortions. Consequently, the performance of the classifier is strongly dependent on the quality of the preceding BSS stage, and the mismatch between the training samples and estimated signals can significantly reduce classification accuracy. To address this limitation, future work could focus on training the classifier using signals that are more representative of the actual outputs of the separation stage. One possible approach is to augment clean training sources with additional infor-

mation, such as additive noise, residual interference, reverberation, or other distortions that imitate typical BSS separated source signals. A more direct approach would be to apply the selected BSS algorithms to training mixtures and use the resulting estimated sources, together with their known ground-truth labels, as additional training or fine-tuning data.

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